



Laser material processing

J Dutta Majumdar & I Manna

To cite this article: J Dutta Majumdar & I Manna (2011) Laser material processing, International Materials Reviews, 56:5-6, 341-388, DOI: [10.1179/1743280411Y.0000000003](https://doi.org/10.1179/1743280411Y.0000000003)

To link to this article: <https://doi.org/10.1179/1743280411Y.0000000003>



Published online: 12 Nov 2013.



Submit your article to this journal [↗](#)



Article views: 3625



View related articles [↗](#)



Citing articles: 82 View citing articles [↗](#)

Laser material processing

J. Dutta Majumdar and I. Manna*

Light amplification by stimulated emission of radiation (laser) is a coherent and monochromatic source of electromagnetic radiation that can propagate in a straight line with negligible divergence. As a result, laser finds diverse applications ranging from mere mundane to most sophisticated uses either for totally commercial or purely scientific purposes, and from life saving to life threatening causes. High power lasers can produce intense heating and perform various manufacturing operations or material processing. The present contribution provides an overview of the application of high power laser only for material processing in engineering applications, and intentionally excludes the scope of application of laser in metrology, biomedical technology, spectroscopy, etc. The manufacturing processes covered have been broadly divided into four major categories, namely, laser assisted forming, joining, machining and surface engineering. Besides discussing the scope and principle of these processes, each section enumerates a detailed update of literature, scientific issues and technological innovations. At the beginning, a brief introduction to different types of lasers and their general applications, fundamentals of laser-matter interaction and classification of laser material processing has been provided. The entire discussion primarily focuses on correlating the properties with processing parameters and microstructure and composition of the material.

Keywords: Laser, Engineering materials, Forming, Joining, Machining, Surface engineering, Microstructure, Properties, Review

Introduction

Light amplification by stimulated emission of radiation or laser is a coherent, parallel, convergent and monochromatic beam of electromagnetic radiation with wavelength in between the ultraviolet (UV) to infrared range.¹ Laser can deliver very low ($\sim$ mW) to extremely high (1–100 kW) focused power with a precise spot size/dimension and spatial/temporal distribution on a given substrate through any intervening medium.^{1–8} Laser has a wide ranging application from printer to pointer, metrology to meteorology, surgery to spectroscopy, isotope separation to invisible surveillance and medical to material treatment.^{5,6} The important properties that enable and justify the use of laser in such a wide spectrum of applications are: spatial and temporal coherence (i.e. phase and amplitude are unique), low divergence (parallel to the optical axis), high continuous or pulsed power density and monochromaticity.

In this contribution, we shall confine ourselves only to laser material processing that involves intense heating of solid targets/components by laser to enable several types of ultrafast, novel and economical processing of materials that distinguish themselves from their conventional counterparts in terms of quality, productivity and efficiency. Before embarking upon reviewing the current

status of different laser material processing, a brief introduction to commercially available lasers and their applications, and classification of laser material processing would be presented. The principle of lasers and physics of laser-matter interaction would also be discussed. Finally, a comprehensive update of different types of laser material processing including the associated scientific and technological aspects of importance would be highlighted. For convenience, the domain of laser assisted material processing or manufacturing has been divided into four sections, i.e. forming, joining, machining and surface engineering. Each process has been summarised below with the current and emerging trends of research and development in that area of laser material processing.

Commercially available laser and its application

The initial foundation of laser theory was laid by Einstein.⁷ In 1960, Maiman⁹ invented the first working ruby laser for which he was awarded the Nobel Prize. Subsequently, several new lasers including semiconductor lasers, Nd:YAG lasers, CO₂ gas lasers, dye lasers and other types of gas lasers were designed and fabricated with better reliability and durability. Table 1 summarises the commercially available lasers, their characteristics and main areas of their application. Depending on the type of laser and wavelength desired, the laser medium could be solid, liquid or gaseous. Different laser types are commonly named according to

Department of Metallurgical and Materials Engineering, Indian Institute of Technology, Kharagpur, W.B. 721302, India

*Corresponding author, email imanna@metal.iitkgp.ernet.in

the state or the physical properties of the active medium. Consequently, there are glass or semiconductor, solid state, liquid and gas lasers. Gas based lasers can be further subdivided into neutral atom lasers, ion lasers, molecular lasers and excimer lasers. The typical commercially available lasers are: solid state or glass laser (Nd:YAG, ruby), semiconductor or diode laser (AlGaAs, GaAsSb and GaAlSb), dye or liquid lasers (solutions of dyes in water/alcohol and other solvents), neutral or atomic gas lasers (He–Ne, Cu or Au vapour), ion lasers (argon (Ar⁺) and krypton (Kr⁺) ion), molecular gas lasers (CO₂ or CO) and excimer laser (XeCl and KrF). The wavelengths of the currently available lasers cover a wide spectral range from the far infrared to the soft X-ray. Laser comprises three principal components, namely, a gain medium, a device for exciting the gain medium and an optical delivery/feedback system.¹⁰ Additional provisions of cooling the mirrors, guiding the beam and manipulating the target are necessary to facilitate material processing.

Laser–matter interaction for material processing

The process of energy deposition from a pulsed/continuous wave (CW) laser beam into the near surface region of a solid involves electronic excitation and deexcitation within an extremely short period of time.¹¹ In other words, the laser–matter interaction within the near surface region achieves extreme heating and cooling rates (10^3 – 10^{10} K s^{−1}), while the total deposited energy (typically 0.1–10 J cm^{−2}) is insufficient to affect, in a significant way, the temperature of the bulk material. This allows the near surface region to be processed under extreme conditions with little effect on the bulk properties.

The initial stage in all laser assisted material processing applications involves the coupling of laser radiation to electrons within the metal. Initially, photon interaction with matter occurs through the excitation of valence and conduction band electrons throughout the wavelength band from infrared (10 μm) to UV (0.2 μm) regions. Absorption of wavelength between 0.2 and 10 μm leads to intraband transition (free electrons only) in metals and interband transition (valence to conduction) in semiconductors. Conversion of the absorbed energy to heat involves: excitation of valence and/or conduction band electrons, excited electron–phonon

interaction within a span of 10^{-11} – 10^{-12} s, electron–electron or electron–plasma interaction and electron–hole recombination within 10^{-9} – 10^{-10} s (Auger process). Since free carrier absorption (by conduction band electrons) is the primary route of energy absorption in metals, beam energy is almost instantaneously transferred to the lattice by electron–phonon interaction. Similarly, transition in semiconductor or polymers having ionic/covalent bonding with energy gap between conduction and valence bands is marginally slower.¹¹

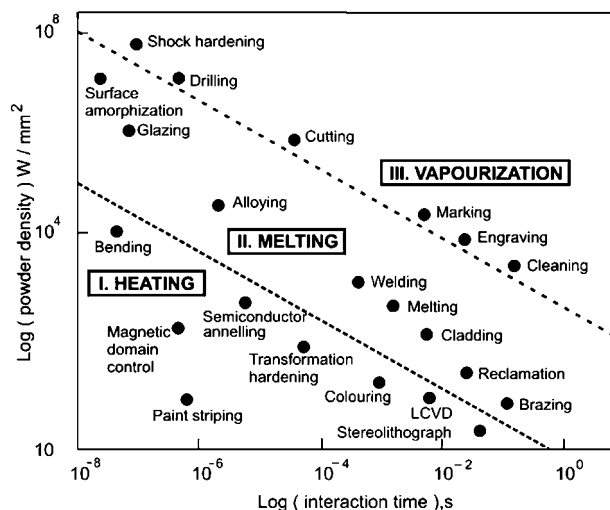
Laser processing of engineering materials

Application of laser to material processing can be grouped into two major classes: applications requiring limited energy/power and causing limited microstructural changes only within a small volume/area without change of state and applications requiring substantial amount of energy to induce the change in state and phase transformation in large volume/area. The first category includes semiconductor annealing, polymer curing, scribing/marketing of integrated circuit substrates, etc. The second type of application encompasses cutting, welding, surface hardening, alloying and cladding. The average power or energy input is relatively low in the first category, while that for the second category is higher as the processes involve single or multiple phase changes within a very short time. Almost all varieties of lasers can perform both types of operations in CW and pulsed mode provided appropriate power/energy density and interaction time for the given wavelength are applied.

The domain for different laser material processing techniques as a function of laser power and interaction time is illustrated in Fig. 1.² The processes are divided into three major classes, namely, heating (without melting/vapourising), melting (no vapourising) and vapourising. It is evident that transformation hardening, bending and magnetic domain control which rely on surface heating without surface melting require low power density laser. On the other hand, surface melting, glazing, cladding, welding and cutting that involve melting require high power density laser. Similarly, cutting, drilling and similar machining operations remove material as vapour hence, need delivery of a substantially high power density within a very short interaction/pulse time. Since all laser material processing

Table 1 Commercially available lasers and their industrial applications

Laser	Discovery	Commercialisation	Wavelength/nm	Application
Ruby	1960	1963	628	Metrology, medical applications, inorganic material processing
Nd:glass	1961	1968	1060	Length and velocity measurement
Diode GaAs/GaAlAs	1962	1965	780–905	Semiconductor processing, biomedical applications, welding
He–Ne	1962		1152	Light pointers, length/velocity measurement, alignment devices
Carbon dioxide	1964	1966	10 200	Material processing – cutting/joining, atomic fusion
Nd:YAG	1964	1966	1064	Material processing, joining, analytical technique
Argon ion	1964	1966	480–515	Powerful light, medical applications
Dye (sodium fluorescein)	1966	1969	535–600	Pollution detection, isotope separation
Copper	1966	1989	511	Isotope separation
Excimer	1975	1976	300–350	Medical application, material processing, colouring
Free electron laser	1971	1997	2000–10 000	Medical surgery, surface modification of polymer



1 Schematic process map in terms of combination of laser power density and interaction time for different types of laser material processing involving either no (only heating) or change of state (melting or vapourisation)¹

operations can be defined by an appropriate combination of power density and interaction time, one is tempted to combine these two into a single scalar parameter as energy density (power density multiplied by time, J mm^{-2}) for the sake of simplification and convenience. However, the exercise is bound to prove futile and not advisable as both the quantum of energy and its temporal and spatial interaction with matter (rather than their product) are crucial to achieve the desired microstructural/phase/state changes and properties for a given material. For instance, application of $10^{-2} \text{ J mm}^{-2}$ energy density may induce either surface hardening (say, for a given material combination of 10^2 W mm^{-2} power density and 10^{-4} s interaction time) or surface melting (say, for a combination of 10^4 W mm^{-2} power density and 10^{-6} s interaction time).

Beam configuration or beam profile also plays an important factor in determining the energy distribution at the interaction zone during laser processing. Four types of beam profiles, namely, Gaussian, multimode, square (or rectangular) and top hat are commonly used for material processing.² A Gaussian beam is most suitable for cutting and welding applications rather than for surface treatment because, being a 'sharp tool,' it tends to vapourise and melt the substrate deeply. In contrast, multimode, top hat and square profiles ('blunt tools') are preferred for surface engineering. Beam shaping is the process of altering an input beam to produce an output beam with a desired spatial or irradiance profile. Beam shaping plays an important role in determining the quality of hardening, welding and laser microprocessing.^{12,13} The beam shaping techniques may broadly be classified into three categories.¹³ The first, and most simple, class of beam shaping is aperturing, which is achieved by using an aperture to allow some portion of the beam power to pass through while the rest of the beam is either reflected or absorbed. A considerable amount of power loss is the major drawback associated with aperturing. In the second type, a field mapper is used to transform an input beam into a desired output beam in a controlled and efficient manner.¹³ This technique could be used to transform a

single mode Gaussian beam into a beam with uniform irradiance.¹³ The third class of laser beam shaping is homogenising or integrating the beam. In this technique, the input beam can be broken up by an array of lenses or facets. Different beam shaping methods may be used for different material processing applications. An elliptical beam may be used for preheating or post-heating during welding.¹⁴ For heat treatment and hardening operations, usually, rectangles, stripes and lines are used. Liu and Kannatey-Asibu¹⁵ used a dual beam (twin spot) configuration to weld with one spot and preheat or post-heat with the other. It was observed that porosity and spatter may be reduced during welding of aluminium with a double focus (twin spot) technique.¹⁶ Furthermore, the shape of the weld penetration profiles may be changed by changing the shape of the beam.¹⁷ Hammond *et al.*¹⁸ used diffractive optical elements to produce a custom power distribution for joining dissimilar materials including combinations of aluminium, Inconel and stainless steel. During hybrid laser-gas metal arc welding technique, application of defocused laser beam was found to increase spreading of the weld pool and to improve the fatigue life.¹⁹ Cho and Farson²⁰ have also simulated and proven the suppression of weld bead humping formation in gas metal arcwelds by adding additional heat input from a defocused laser beam. Others have used oblong spots, twin spots or other custom shapes to reduce weld pool humping at high speeds, decrease porosity, reduce defects from coated materials, alter the penetration profile or improve weld quality.²¹

In the following sections, we will now review the individual classes of laser material processing and the current status of understanding.

Laser assisted forming

One of the major goals of material processing is to produce finished products of desired dimension, shape, geometry, design and properties. Laser material processing offers a unique possibility of manufacturing finished components directly from the raw materials without any elaborate intermediate operation.^{1-4,22} Among the several types of laser assisted forming processes in vogue, the major and successful examples include laser assisted bending, rapid prototyping/manufacturing and reclamation/repairing.¹⁻⁴ These processes distinguish themselves from other laser material processing methods in their proclaimed objective of single step manufacturing of a finished or semifinished product than contributing towards any other intermediate processing steps such as machining, joining or surface engineering. For brevity, we will address all these laser assisted versions of otherwise conventional manufacturing processes as laser forming. In the present section, laser assisted manufacturing techniques such as laser bending and direct laser manufacturing techniques would be discussed in details.

Laser assisted bending

Laser assisted bending involves modifying the curvature of sheet metal by thermal residual stresses induced by laser assisted heating without any externally applied mechanical forces.²³⁻²⁹ Flexibility of non-contact processing, amenability to various types of materials and direct manufacturing of components with diverse shape/

geometry, properties and composition with high precision/productivity are the advantages associated with laser assisted bending. Bending angle and properties of the bent zone may be controlled by controlling parameters such as laser beam radius, power density, interaction/pulse time, material properties (thermal, physical or chemical) and dimension/geometry of the workpiece (thickness, curvature, etc.). Bending of strong and difficult to bend metals (body centred cubic, body centred cubic or hexagonal close packed and hexagonal close packed), intermetallics, composites and ceramics has been an important motivation for the increasing interest in laser assisted bending. The success of laser bending of semiconductor and polymeric sheets is of great interest to the semiconductor and packaging industry. There are three mechanisms of laser assisted bending, i.e. temperature gradient mechanism, buckling mechanism and upsetting mechanism.^{25–29} Many applications involve a complex combination of these mechanisms rather than only one of them.

Namba^{23,24} was one of the first few to attempt and report the high degree of flexibility of laser assisted bending of AISI 1045 and AISI 304 stainless steel sheets by using a defocused CO₂ laser. Subsequently, several studies on laser bending of steel and other metallic materials have been conducted.^{25–28,30} Vollertsen and Geiger³¹ described the possibility of laser assisted straightening of welded component in a car body to reduce distortion arising out of welding by numerical modelling and subsequent experimental study. Hennige *et al.*³² developed a hardware for closed loop controlled laser bending that could produce an accuracy of 0.2° for 10° bending angle, which corresponds to the range of scattering of one single bending operation. Depending on the number of necessary steps to achieve a given bending angle, different types of control strategies could be performed. Geiger³³ described hybrid forming, a combination of conventional forming followed by net shaping by laser forming to combine the speed of conventional forming with the accuracy of laser forming without any special set-up. However, a detailed process planning is required for the large scale application of the hybrid process. Magee and de Vin³⁴ demonstrated the possibility of accurate bending of mild steel sheet with 0.8 mm thickness using the hybrid technique of bending and subsequently, cutting using the same set-up.

Laser assisted deep drawing is a non-contact process in which a laser beam is used to heat the wire to a critical temperature in order to make the wire plastic and easily deformable. High power laser has been successfully employed for drawing of several metals and alloys. Under optimum processing conditions, nickel wire was found to exhibit an increased drawability as compared with conventional annealing.³⁵ Drawing limit (in terms of area reduction ratio, i.e. final/initial cross-sectional area) was found to increase to a maximum value and then decrease with increasing incident laser power.³⁵ This behaviour is attributed to the limit of maximum drawability and plasticity of nickel at a given elevated temperature range. Furthermore, brittle materials can be successfully drawn without any rupture or microcrack using laser. Schuöcker³⁶ developed a mathematical model to predict the temperature and stress distribution during laser assisted deep drawing, showing a considerable reduction in drawing force.

Deformation behaviour of sheet metal (Ti–6Al–4V) by curved laser irradiated path was studied by Chen *et al.*³⁷ using finite element method. Temperature, displacement, stress and strain fields were determined using this model followed by analysis of deformation behaviour using thermomechanical principle. The simulated results showed that three-dimensional deformation could occur only on one side of a scanning path where the rigid constraint was relatively lower. On this side, the sheet is slightly extended and thinned down. On the other side of the scanning path, the sheet only produced slight rigid rotation. This is very different from the linear laser bending, which is characteristic of symmetrical bending related deformation on either side. Wu and Ji³⁸ studied the deformation stress field during the laser bending of sheet metal using three-dimensional thermo–elastoplastic finite element analysis based on the reduced integration method. The stages of laser bending and predicted displacement fields agreed well with the experimental data. Marya and Edwards³⁹ analysed the laser bending of two titanium alloy sheets, Ti–6Al–2Sn–4Zr–2Mo (near alpha alloy) and Ti–15V–3Al–3Cr–3Sn (beta alloy) using a conduction model with a travelling Gaussian heat source. Temperature and bending angle were predicted and correlated with process parameters. Bending was found to initiate at 0.48 of the melting temperature and attained a maximum value at ~0.65 of the melting temperature. Bending behaviour of Ti using a CW CO₂ laser was investigated by Okuda *et al.*⁴⁰ who found that bending angle would primarily depend on laser power and spot diameter. The bending direction tended to change from the laser irradiation side to its opposite side when large laser spot diameter was applied.⁴⁰

Cheng and Lin⁴¹ developed an analytical model which showed an increase in angle of bending with increasing laser power, decreasing scan speed and decreasing sheet thickness. Kovacevic *et al.*⁴² developed a three-dimensional finite element simulation including a non-linear transient and indirectly coupled thermal–structural analysis accounting for the temperature dependence of the thermal and mechanical properties of the materials. The bending angle, stress–strain relation, temperature and residual stress distribution were predicted by the simulation for different materials, sheet thicknesses, scanning speeds and laser powers. The experimental results were suitably compared with the theoretical prediction. A selected number of important studies on the mechanism of laser bending, process description and theoretical modelling have been reported elsewhere.^{28,43–54}

Laser assisted bending of ferrous metals and alloys

Iron and steel happen to be the earliest and most widely explored material to study the usefulness and mechanism of laser assisted bending.^{55–74} Effects of the plastic anisotropy in cold rolled mild steel on bending deformation during the laser forming process have been investigated experimentally and numerically by Peng and Lawrence.⁵⁵ The variables considered were laser power, scanning speed and number of scans for sheets with different degrees of reduction by prior cold rolling. The simulation results are consistent with the experimental observations.⁵⁵ Wu and Zheng⁵⁶ studied the effect of process parameters on bending angle in laser assisted bending of AISI 1008 steel. Bending angle was found to be directly proportional to the laser power and

number of repetition and inversely proportional to the scan speed and beam size/diameter.⁵⁶ Yau *et al.*⁵⁷ used a low power Nd:YAG laser to study the bending behaviour of a thin alloy steel strip used for integrated circuit lead frames. Bending occurred at the threshold line energy of 0.14 J mm^{-1} , beyond which a linear relationship between bending angle and laser energy was observed and attributed to the requirement of a critical temperature for thermal upsetting or permanent change of shape.⁵⁸ Lawrence *et al.*⁵⁹ studied bending of mild steel sheets using a 2.5 kW high power diode laser. The bending angle varied from 0.5° to 40° and increased with increasing applied power, increasing number of passes and decreasing scan speed. It was found that minimum threshold energy was necessary to initiate bending. However, the bending angle did not show appreciable change at very high input energy. Yoshioka *et al.*⁶⁰ studied the bending behaviour of AISI 304 austenitic stainless steel wire (0.1 mm diameter) to develop complex frame structures such as zigzag or crank wire frames with an Nd:YAG laser offering $0.02\text{--}0.2 \text{ J}$ energy. Dutta Majumdar *et al.*⁶¹ carried out a detailed investigation on laser assisted bending of AISI 304 stainless steel sheet of different thicknesses by CW CO_2 laser to determine the extent and rate of bending as a function of laser parameters. The rate of bending increased with increasing applied power density attributed to increased material flow at a higher incident energy and hence, increased thermal distortion associated with the process. However, an optimum range of laser power should be selected so that the applied power density is capable of bending without excess melting, evaporation or crater formation. Wang *et al.*⁶² studied the thermophysical process of bending of St C45 steel (C: 0.08, Si: 0.3–0.1, Mn: <0.4, P: <0.025 and S: <0.025) by a CW CO_2 laser. The dynamic microdeformation of the specimen was measured using a laser beam reflex amplifier system. The final deformation angle was found to depend on a combined action of thermal strain and phase transformation strain.

Tube bending is important in manufacturing of boilers, engines, heat exchangers, air conditioner, tube and pipe products. Li and Yao⁶³ established the mechanism of laser bending of tube. It was found that bending is primarily achieved through thickening of the scanned region instead of thinning of the unscanned region, and the scanned region assumes a slightly protruded shape. The absence of appreciable wall thinning is one of the major advantages of laser bent tubes. The bending radius increased with the beam diameter and the ratio of the tube outer diameter to the wall thickness. Furthermore, the bending efficiency increased with the maximum scanning angle up to a critical value. Hao and Li⁶⁴ developed a numerical model to establish a relationship between the bending angle and processing parameters.

Laser assisted bending of non-ferrous metals and alloys

Liu *et al.*⁷⁵ studied the deformation behaviour of the SiC reinforced 6092 aluminium matrix composite using a 2 kW Nd:YAG laser. The bending angle was found to decrease with increasing scan speed because of the decrease in interaction time between the laser beam and composite. On the other hand, the bending angle increases with increasing applied power. Under a suitable condition for bending, a linear relationship

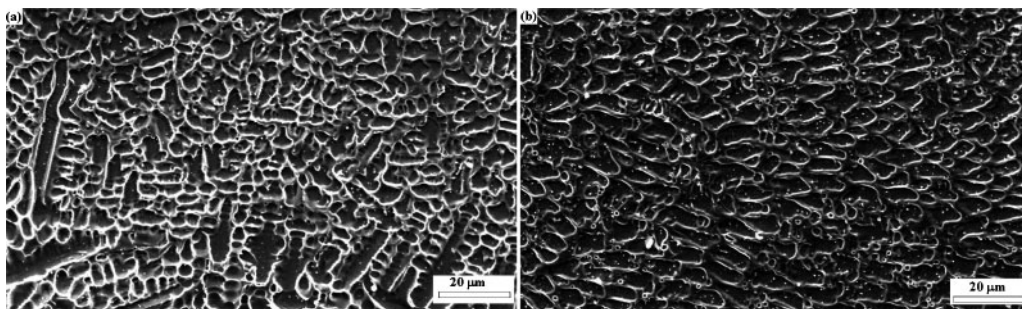
between the bending angle and the number of irradiation passes was observed. Ramos *et al.*⁷⁶ investigated the microstructures of Alclad 2024-T3 Al–Cu alloy following bending by a CO_2 laser. It was observed that the irradiated zone experienced different stages of thermal annealing including recovery (subgrain formation), recrystallisation and grain growth depending on the deposited thermal energy. Recrystallised zone suffered partial melting at the grain boundary corners or triple points leading to precipitation along grain boundaries on cooling. Yau *et al.*⁷⁷ carried out laser bending of Al based metal matrix composite (A2009/SiC) by a high power (2 kW) CW Nd:YAG laser and determined the optimum processing conditions for attaining the maximum bending angle. Similar studies on deformation behaviour of laser assisted bending of aluminium based metal matrix composite (Al 6013/SiC_p) sheets showed that bending could be achieved only under a narrow processing range and increased with increasing laser power, decreasing processing velocity and increasing number of irradiations.⁷⁸ In another study, it was observed that bending angle was inversely proportional to the volume fraction of SiC.⁷⁹ Magee *et al.*⁸⁰ studied the bending behaviour of AlCuMg alloy and observed that bending rate/angle primarily depended on temperature and decreased with the number of pulses due to material accumulation at the bend.

Laser assisted bending is a unique technique to bend the difficult to deform metals and alloys. Okuda *et al.*⁸¹ studied the deformation mechanism of Mg alloy by finite element modelling. Chan *et al.*⁸² compared the deformation behaviour of chromium sheet having limited room temperature ductility subjected to roll compression and laser bending technique. In roll compression test, a bending angle of 90° was achieved without fracture at a temperature of 100°C . However, the maximum bending angle achieved by non-contact laser bending was 23.5° and the same increased with increasing applied power density and number of irradiation. In spite of its poor room temperature ductility, Ti_3Al based intermetallic alloy was reported to be laser bent by Chan and Liang⁸³ primarily due to the differential degree of thermal expansion of the intermetallic across the thickness.

Laser assisted bending has been studied in Ti and its alloys.^{83–90} Magee *et al.*^{84–86} studied the influence of process variables on laser bending of $\alpha + \beta$ Ti alloy (Ti–6Al–4V) using a CW CO_2 laser. Blake *et al.*^{87,88} studied the effect of beam divergence, feedrate, laser power and pulse width on the angle and range in laser assisted bending of Ti–6Al–4V and Ti–15V–3Cr–3Sn–3Al using a 400 W pulsed Nd:YAG and a 50 W desktop CO_2 laser. Similarly, Maher *et al.*⁸⁹ and Hatayama and Osawa⁹⁰ investigated the effect of process variables on the bending behaviour of pure Ti. Marya and Edwards⁹¹ studied the interactions between process variables, materials properties and residual angular distortions (bending angle) during laser assisted bending of Ti–6Al–2Sn–4Zr–2Mo and Ti–15V–3Al–3Cr–3Sn alloys.

Laser rapid prototyping

One of the most recent applications of laser in material processing is the development of small, complex and intricate components by coupling laser with computer controlled positioning stages and computer aided engineering design.^{2,92–96} This technology is based on



2 Scanning electron micrographs of top surface of laser fabricated AISI 316L stainless steel with power densities of *a* 0.031 kW mm⁻² and *b* 0.091 kW mm⁻² and at scan speed of 2.5 mm s⁻¹, and powder feedrate of 136 mg s⁻¹ (Ref. 115)

repetitive deposition and processing of material layers known as additive freeform fabrication technique.^{97–100} The advantages offered by these techniques include the capabilities to produce components of intricate shape with a greater accuracy, faster processing speed, economy in energy and material consumption, etc.

Laser engineered net shaping is a process in which near net shape metal structures are built from powdered metal layer by layer from computer generated designs. Zhang *et al.*¹⁰¹ discussed the characteristics of laser engineered net shaping, directed light fabrication and shape deposition manufacturing. The fundamental of selective laser sintering has been studied by Goode.¹⁰² Dai and Shaw¹⁰³ developed a three-dimensional thermomechanical finite element model to investigate the transient temperature and residual stress fields in laser assisted fabricated part. The model system chosen was 93Cu+7Sn and NiMo mixed with 70%(MoW)2NiB2:30%NiMo. McAlea *et al.*¹⁰⁴ discussed the laser based rapid prototyping technique and compared with the sand and investment casting technique. The possibilities of application of laser assisted deposition in fabrication of part in packaging industry and microelectronic component have been discussed by Bauer *et al.*¹⁰⁵ and Micheli and Boyd¹⁰⁶ respectively. Griffith *et al.*¹⁰⁷ described the use of contact and imaging techniques to monitor the thermal signature during laser engineered net shaping. The understanding of solidification behaviour, residual stress and microstructural evolution with respect to thermal behaviour was discussed.

Metal rapid prototypes and fabrication of moulds using hybrid processes of selective laser cladding and milling techniques have been studied by Jeng and Lin.¹⁰⁸ The selective cladding process was employed to build up the material layer by layer, and the milling operation was employed to level the top surface of the clad to increase the accumulation of powder and accurately control the clad height. Similarly, the fabrications of three-dimensional structure using microelectrodischarge machining and laser assembly and application of laser welding of wire, subsequent deposition and milling for rapid prototyping were discussed by Kuo *et al.*¹⁰⁹ and Choi *et al.*¹¹⁰ respectively.

Laser assisted rapid manufacturing of ferrous components

Laser assisted rapid manufacturing has been extensively studied in iron and steel.^{108–120} Direct laser sintering of iron, Cr–V tool steel, 7Fe–4.3C, 97Al–3Fe, 93.3Cu–2.8Sn–3.9C and 93.4Fe–1.6C–2.5Cu–2.5MoS₂ alloys was undertaken and the effect of process parameters on sintering behaviour was studied.¹¹¹ Components of different shapes from AISI 304 stainless steel were

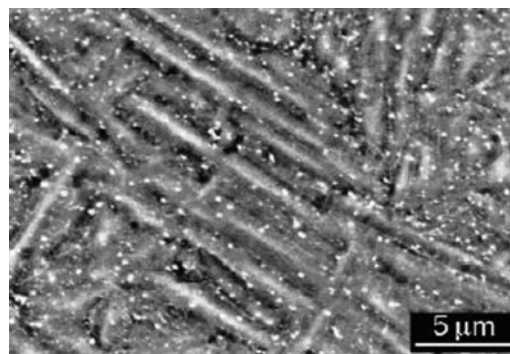
fabricated by Jeng *et al.*¹¹² using selective laser cladding technique using a Rofin Sinar RS820 1500 W CO₂ laser. The effect of process parameters on the clad height and surface quality has been studied for direct laser deposited AISI 316L stainless steel using diode laser and CW CO₂ lasers.^{113,114} Dutta Majumdar *et al.*^{115,116} observed that the morphology and degree of fineness of the microstructure of AISI 316L components fabricated using a 1.5 kW CW diode laser were directly related to the laser parameters. The range of grain size (maximum to minimum) was taken as a measure of homogeneity. It was found that the range of grain size and microporosity reduced to a minimum with increasing scan speed. Figure 2*a* and *b* shows the microstructures of the laser fabricated AISI 316L stainless steel with power densities of 0.031 kW mm⁻² (Fig. 2*a*) and 0.091 kW mm⁻² (Fig. 2*b*) and at a scan speed of 5 mm s⁻¹, with a powder feedrate of 136 mg s⁻¹ (within optimum processing region).¹¹⁵ It is apparent that the grain size is coarser at higher power density and the resultant morphology varied with laser parameters (mixture of cellular and dendritic at lower and predominantly coarse cellular at higher power density). The microhardness is uniform throughout the cross-section along the wall height with a marginally higher value near the substrate and a lower value at the intermediate region. The marginally higher value of microhardness near the substrate region is mainly because of the refinement of microstructure due to a high quenching rate from the underlying substrate. On the other hand, the lower level of microhardness at the intermediate region is attributed to the grain coarsening effect. Application of a higher power density decreases the average microhardness attributed to the coarsening of grains revealed from a detailed microstructural investigation.¹¹⁵ Similarly, application of a higher scan speed increases the average microhardness of the fabricated layer. With increasing scan speed, due to a shorter time of interaction, a low energy is supplied during melting, resulting in the refinement of grains, and hence, an increase in the average microhardness. The effect of powder flowrate on the microhardness does not, however, show any specific trend. From the variation in microhardness with laser parameters, it may be concluded that the hardening of the formed parts is mainly because of the grain refinement and for an improved microhardness, a combination of low power and high scan speed should be chosen. Furthermore, the critical potential for pit formation E_{PPI} [550 mV(SCE)] was superior to that of the conventionally processed AISI 316L stainless steel substrate [445 mV(SCE)] also due to the pronounced uniformity and refinement of grain size.

Laser assisted rapid manufacturing of non-ferrous components

Metallic foam is increasingly becoming popular due to its unique combination of low density and high compressive strength.¹²¹ Aluminium foam has evoked a special interest due to its light weight structure and possibilities of various structural applications in automotive, aerospace and allied industries. However, manufacturing techniques associated with aluminium foam limit the application of it in a wider scale. Kathuria¹²² attempted to fabricate Al–7%Si foam using Nd:YAG laser and obtained a localised and unidirectional expansion of the foam in the direction of laser irradiation with as high as 60% porosity and closed cell structure. The synthesis involved mixing the alloy powder with a foaming agent, cold isostatic pressing of the mixture to form a foamable sandwich precursor and heating it up to its melting point by a high power laser beam. Laser irradiation is useful in inducing rapid decomposition of the foaming agent into hydrogen or some other gaseous product and allowing precipitation of intermetallic phases to strengthen the matrix phase.

Tang *et al.*¹²³ and Wang *et al.*¹²⁴ carried out selected laser sintering of Cu based alloy. Khaing *et al.*¹²⁵ developed a three-dimensional soft and porous structure from a mixture of Ni, bronze and Cu powders using an EOSINT M250 laser sintering machine. Kathuria¹²⁶ investigated the selected laser sintering of three-dimensional components made of Co, Ni and Cu alloys. A list of similar studies on the development of Cu and its alloys by laser assisted rapid manufacturing have been reported elsewhere.^{127–129}

Ti and its alloys are widely used as aerospace components and biomedical implants. Hayano¹³⁰ and Tolochko *et al.*¹³¹ fabricated dental root implants made of Ti by selected laser sintering. Gao *et al.*,¹³² Mei *et al.*,¹³³ Arcella and Froes¹³⁴ and Abbott and Arcella¹³⁵ attempted the fabrication of fully dense and rapidly solidified Ti–6Al–4V components by direct laser deposition. Srivastava *et al.*^{136,137} developed TiAl alloy by direct laser fabrication using gas atomised Ti48Al2Mn2Nb powder as feedstock and a VFA 600 W CO₂ laser unit. The microstructure was heterogeneous and extremely fine. A post-heat treatment was found to improve the microstructural and compositional homogeneity. Banerjee *et al.*^{138–140} developed Ti–6Al–4V–TiB, Ti–25%V and Ti–TiB composites using laser engineered net shaping process. Tomochika *et al.*¹⁴¹ attempted to synthesise NiTi intermetallic compound by laser assisted melting of a combination of pure Ti (300 µm diameter) and pure Ni wire (250–350 µm diameters) followed by high pressure Ar gas assisted atomisation and coating on aluminium substrate. The particle size of the atomised powder ranged from 10 to 125 µm and the thickness of the coating was 3 mm. The sintered compact of equiatomic blend of Ti and Ni was single phase NiTi martensite with martensitic M_f and austenitic finish A_f temperatures of 320 and 365 K respectively. Zhang *et al.*¹⁴² developed TiAl intermetallics by laser engineered net shaping using a CW Nd:YAG laser. Wu and Mei¹⁴³ developed Ti and TiAl alloy parts with various geometries by direct laser fabrication. Tang *et al.*¹⁴⁴ prepared Ti/TiN *in situ* composites by laser induced reaction between Ti powder and N₂ gas and evaluated their mechanical properties. For complex alloy systems, the microstructure of the direct laser clad component was found to be



3 Scanning electron micrograph of direct laser deposited Ti–8Al–1Er¹⁴⁵

different from the single clad layer and varied with the numbers of passes. Figure 3 shows the scanning electron micrograph of the top surface of direct laser clad Ti–8Al–1Er samples after multiclad deposition.¹⁴⁵ The microstructure consists of acicular α' martensite with the presence of very fine and homogeneously distributed Er rich precipitations.

Karlsen *et al.*¹⁴⁶ developed the Co–Cr–Mo superalloy for bioimplant with improved biocompatibility. Ni and Co based superalloys were fabricated by direct laser metal deposition technique.^{146–148} The tensile strength of the laser consolidated Co based superalloy was reported to be 50% higher than that of the as cast or powder fabricated components attributed to the exceptionally fine columnar dendrite microstructure produced by the rapid solidification.¹⁴⁷

A new rapid prototyping method has been developed by combining selective laser sintering and gel casting technique for the fabrication of complex shaped Al₂O₃ ceramic parts.¹⁴⁹ The process involves aqueous gel casting to form high mechanical strength Al₂O₃ based green body by *in situ* polymerisation of Al₂O₃ slurry containing monomer and cross-linker, and sintering of the green compact by selective laser sintering machine. Larrea *et al.*¹⁵⁰ described the procedure for preparing large surfaces of eutectic composites of Al₂O₃–ZrO₂, with thickness up to 250 mm, using a modified laser zone melting. A ceramic precursor was prepared by compaction and sintering (at 1500°C) of powder blends with composition of 37 mol.-%ZrO₂ and 63 mol.-%Al₂O₃. The surface of a ceramic precursor is scanned with a rectangular CO₂ laser beam of 20 × 0.5 mm size at a scan speed of 36–72 mm h^{–1}, which induces surface melting. The microstructure consisted of fine, alternating and interpenetrating Al₂O₃ and ZrO₂ single crystal lamellae.

Laser can be a useful tool for *in situ* rapid fabrication of composite components such as cutting tools, shear blades, etc. Lu *et al.*¹⁵¹ fabricated TiC dispersed Cu–Ti–C and Cu–Ni–Ti–C composites by laser scanning (using a CO₂ laser) of ball milled powder mixtures of Cu–Ti–C and Cu–Ni–Ti–C. Liu and DuPont¹⁵² developed crack free functionally graded TiC/Ti composite materials by laser engineered net shaping with compositions changing from pure Ti to ~95 vol.-%TiC. Kathuria¹⁵³ fabricated porous components of 93Cu + 7Sn and 70(MoW)2NiB2:30NiMo alloy powders using Nd:YAG laser. Wang *et al.*¹⁵⁴ fabricated W/W₂Ni₃Si ternary *in situ* metal silicide matrix composites using direct laser deposition. Duan and Wang¹⁵⁵ developed rapidly solidified Cr₃Si metal silicide '*in situ*'

composites by laser melting of Cr–Si–Ni alloy powders. Cheng *et al.*¹⁵⁶ studied the influence of process parameters on the structure and integrity of Al/SiC composite by direct laser deposition.

An intelligent laser processing based on size dependent optical absorption coefficient of metal particles has enabled fabrication of almost monodispersed metallic nanoparticles on chosen substrate.¹⁵⁷ Using this method, Ag clusters with mean diameters of 10 nm and size distributions of 0.13 have been prepared.¹⁵⁷

Irradiation with the second harmonic generation of 1.06 μm Q switched Nd:YAG laser on powdered non-linear crystals suspended in a photopolymeric solution enabled high resolution rapid prototyping.¹⁵⁸ One such structure developed by this method was a three-dimensional periodic photonic bandgap structure of aluminium oxide that consisted of layers of parallel rods forming a face centred tetragonal lattice.¹⁵⁹ A similar laser driven direct write deposition technique (from trimethylamine-ane and oxygen precursors) was successfully utilised to fabricate a three-dimensional microstructure consisting of aluminium oxide and aluminium.¹⁶⁰ These laser deposited rapid prototype ceramic components are useful as micromechanical actuators such as microtweezers and micromotors.

Summary and future scope

Laser forming offers a wide range of direct or single step, contact less and novel methods of fabricating finished products/components of metallic, ceramic and polymeric origin. While bending is confined to only dimensional changes, material addition continuously, layer by layer or in sequence is apart of laser assisted direct manufacturing processes. The possibilities offered by laser forming include bending, rapid prototyping and direct manufacturing. Laser bending is attractive to both automobile and aircraft industry because of high precision and production rate achievable in this process without application of mechanical forces. Laser bending that primarily depends upon 'temperature gradient' and 'buckling' mechanisms is routinely applied in tailoring the curvature of aluminium, iron and titanium based metals and alloys. Although the process is commercially exploited in manufacturing, certain fundamental aspects need to be properly understood, particularly with regard to the precise mechanism of laser bending: Does bending necessitate partial melting at the laser irradiation zone? Why does thickness increase along the bend edge? Why does bending rate decrease with number of pass? What microstructural features (grain size, texture, etc.) accompany and affect bending? How does prior microstructural state influence the angle of bending?

Both laser assisted manufacturing and rapid prototyping are major laser forming processes that have found commercialisation in many applications. The main reasons for interest in these processes stem from the scope of direct (one step) manufacturing of round or square sections, hollow tubes and more complex geometry with the same machine and identical fixture. Solid parts are usually made in layers. Thus, the interfaces of the consecutive layers remain the weakest point. Investigations are warranted to predict the stresses generated at the edges and corners, surface roughness/contour and compositional distribution in the solid object. Development or epitaxial repair of single crystal superalloy component will be a real breakthrough. In this regard, more comprehensive treatments of

heat and mass transfer in specific applications are warranted to establish the reproducibility of the laser forming processes.

Laser joining

Laser joining is a directed energy beam assisted fusion joining technique that involves the application of a high power laser beam as a source of heat to fuse and join two solids of similar or dissimilar natures. The advantages of laser joining over conventional fusion or arc welding processes include high welding speed, narrow heat affected zone (HAZ), low distortion, ease of automation, ability for single pass joining of thick sections and better design flexibility with controlled bead size.² The special interest in laser joining lies in its capability of fusion joining of dissimilar materials with different section thicknesses, compositions and/or physical/chemical properties for industry scale applications particularly in automotive and aerospace industry.^{161–168} Laser joining encompasses welding, brazing, soldering and microwelding. Joining of materials requires a laser source capable of delivering high power density with precision, reliability, flexibility and overall economy. Hence, pulsed or CW Nd:YAG or CO₂ laser (very seldom ruby laser) and recently the diode lasers are the commonly used lasers for joining. The main process variables in laser assisted joining are laser power, beam diameter, beam configuration, travel speed of the workpiece, substrate condition (roughness and temperature), filler type/feedrate, alloy composition and thermophysical properties (melting temperature, specific heat, density and thermal conductivity) of the workpiece. Table 2 presents a ready reference for laser assisted joining of iron based and non-ferrous materials. These studies are selected primarily to highlight recent advances and outline the unresolved issues in using laser as a tool for joining of materials. Laser joining is applicable to both inorganic/organic and similar/dissimilar materials with an extremely high precision, versatility and productivity that can only be matched by electron beam welding. Moreover, laser assisted joining can be done in atmosphere (with proper shrouding), unlike the vacuum processing needed in electron beam welding.

Laser welding

Laser welding, because of the sheer volume/extent of work and advancement over the years, constitutes the most important operations among the laser joining processes.^{2,161–185} There are two fundamental modes of laser welding depending on the beam power/configuration and its focus with respect to the workpiece: conduction welding and keyhole or penetration welding (Fig. 4a and b). Conduction welding occurs when the beam is out of focus (above the surface) and the power density is low/insufficient to cause boiling at the given welding speed. In deep penetration or keyhole welding, the beam focus lies below the surface so that sufficient energy per unit area is available to cause evaporation and formation of a hole in the melt pool. The 'keyhole' behaves like an optical black body in that the radiation enters the hole and is subjected to multiple reflections before being able to escape. The transition from conduction mode to deep penetration mode occurs with increasing laser intensity and duration of laser pulse applied to the workpiece.

Table 2 Summary of selected recent studies on laser joining of materials**a. Laser assisted welding of ferrous metals and alloys**

Scope/material	Objectives/outcome	Reference	Year
AISI 304 stainless steel	Fatigue behaviour of laser welded specimens in air, hydrogen and Ar using a CW CO ₂ laser	Tsay <i>et al.</i> ²²⁸	2003
Austenitic and duplex stainless steel	Study of microstructures of the welded zone using Nd:YAG laser	McPherson <i>et al.</i> ²²¹	2003
Duplex stainless steel 22Cr–5Ni–3Mo (UNS S32205)	Optimum conditions to minimise the δ -ferrite content in weld zone formed by CO ₂ laser	Capello <i>et al.</i> ²²⁹	2003
AISI 304 stainless steel	A finite element based heat transfer model to predict the weld bead shape and thermal distortion using Nd:YAG laser	Chang and Na ²³⁰	2002
Ni+Au–Ni plated steel	Effect of Au/Ni plating on strength of welded steel sheets using pulsed Nd:YAG laser	Biro <i>et al.</i> ¹⁷²	2001
Steel	Effect of temperature dependence of surface tension on welding of steel using CW laser	Fuhrich <i>et al.</i> ¹⁷³	2001
AISI 304 stainless steel	Extremely narrow welding seam with fully austenitic microstructure in laser welded AISI 304 stainless steel using photolytic iodine laser	Farid and Molian ¹⁷⁴	2000
Galvanised steel	Relation between defects in welding with weld speed, average power density and pulse duration	Tzeng ²³¹	2000
High carbon steel	A crack free, stronger and wider weld pool by CO ₂ laser welding	Ng and Watson ¹⁷⁵	1999
Plain C and AISI 304 stainless steel	Welding by feeding of metallic wire	Sun <i>et al.</i> ²²⁴	1998
Low carbon galvanised steel	Effect of shielding gas on weld penetration and formability	Chung <i>et al.</i> ²¹⁸	1999
Mild steel	Strength of welded joint made by laser spot welding and resistance welding	Yang and Lee ²¹⁶	1999
AISI 304 stainless steel and free cutting steel	Laser welding for fabrication of hydraulic valves	Li and Fontana ²³²	1998
AISI 304 stainless steel	Combination of pulsed beam with high peak power and modulated CW beam to obtain deep penetration	Narikiyo <i>et al.</i> ²³³	1998
High strength low alloy steel 15–5PH (0.04C–14.8Cr–4.77Ni–3.29Cu–0.73Mn–0.53Si)	Fatigue behaviour and cracks in laser welded steel	Onoro and Ranninger ²³⁴	1997
Austenitic stainless steel	Non-equilibrium structure and phase analysis	Kusuda <i>et al.</i> ²³⁵	1997
Austenitic stainless steel	Influence of high speed and low power on welding by CW CO ₂ laser	El-Batahgy ¹⁷⁶	1997
Austenitic stainless steel	Relation between weld zone size and shape with applied power and scan speed	El-Batahgy ²²²	1997
Stainless steel and titanium	Plasma plume temperature and equilibrium	Szymanski <i>et al.</i> ¹⁷⁷	1997
Stainless less, Kovar, gold	Influence of power density on defect/hole and centreline cracks	Cheng <i>et al.</i> ¹⁷⁸	1996
Stainless steel	Effect of gravity on Cr distribution	Wang and Tandon ¹⁷⁹	1995
Tritiated AISI 316L stainless steel	Segregation of tritium in the weld and HAZ	Roustita <i>et al.</i> ²³⁶	1994
Electroaluminised steel sheet	Formation of Zn and Fe rich intermetallics in the weld zone	Akhter <i>et al.</i> ²³⁷	1990

b. Laser assisted welding of non-ferrous metals and alloys**Laser welding of Al/Al alloys**

Material	Objectives/outcome	Reference	Year
5182 and 5752 Al alloys	A model to predict fusion zone geometry for keyhole mode of LBW with experimental validation using a CW Nd:YAG laser	Zhao and DebRoy ²⁴³	2003
AA 5083-O and A356-T6 aluminium alloys	Effect of single beam and dual beam welding on the microhardness and composition	Haboudou <i>et al.</i> ²⁴⁷	2003
Al	Effect of application of external electrical current on laser welding using a TLF3000 CO ₂ laser and fluid flow	Xiao <i>et al.</i> ¹⁸⁷	2001
C95800 Ni–Al bronze	Effect of welding speed on weld quality and microstructure during diode pumped Nd:YAG laser assisted welding	Leong <i>et al.</i> ¹⁶⁷	2000
6061-T6 Al–Mg–Si alloy	A kinetic analysis for dissolution of Mg ₂ Si during laser welding using a 2.5 kW CW CO ₂ laser	Hirose <i>et al.</i> ¹⁶⁸	1999
AA 1100 Al alloy	Conditions for pore free, conduction pool geometry welding during pulsed Nd:YAG laser	Weckman <i>et al.</i> ¹⁶⁹	1997
SiC dispersed Al–Cu–Mg based metal matrix composite	Carbide formation kinetics with respect to laser intensity and pulse duration during welding using a pulsed Nd:YAG laser	Yue <i>et al.</i> ²⁴⁹	1997
8090 Al–Li alloy	Studies on fusion zone dimension, solute evaporation, hardness, porosity and tensile properties of the weld zone welded using 3 kW CW CO ₂ laser	Lee <i>et al.</i> ¹⁷⁰	1996
Al–Fe–V–Si alloy	Formation of Si/Fe/V rich Al ₄ Fe type precipitates and Si/Fe rich phase on boundaries of CW CO ₂ laser welded alloy	Whitaker and McCartney ¹⁷¹	1995

Table 2 Continued

Laser welding of Ti based materials			
Material	Objectives/outcome	Reference	Year
Ti-24Al-17Nb	Studies on weld microstructure of CO ₂ laser welded sample	Wu <i>et al.</i> ²⁵³	2002
TiNi shape memory alloys	Influence of laser parameters on M_s temperature and ductility, amount of β_2 phase and strength related to shape memory effect	Hsu <i>et al.</i> ¹⁸⁰	2001
Ti-6Al-4V alloy	Influence of power density on keyhole formation and melt depth during pulsed Nd:YAG laser welding	Perret <i>et al.</i> ¹⁸¹	2001
Ti	Comparison of electron beam, laser and TIG welded Ti	Qi <i>et al.</i> ²⁵¹	2000
Ti-6Al-4V and Ti-6Al-2Sn-4Zr-2Mo	Butt joint (using filler wire) with a CW CO ₂ laser and autogenous welding by pulsed Nd:YAG laser	Li <i>et al.</i> ²⁵⁵	1997
Ti-6Al-4V	Crack initiating mechanism due to laser welding using a CW CO ₂ laser	Tsay and Tsay ¹⁸²	1997
TiAl	Joining by self-propagating high temperature synthesis reaction using a 2.5 kW CW CO ₂ laser	Uenishi and Kobayashi ²⁵⁴	1996
SiC-fibre reinforced Ti alloy	Influence of scarf angle on butt and scarf joints using a CW CO ₂ laser	Hirose <i>et al.</i> ¹⁸³	1995

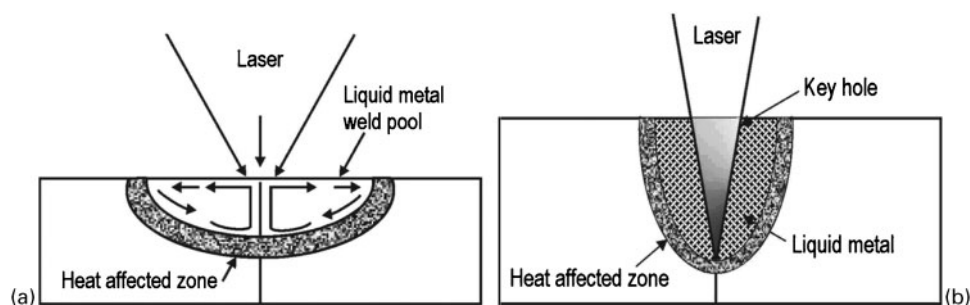
The welding efficiency can be expressed in terms of the power (or energy) transfer coefficient η , where η is the ratio between the laser power absorbed by the workpiece and the incident laser power. η is usually very small but can approach unity once a keyhole has been established. The melting efficiency or melting ratio ε is given by

$$\varepsilon = \frac{v d W \Delta H_m}{P} \quad (1)$$

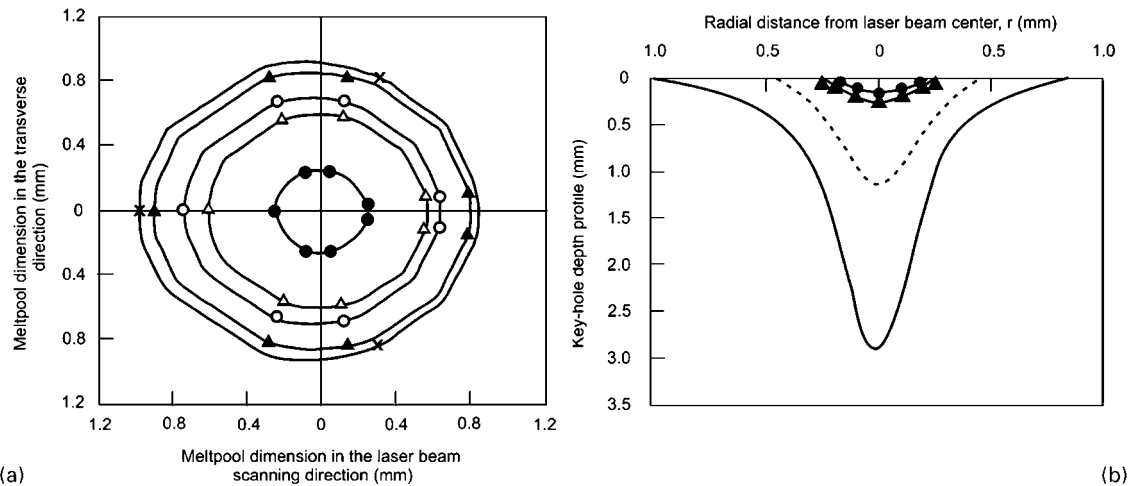
where ε is the rate of melting, P is the incident laser power, v is the welding speed, d is the sheet thickness, W is the beam width and ΔH_m is the enthalpy or heat content of the metal at the melting temperature. The maximum values of ε are 0.48 and 0.37 for penetration welds and conduction welds respectively.¹⁶⁶

It is apparent that ε never approaches unity even when $\eta \approx 1$. Both η and ε can be enhanced under the keyhole welding condition if the absorption coefficient can be increased. This can be accomplished by application of absorbent coating, using smaller wavelength laser light, surface roughening or texturing, preheating, tailoring of temporal irradiation profile and/or oxidation/nitriding. Among various process parameters, the quality and properties of laser weld depend on laser pulse time and power density, laser spot diameter/penetration, traverse speed of the laser beam across the substrate surface, shielding gas, melt area, melting ratio and material properties such as absorptivity, specific heat, density, etc. For details about the effect of process parameters on laser welding, the reader may refer to Duley.¹⁶⁶

The combination of laser beam with metal inert gas or tungsten inert gas (TIG) arc (so called hybrid technique) seems promising from the viewpoint of bead formation except that occurrence of large blowholes and voids can pose serious problems. Recently, Kern *et al.*¹⁸⁶ have detected an intrinsic current in the melt pool driven by thermoelectric potential. It is shown that the application of external magnetic field in laser welding of steel can modify the shape of seam cross-section, level of porosity and bead geometry at higher speed of welding. Xiao *et al.*¹⁸⁷ modified the magnetofluidynamic approach by applying an electrical current through an external power source through a 2.4 mm diameter tungsten electrode during laser welding using a TLF5000 CO₂ laser and a TLC 100 five axis laser processing machine. Bead on plate welds were made on 5 mm thick plates of Al. Helium with a flowrate of 25 L min⁻¹ was supplied by a side pipe with an inner diameter of 6 mm to suppress formation of laser induced plasma and protect the weld pool from the atmosphere. The electrical current flowing in the weld pool induced an azimuthal magnetic field that was proportional to the current density resulting in electromagnetic forces proportional to the square of current density. Because of the divergence of the current lines from the electrode tip to the workpiece, the current density distribution in the weld pool turned non-uniform such that the current density close to the electrode tip and keyhole front was larger than that at the bottom and rear of the melt pool. Accordingly, the molten metal in front of the keyhole was accelerated towards the



4 Schematic of a conduction melt pool (semicircular) and b deep penetration (keyhole) fusion zone and weld¹⁶⁶



5 **a** melt pool size and shape at substrate surface under quasi-steady state condition and **b** depth of keyholes formed during laser welding for incident laser power $P=1850$ W, laser beam scanning speed $v=2.12$ cm s⁻¹ and absorptivity $A=8\%$ (Ref. 197)

bottom that produced additional heat to the region. The weld depth, therefore, increased, while the width decreased. It was concluded that an external current can significantly influence the fluid flow of the weld pool and shape of the seam cross-section in laser welding of aluminium, to result in improved flexibility.

Hirsch *et al.*¹⁸⁸ developed a finite element model of heat and mass transfer to predict the weld pool geometry and weld depth during CO₂ laser welding of ceramics and compared the predicted result with the experimentally observed values. The predicted weld pool shape agreed well with the experimentally observed one. Several investigations address similar issues of modelling the heat/mass transfer and fluid flow during laser welding to predict thermal history, shape and size of the weld pool.^{187–195}

During deep penetration laser welding, the plasma created over the keyhole absorbs beam energy and reduces power efficiency. Thermal movement of laser induced plasma including the principle and feasibility of controlling the plasma by electric and magnetic fields was theoretically and experimentally analysed by Peng *et al.*¹⁹⁶ An experimental procedure involving raising the nozzle during laser welding was used to evaluate the effect of increasing the power efficiency by driving away the charged particles. The power efficiency increased with increasing magnetic field intensity. The power efficiency appears to reach the highest value at an optimal electric and magnetic field intensity.

Kar and Mazumder¹⁹⁷ developed a model to describe the laser keyhole welding to determine the free surface velocity and temperature distributions as well as the shape of the keyhole. The velocity and temperature fields in the molten material can be obtained by considering the following momentum, energy and mass conservation equations

$$\rho l \left[\frac{\delta u_1}{\delta t} + (u_1 \nabla) u_1 \right] = \rho_1 g - \nabla p_1 + \mu_1 \nabla^2 u_1 \quad (2)$$

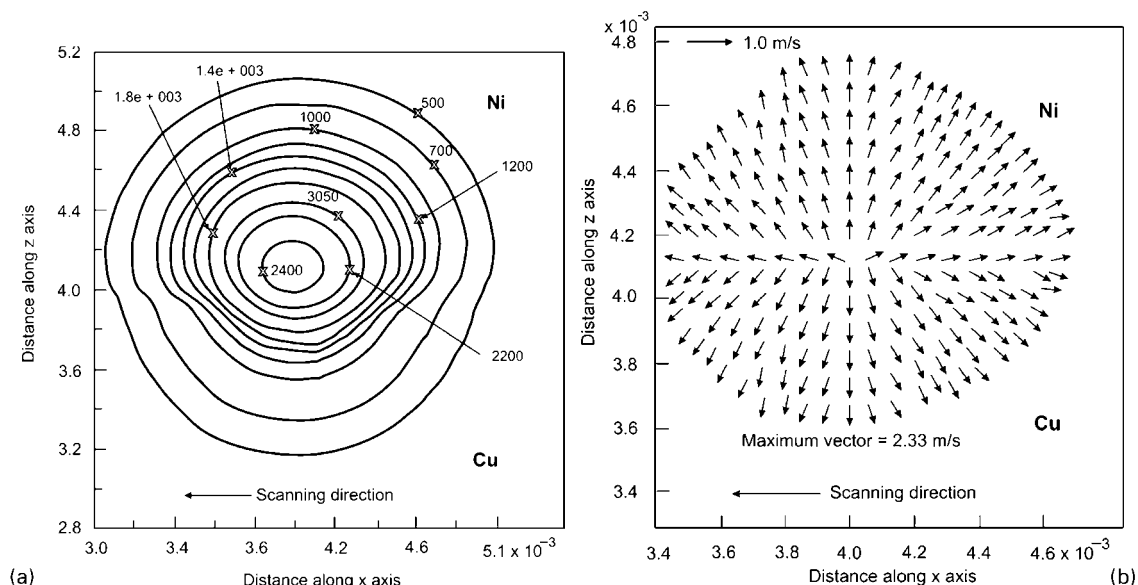
$$\rho_1 c_{pl} \left[\frac{\delta T_1}{\delta t} + (u_1 \nabla) T_1 \right] = k_1 \nabla^2 T_1 \quad (3)$$

$$\nabla u_1 = 0 \quad (4)$$

where ρ_1 , μ_1 , c_{pl} , k_1 , u_1 , g , p_1 and T_1 are the density, viscosity, specific heat at constant pressure, thermal conductivity, velocity vector, acceleration due to gravity, pressure and temperature of the molten material respectively. t and ∇ are the time variable and differential operator respectively.

Figure 5a shows the shape of the weld pool at the top surface of the workpiece. The intersection of the vertical and horizontal lines in the middle of this figure indicates the centre of the laser beam which moves to the right along the horizontal line. The innermost curve in this figure is the liquid/vapour interface which represents the size of the mouth of the keyhole at the substrate surface. The mouth of the keyhole is found to be almost similar for the welding parameters used for deriving the weld pool geometry (Fig. 5a). However, the outer four curves are the solid/liquid interfaces representing the sizes of the melt pool at the substrate surface, indicating that weld pools of different sizes can be obtained for different welding conditions. These four curves also show that the leading solid/liquid interface is closer to the centre of the laser beam than the trailing solid/liquid interface, which is behind the laser beam opposite to the scanning direction. This effect arises because continuous heating occurs ahead of the laser beam along the direction of scanning while cooling takes place behind the beam in the opposite direction. Figure 5b has been obtained by plotting the solid/liquid and liquid/vapour interfaces on the vertical plane that coincides with the laser beam scanning direction and indicates that the depths increase with increasing welding time until the quasi-steady state is reached. Furthermore, the depth of the melt pool increases more rapidly than the depth of the keyhole as the welding time increases.

Rai *et al.*¹⁹⁸ presented a convective heat transfer model to calculate temperature and fluid flow fields during keyhole mode laser welding for different mechanisms of heat transfer in the weld pool. A vorticity based turbulence model has been used to estimate the values of effective viscosity and thermal conductivity in the melt. The model is used to calculate the temperature and velocity fields, weld pool geometry, cooling rate and solidification parameters for welding of AISI 304 stainless steel and Al 5754 aluminium alloy.



6 *a* isotherms along the top surface of laser welded Cu with Ni after 25 ms from turbulent simulation (temperatures are shown in °C). In the contour labels, 1×10^3 indicates 1000 units. *b* top surface velocity vector in the molten pool after 25 ms in turbulent simulation. In moving coordinate system laser torch located at $x=4$ mm and $z=3.94$ mm. At time $t=0$ s, $z=3.94$ mm indicates the separation line between Cu and Ni workpieces²⁰⁰

The morphology of the solidification front during welding may be predicted from the ratio of thermal gradient G to solidification rate R_s . The criterion for plane front instability based on constitutional supercooling is given by the following relation¹⁹⁹

$$G/R_s < \Delta T_E/D_L \quad (5)$$

where ΔT_E represents the temperature difference between the solidus and liquidus temperatures of the alloy and D_L is the diffusivity of a solute in the liquid weld metal. $\Delta T_E/D_L$ for aluminium alloy is found to be of the order of 10^3 . Figure 5*b* shows the variation in G/R_s with net energy input for aluminium alloy. It is observed that G/R_s increases with increasing net energy input, which indicates that the solidification microstructure will become less dendritic with decreasing welding velocity.

Computational modelling of welding of dissimilar metals has been extensively studied by several investigators.^{200,201} The effect of turbulence on momentum, heat and mass transfer during laser welding of a copper–nickel dissimilar couple has been studied by carrying out three-dimensional unsteady Reynolds averaged Navier–Stokes simulations.²⁰⁰ Figure 6*a* shows the isotherms on the top surface, as obtained from turbulent simulations showing the shifting of the maximum temperature location towards the nickel side. A combined consequence of the differences in thermal diffusivities of Cu and Ni and their melting temperature causes the convective transport of molten pool in a complex manner. The velocity vector plots for turbulent simulation of the molten metals are depicted in Fig. 6*b*, where it is evident that in the case of turbulent transport, the mean thermal and concentration gradients decrease due to an increased diffusion strength.

This decreases the strength of Marangoni convection in turbulent flow eventually. Furthermore, a greater degree of momentum diffusion in the turbulent pool results in a decrease in the maximum velocity magnitude, as is evident from Fig. 6*b*. As a result, the mean advection of heat from the centre of the pool towards its

peripheries is weaker than that in the case of laminar transport, which tends to decrease the pool width. The effects of turbulence are found to be most prominent on the species transport in the molten pool. The composition distribution in turbulent simulation is found to be more uniform than that obtained in the simulation without turbulent transport. The nickel composition distribution predictions, as obtained from the present turbulence model based simulations, are also found to be in good agreement with the corresponding experimental results.

A large variety of metals and alloys have been welded with pulsed or CW lasers. The earliest study on deep penetration laser beam welding (LBW) was earlier outlined by Brown and Banas.²⁰² The potential scope of the application of pulsed laser welding was earlier outlined by Duley,²⁰³ Ready,²⁰⁴ Berretta *et al.*²⁰⁵ and Schwartz.²⁰⁶ Li and Gobbi²⁰⁷ and Schubert *et al.*²⁰⁸ gave a brief overview on the potential applications of high power CO₂ and Nd:YAG lasers to create lightweight structures in the automotive, aerospace and aeronautic industries. In this context, the welding behaviours of Ti–6Al–4V (for aeronautic application), Inconel 718 (exhaust frame) and Hastelloy X (combustion liner) have been discussed.

Owing to a large thermal gradient associated with laser assisted welding, the generation of residual stress leads to the formation of distortions and defects in the welded joint. Postacioglu *et al.*^{209,210} predicted the distortion generated in a weld pool using linear thermoelastic theory. Niordson²¹¹ analysed the fracture toughness of steel joined by a laser welding in terms of steady state crack growth along the weld. Laser welding gives a high mismatch of yield strength between the weld zone and base material, due to the fast thermal cycle that the material undergoes in welding. Fracture is modelled by using a cohesive zone criterion in front of the crack tip along the fracture zone. It is found that a thinner laser weld gives higher interface toughness. Furthermore, it is shown that the preferred path of the crack is in the base

material slightly outside the weld, a phenomenon also observed in experiments.

In the following subsections, a brief overview of the studies on laser assisted welding of specific ferrous and non-ferrous alloys has been discussed.

Laser assisted welding of Fe based materials

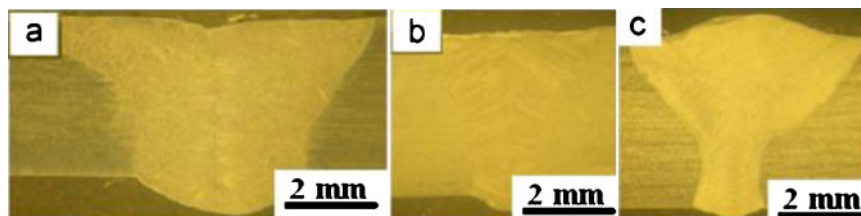
Laser assisted welding has been successfully implemented in a large number of ferrous alloys.^{212–237} Baarden *et al.*²¹² carried out laser welding of rimmed steel for automotive application. Laser assisted welding of several grades of stainless steel was attempted for applications in power plant and chemical industries.^{213,214} High speed laser welding of tin plate and tin free steel for container industries was reported by Mazumder and Steen.²¹⁵ Until recently, automobile manufacturing extensively used resistance spot welding. However, resistance spot welding requires many welding machines for parts with geometrical and structural complexity and hence, is now substituted with laser spot welding, which seems advantageous over the former technique both in terms of low cycle fatigue strength and residual stress.²¹⁶ Yang and Lee²¹⁶ measured the low cycle fatigue strength and residual stress of laser spot welded mild steel (using a CW CO₂ laser) and compared the weld strength with that welded by resistance spot welding. Finite element modelling revealed that the residual tensile stress was distributed over the inner and outer rims of the fusion zone. Banas²¹⁷ conducted high power laser welding experiments with austenitic stainless steel and obtained a dense and defect free weld zone, as revealed by radiographic inspection. The welded joint strength was equivalent to that of the base metal. Chung *et al.*²¹⁸ evaluated the influence of shielding gas (helium, carbon dioxide, argon, nitrogen and 50% argon + 50% nitrogen) on CO₂ laser weldability of low carbon 1.5 mm thick automotive galvanised steel. Helium and 50% argon + 50% nitrogen were the best shielding gases while pure argon as shielding gas was most effective in terms of penetration and formability for welding with high power CO₂ laser welding. The shielding gas determines the period, size and shape of the plasma created by laser irradiation depending on the ionisation potential, dissociation potential, thermal conductivity and atomic weight of shielding gas. Malek Ghaini *et al.*²¹⁹ studied the influence of laser pulse energy, duration and travel speed on weld dimension, microstructure and hardness during overlapping laser assisted bead on plate spot welding of low carbon steel using a pulsed Nd:YAG laser. It was found that significant variations in the microstructure and hardness of weld within a weld spot could arise in full penetration welding due to the rapid solidification process associated with the technique.

Kwok *et al.*²²⁰ fabricated austenitic (S30400 and S31603), duplex (S31803) and super duplex (S32760) stainless steels by laser penetration welding with a CW Nd:YAG laser in an argon atmosphere. While the laser welded S31603 retained the original austenitic structure, the laser welding of S30400 contained austenite as the major phase and δ -ferrite as the minor phase. On the other hand, a slight change of δ -ferrite to austenite ratio was found in both the laser welds of S31803 and S32760, with austenite present at the δ -ferrite grain boundaries. The welds exhibited passivity but their pitting corrosion resistance deteriorated as compared with that of the unwelded specimens. The decrease in pitting corrosion resistance of the welds was attributed

to the microsegregation in the weld zone of S31603, and to the presence of δ -ferrite in S30400. For the duplex grades S31803 and S32760, the change of the ferrite/austenite phase balance in the weld zone could be the cause of the decrease in corrosion resistance.

McPherson *et al.*²²¹ achieved autogenous welding of austenitic stainless steel and duplex steel using an Nd:YAG laser and compared the results with that obtained by submerged arc welding. Laser welding produced higher hardness in the weld zone and prevented distortion. El-Batahgy²²² studied the welding behaviour of AISI 304L, 316L and 347 stainless steels with different sheet thicknesses. The fusion zone shape and final solidification microstructure were evaluated as a function of laser parameters. Both bead on plate and autogenous butt weld joints were made using a carbon dioxide laser with a maximum output power of 5 kW in CW mode. Depending on the base metal thickness, parameters such as laser power, welding speed, defocusing distance and shielding gas should be carefully selected so that weld joints having complete penetration, minimum fusion zone size and acceptable weld profile are produced. Heat input as a function of both laser power and welding speed has almost no effect on both the type of microstructure and mechanical properties of welds. The microstructure of all laser welds was always austenitic with about 2–3% δ -ferrite. However, the lower the laser power and/or the higher the welding speed, the finer the solidification structure. Primary ferrite or mixed mode solidification resulted in crack free welds. Chan *et al.*²²³ carried out welding of 0.5, 0.6, 0.8 and 1.0 mm thick steel sheets to form tailored weld blanks with different thickness ratios of 2 (0.5/1.0 mm), 1.67 (0.6/1.0 mm) and 1.25 (0.8/1.0 mm) using a Nd:YAG laser. Subsequent formability and uniaxial tensile tests showed that tailored weld blanks of different dimensions and cutoff radii yielded different major and minor strain values. Furthermore, the forming limit diagrams of the tailor welded blanks indicated that both forming limit and minimum major strain value decreased as the thickness ratio would increase. This implied that the higher the thickness ratio, the lower the formability of the tailored weld blanks.²²³

The welding of low alloy 13CrMo44 steel to AISI 347 stainless steel was carried out using 1.2 mm diameter ENiCrMo-3 nickel based wire as filler.²²⁴ The joints were in the tube form with an outer diameter of 43.5 mm and a wall thickness of 4.5 mm. Mixing behaviour of the filler with the base metal was evaluated by measuring Cr and Ni contents in the weld metal. Composition along the weld zone was uniform producing uniform hardness throughout. Costa *et al.*²²⁵ studied the weldability of hard metal (K40, 88%WC–12%Co and K10, 94.4%WC–5.6%Co) to steel with a high power CW CO₂ laser or Nd:YAG laser and pulsed Nd:YAG laser. Weld bead size, microstructure, bending strength and hardness were evaluated to correlate the mechanical properties with the concerned welding parameters to obtain full penetration and high strength joints. K40 (88%WC–12%Co) quality tips have higher resistance after welding, compared with that in K10 samples, that contained cracks after welding. It appears that diffusion of tungsten to the bead during welding leads to the formation of several carbides in the martensitic matrix and induces higher bead hardness. In spite of having eutectic structure with multiple carbides, the weld bead did not develop cracks. It was concluded



7 Macrostructure of weld zone formed by a TIG welding, b laser welding and c LATIG hybrid welding techniques²⁴⁰

that laser welding particularly with CW Nd:YAG laser was an effective joining technique for hard metals, *vis-à-vis* brazing or mechanical clamping, for manufacturing cutting tools with small weld beads and HAZ and a minimum residual stress.

Electrogalvanised steel sheet with zinc coating (thickness of 7.5 μm) was laser welded using a CW CO₂ laser with 23 mm beam diameter having near Gaussian beam configuration. Subsequent electrochemical tests in 1 N NaCl solution revealed the presence of Zn-Fe intermetallic phases in the HAZ that accounted for the improved adhesion and better sacrificial protection against corrosion.²²⁶ Iqbal *et al.*²²⁷ developed a new method of using two tandem laser beams for lap welding of galvanised steel sheets. This involves a precursor beam and a higher power actual beam, generated independently or otherwise split from the same source. The first beam cuts a slot, thus making an exit path for the zinc vapours, while the second beam performs the desired welding. The process is easier as it does not need any preprocessing, prearranging with additional components or contour limitations and the welders may proceed with the jobs as per non-coated steel and applying the technique, the weld zone can get rid of all or a part of the zinc vapour porosity at lower or higher speed respectively. The experimental results seem promising showing total absence of zinc in the welds. However, the selection of proper source, beam alignment (between weld and cut) and proper sheet clamping are important.

Hybrid laser welding is a technique which combines both the laser welding and an arc welding process, to weld materials. By application of the technique, the beneficial effects of both the techniques are achieved. Based on the combinations of specific laser source and other welding sources, there are a number of hybrid laser welding. The advantages of laser hybrid welding over the individual welding technique include flexibility in welding of a thicker section, a high welding speed, less distortion, less weld defects and flexibility of bridging of a larger gap.^{238,239} A detailed study of the microstructures of welded zones developed by TIG welding, laser welding and laser-TIG (LATIG) hybrid welding was undertaken by Yan *et al.*²⁴⁰ It was observed that all the welding techniques were capable of producing full penetration joints. Figure 7 shows the macrostructures of the weld zone. From Fig. 7, it is evident that the zone is free from cracks and porosities. The shape and size of the weld zone were, however, different in three cases. Different widths and fusion areas of different weld joints are attributed to the different welding speeds and heat inputs used in the different welding methods. Figure 8 shows the microstructures of the fusion zones of the three joints at high magnification. All joints are composed of dark δ -Fe dendritic structure in austenite

matrix. However, different dendrite sizes can be observed in the three joints. The δ -Fe dendrite size in TIG joint is obviously larger than that in laser and hybrid joints. The primary and secondary dendrite arm spacings of TIG joint are about 10–14 and 3–7 mm respectively. The laser joint shows the smallest dendrite size in all joints. The primary dendrite arm spacings of laser and hybrid joints are about 2–5 and 4–8 mm respectively. Furthermore, the δ -Fe in laser hybrid joint exists as network morphology in fusion zone, while exists as columnar structure in laser joint.

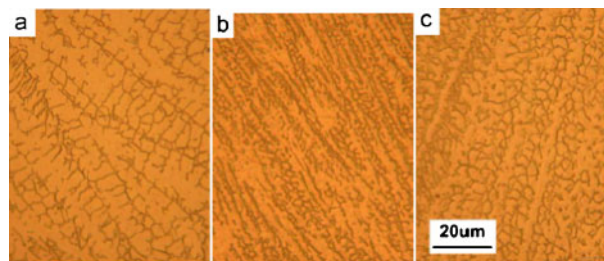
The tensile strengths of TIG, laser and hybrid joints are 560, 733 and 683 MPa respectively. The TIG joint exists as cup-cone shaped fracture, while the laser and hybrid joints exist as pure shear fracture. The laser welding and hybrid welding are suitable for welding 304 stainless steel in industry application owing to the high welding speed and excellent mechanical properties.

Gao *et al.*²⁴¹ studied the microstructural characteristics of mild steel weld joints developed by CO₂ laser-metal inert gas hybrid welding.

Laser assisted welding of non-ferrous metals

Compared with the number of studies on welding of titanium, plain carbon and alloy steels using high power lasers, the welding of aluminium did not attract the same level of research attention mainly because aluminium alloys pose several difficulties in welding due to the high surface reflectivity of laser and affinity for oxygen. Nevertheless, the scope and relevant issues of laser assisted welding of lightweight metals such as Al, Mg, Ti and their alloys for aerospace applications have been reviewed by Schubart *et al.*²⁰⁸

Both conduction mode welding and keyhole mode welding are possible in aluminium.²⁴² This porosity problem can be reasonably eliminated by surface milling and vacuum annealing.²⁴² Zhao and DebRoy²⁴³ have developed a model to predict the fusion zone geometry for keyhole mode LBW of 5182 and 5752 Al alloys using a CW Nd:YAG laser and compared the results with



8 Microstructure (at high magnification) of weld zone formed by a TIG welding, b laser welding and c LATIG hybrid welding techniques²⁴⁰

experimental results. Dousinger *et al.*²⁴⁴ described the factors relevant to deep penetration welding, process efficiency and process stability for welding of aluminium alloys.

Autogenous 'bead on plate' laser beam welding of Al alloys by a 3 kW CO₂ laser under Ar or N₂ atmosphere is possible in the 700–1300 W power range and 1500–9000 mm min⁻¹ scan speed with the focus located at 1–3 mm below the surface.²⁴⁵ Though bead on plate experiments do not represent a true joining exercise, these studies are helpful to correlate the dimension and geometry of melt pool and HAZ with laser and material parameters. The effects of different gases were evaluated in terms of weld line appearance, fusion zone dimension, solute evaporation, microhardness, post-weld tensile properties and porosity distribution. In comparison with the electron beam welding, laser welding yielded a higher fusion zone depth to width ratio, cooling rate and porosity, and a lower solute loss and post-weld tensile strain.²⁴⁵ A similar investigation on the microstructural evaluation following autogenous bead on plate CO₂ laser welding of an Al-8.5Fe-1.2V-1.7Si alloy (wt-%) on 2 mm thick sheet showed that the fusion zone microstructure consisted of faceted precipitates of ~10 µm embedded in cellular-dendritic α-Al matrix with sub-micrometre intercellular phases.²⁴⁶

Continuous wave Nd:YAG laser welding of AA 5083-O wrought aluminium alloy with a high Mg content (4.5%) and A356 cast aluminium alloy with 7%Si and a cast oxide layer was carried out by Haboudou *et al.*²⁴⁷ It was concluded that surface preparation as well as dual beam welding are adequate for reducing porosity in laser assemblies. Braun²⁴⁸ did a detailed study on butt welding of a 6013 aluminium alloy sheet using a 3 kW Nd:YAG laser with two kinds of filler metals: gas atomised powders of aluminium alloys AlMg5, AlSi12, AlSi12Mg5 and AlSi10Mg, as well as powder mixtures of Al and Si and the binary alloys Al-5Mg, Al-5Zr, Al-5Cr and Al-10Mn. The weld region exhibited a dendritic/cellular structure in the fusion zone and a partially melted zone beside the fusion boundaries. The effect of filler wire composition and post-weld treatment on the microstructure, hardness and corrosion behaviour was established to optimise the process parameters. Yue *et al.*²⁴⁹ studied the welding behaviour of a SiC particulate reinforced 2124 Al alloy composite using a pulsed Nd:YAG laser. The influences of laser parameters, i.e. laser intensity, pulse duration and focus position on the depth and size of fusion zone were investigated. Tendency to crack formation could be minimised at an applied power density of $<4 \times 10^9 \text{ W m}^{-2}$ and a pulse duration of 8–10 ms. Wang *et al.*²⁵⁰ have demonstrated the versatility of laser welding by carrying out *in situ* weld alloying and LBW to join SiC reinforced 6061 Al metal matrix composite using titanium as filler alloying element. Microstructural studies show that the detrimental needle-like aluminium carbides are completely eliminated and the central fusion joint consists of TiC, Ti₃Si₃ and Al₃Ti along with some large pores.

A comparative study of the microstructures and properties of 0.5 mm thick welded sheets of commercial purity titanium by high vacuum electron beam welding, CO₂ LBW and TIG welding processes showed that the electron beam welding is more suitable for defect free welding of commercial purity titanium.²⁵¹ The mechanism of keyhole formation during pulsed Nd:YAG laser

interaction with Ti-6Al-4V alloy was established by Perret *et al.*²⁵²

Wu *et al.*²⁵³ welded a Ti-24Al-17Nb (at-%) alloy with a fast axial flow type CO₂ laser of 3.0 kW rated power. The weld microstructure consisted primarily of ordered β phase (namely β₂ phase) and was independent of the laser parameters. The tensile strength and ductility of the joints were not affected. Uenishi and Kobayashi²⁵⁴ attempted fusion welding of Ti-46Al-2Mo using a 2.5 kW CW CO₂ laser and established the parameters for crack free welding. A detailed study of laser welding of Ti-6Al-4V and Ti-6Al-2Sn-4Zr-2Mo sheets using CO₂ and Nd:YAG lasers showed that the porosity level could be minimised by CO₂ welding with filler wire.²⁵⁵ In this regard, it is relevant to note that marginal deteriorations in tensile strength and fracture toughness were observed in laser welded of Ni-51 at-%Ti alloys using Nd:YAG laser.²⁵⁶

Spot welding of Ta plates was performed with a pulsed Nd:YAG laser (50 W) coupled with a step index fibre. A series of spot welds were achieved with a peak power of 3 kW (i.e. a power density of $42 \times 10^9 \text{ W m}^{-2}$) and pulse duration between 1 and 16 ms.²⁵⁷ The presence of small and large keyholes suggested that the keyhole geometry and solidification time primarily determined whether volume defects (porosity) would form and optimisation of process parameters could minimise the tendency of defect formation.

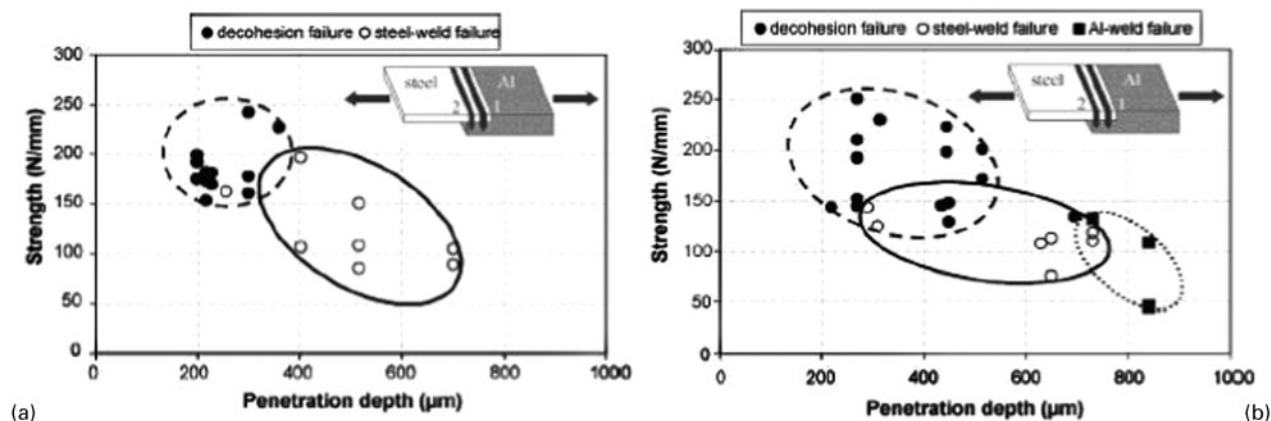
The optimum process parameters during pulsed laser welding (using 1.6 kW pulsed YAG laser) of V-4Cr-4Ti alloy was determined.²⁵⁸ Laser beam welding of Hastelloy X was performed using a 1.2 and 6 kW CO₂ laser with lap and butt joints.²⁵⁹ Cracks in the fusion zone were observed which could be eliminated by changing the beam diameter and alignment.

Laser welding process was introduced into dentistry by the end of 1980s causing a great impact on development of cheaper and smaller equipment using simpler technique. Ag-Pd-Au-Cu alloy used for dental implants was welded using a Dentaureum DL 20002S Nd:YAG laser at a beam power of 6.08 kW and an interaction time of 14 ms, producing a welding energy of ~85.12 J, leading to the formation of refined microstructure with a high corrosion rate.²⁶⁰

Song *et al.*²⁶¹ studied the weldability of AZ31B magnesium alloy sheets by laser-arc hybrid welding process. A sound overlap welded joint of magnesium alloy AZ31B sheets without defects was obtained. Liu *et al.*²⁶² compared the weldability of AZ31B magnesium alloy using hybrid LATIG welding, LBW and TIG welding and found that the welding speed of LATIG was higher than that of TIG but comparable with that of LBW, and the penetration of LATIG was twice that of TIG and four times that of LBW. In addition, arc stability was better in hybrid LATIG welding than TIG welding, especially at high welding speed and low TIG current. It was found that the HAZ of joint was only observed in TIG welding, and the size of the grains was evidently coarse.

Kim *et al.*²⁶³ studied the welding of Cu₅₄Ni₆Zr₂₂Ti₁₈ metallic glass using pulsed Nd:YAG laser. It was observed that crystallisation could be suppressed both in weldment and HAZ at a welding speed of 60 mm min⁻¹.

Laser microwelding of AlMg₃ with a thickness of 0.2 mm and AlMg_{4.5} Mg foils of 1 mm in thickness was



9 Linear transverse tensile strength versus penetration depth for joints welded with *a* 150 mm length focal lens and *b* 200 mm focal lens. The presence of three failure mode may be noted: decohesion mode with failure on the weld/aluminium interface of the two joints, a weld and steel/weld interface failure mode in joint 2 and an aluminium/weld interface failure in the lower part of the joint 1 (Ref. 267)

conducted using a pulsed Nd:YAG laser and the effects of pulse shaping on the welding shape and microstructures were investigated experimentally with numerical studies.²⁶⁴

Laser assisted welding of dissimilar materials

Joints between dissimilar metals are commonly encountered in the power generation, chemical/petrochemical plants, nuclear and electronics industries for the purpose of tailoring component properties or weight reduction. More recently, laser welding technologies have been successfully used to manufacture hybrid microsystems consisting of different materials.²⁶⁵ Sun and Ion²⁶⁶ presented a brief overview of laser welding of dissimilar metal combinations, the theory behind the weldability materials combination and highlighted the above benefits in a number of practical applications.

Sierra *et al.*²⁶⁷ studied the weldability of low carbon steel to 6000 series aluminium alloy in keyhole welding mode using a CW Nd:YAG laser. Defect free welding can be obtained on overlapping aluminium and steel joints with steel on aluminium with the weld zone composed of solid solution of aluminium in iron and FeAl phases when the penetration of steel in aluminium was below 500 μm . Embrittlement at the joining zone was observed, mainly located on the weld/aluminium interfaces comprising Fe_2Al_5 and/or FeAl_3 phases with thicknesses between 5 and 20 μm . Figure 9*a* and *b* shows the variation in linear failure strength (N mm^{-1}) which corresponds to the strength divided by specimen width as a function of penetration depth for 150 and 200 mm focal lengths. It is apparent that the general strength decreases with increasing penetration with a maximum strength close to 250 N mm^{-1} obtained for 200–400 μm penetration depths. Increasing penetration depth leads to a change in the assembly and creation of weak points (in the weld and on the steel/weld interfaces) and induces a severe decrease in strength. Welds obtained in an aluminium on steel configuration always exhibited cracks at the interfaces.

Majumdar *et al.*²⁶⁸ developed a crack free weld between a commercially available Ti alloy (Ti–6 wt-%Al–4 wt-%V) and a wrought Al alloy (Al–1 wt-%Mg–0.9 wt-%Si). Intermetallic compounds (mainly TiAl and Ti_3Al) were formed in the fusion zone depending on the amount of Al and Ti melted by the laser. These intermetallic phases are very

brittle and the solid state cracks are formed near the Al side of the interface because of the stress developed after the solidification. In order to minimise the dissolution of Al in the fusion zone and to increase the toughness of the intermetallic phases, Nb foil is added as a buffer between the Ti alloy and Al alloy workpieces. It is observed that the partially melted Nb acts as a barrier to dissolve Al in the fusion zone and facilitates a good joining condition for welding of Ti alloy with Al alloy.

Liu *et al.*²⁶⁹ developed a new kind of hybrid joining technique called ‘laser (continuous) weld bonding’ as an alternative to laser welding and adhesive bonding. With this method, a good joint between AZ31B Mg alloy and A6061 Al alloy was acquired.

Mai and Spowage²⁷⁰ studied welding of steel to Kovar, copper to steel and copper to aluminium using a 350 W pulsed Nd:YAG laser. Laser welding of steel and Kovar resulted in significant intermixing of both materials within the fusion zone. Although the steel plate was completely melted, a complete metallurgical bond could not be obtained at the interface between the Cu plate and the weld metal, mainly due to the high thermal conductivity of copper. The weldability of copper to aluminium is regarded as relatively good. The Cu–Al phase diagram shows a wide range of Cu–Al phases that may be formed. In addition, non-equilibrium cooling conditions are known to promote the formation of a series of metastable phases. Under optimum processing parameters, crack free welds between copper and Al 4047 were obtained.

Kolodziejczak and Kalita²⁷¹ carried out keyhole LBW of dissimilar magnesium alloys AZ91 and AM50 with thicknesses of 4.5 mm using a CO_2 laser with a maximum power of 2.5 kW using helium as shielding gas. The microstructures, microhardness, tensile strength, bend strength and corrosion resistance of the welded zone were reported.

Qi and Song²⁷² reported on the role of Ni interlayer on the transformation of bonding mode of the joint between magnesium and mild steel developed by hybrid LATIG welding technique. The formation of Mg_2Ni intermetallic was reported at the interface between the magnesium alloy and nickel which was concluded to provide the metallurgical bonding of welding. On the other hand, the solid solution of Ni in Fe was detected

along the side face of molten pool and accumulated at the bottom.

Liu and Zhao²⁷³ reported the application of laser–gas tungsten arc hybrid welding technique for successful joining of AZ31B Mg alloy and 304 stainless steel. The mechanism of bonding was the chemical interaction between magnesium and iron. However, the formation of metallic oxide was responsible for the initiation of fracture during tensile testing.

Laser brazing

Like welding, laser is widely used in brazing non-ferrous metals/alloys and ceramics/glasses. Schubert *et al.*²⁷⁴ have shown that a 1.4 kW diode laser (808 nm) with a rectangular spot can be very useful in cladding and brazing.

Biro *et al.*¹⁷² have studied the effects of Ni and Au/Ni plating on laser brazing/welding of 200 µm thick aluminium, nickel, Kovar and cold rolled steel sheets in the lap joint configuration using a pulsed Nd:YAG laser and a range of weld process conditions. The strength of the joint was equivalent to that of the annealed base material. Significant gas porosity was observed at the interface between the Al weld pool and the unmelted Ni and Au/Ni plating layers. However, porosity did not affect the tensile shear strengths of the Al joints. Au/Ni braze increased the strength of the Au/Ni plated Ni specimens to that of the base material.

Laser soldering

Laser soldering of semiconductor devices needs extreme care and perfection. However, the precision of laser soldering is far superior to conventional soldering. For instance, 215 µm wide tape automated bonding leads could be soldered to a device with 430 µm centre to centre spacing by 98 ms diode pumped Nd:YAG laser at 5.8 W/lead laser power. As conventional soldering, one of the major concerns with laser soldering remains the mechanical property, particularly creep resistance at elevated temperatures. Guo *et al.*²⁷⁵ have studied the creep behaviour in Cu and Ag particle reinforced composite and eutectic Sn–3.5Ag and Sn–4.0Ag–0.5Cu non-composite Pb free solder joints at 25, 65 and 105°C (representing homologous temperatures ranging from 0.61 to 0.78). Creep resistance in composite solder joints was significantly improved with Cu particle reinforcements, but not with Ag reinforcement. The strain at the onset of tertiary creep for Cu and Ag reinforced composite solder joints was typically lower compared with non-composite solder joints. The activation energies for creep were similar for all the solder materials. Brandner *et al.*²⁷⁶ have explored soldering of thin or narrow Cu based electrical joints with solid state (1064 nm) and diode (808 nm) lasers to identify the mechanism of energy coupling, model the temperature rise and determine the process window. Laser soldering may be more beneficial for secondary fluxless reflow solder bumping than soldering in a furnace.

Berkowitz and Walvoord²⁷⁷ have studied laser soldering of fine pitch printed wiring boards using a CW diode laser. The success of laser soldering of such semiconductor devices depends on the selection of laser, optimisation of laser parameters, surface preparation/cleaning, lead form configuration, solder selection and soldering process control.

Laser tissue joining

Laser has now opened new possibilities of joining organic substances including human and animal tissues and organs. Laser tissue welding uses laser energy to induce thermal changes in the molecular structure of the tissues, though low strength anastomoses and thermal damage of tissue are major sources of concern. On the other hand, laser tissue soldering is a bonding technique in which a protein solder is applied to the tissue surfaces to be joined, and laser energy is used to bond the solder to the tissue surfaces. The addition of protein solders to augment tissue repair procedures significantly reduces the problems of low strength and thermal damage associated with laser tissue welding. McNally *et al.*²⁷⁸ have attempted to determine the optimal solder and laser parameters for tissue repair in terms of tensile strength, temperature rise and damage and the microscopic nature of the bonds formed. An *in vitro* study was performed using an 808 nm diode laser in conjunction with indocyanine green doped albumin protein solders to repair bovine aorta specimens. Liquid and solid protein solders prepared from 25 and 60% bovine serum albumins were compared. Increasing the laser irradiance or surface temperature resulted in an increased severity of histological injury. The strongest repairs were produced with an irradiance of 6.4 W cm⁻² using a solid protein solder composed of 60% bovine serum albumin and 0.25 mg mL⁻¹ indocyanine green with a surface temperature around 85 ± 5°C and thermal gradient of ~15°C across 150 µm thick solder strips. Post-soldering histological examination showed negligible evidence of collateral thermal damage to the underlying tissue. Microstructural study revealed albumin within the tissue collagen matrix and subsequent fusion with the collagen as the mechanism for laser tissue soldering. Thus, laser tissue soldering may be effective for producing repairs with improved tensile strength and minimal collateral thermal damage over conventional tissue welding. The tensile strengths of the welded shell membrane of hen eggs and the Wistar rat skin showed significant improvement using Forsterite and Cunyte tunable lasers with a range of wavelength from 1150 to 1500 nm.²⁷⁹

Summary and future scope

Laser joining is one of the earliest recorded applications of laser material processing, which is still considered a major field of useful application of laser. The principle reason for this popularity is the fact that heating by laser irradiation is possible for all kinds of engineering materials irrespective of the chemistry, state, bonding or size/geometry. This contactless direct heating is obviously a big advantage in joining a component with another of the same or different type by using laser just as a clean source of heating without the risk of any chemical reaction or change of the irradiated material. Obviously, joining special steels and alloys constitutes the main demand for laser welding or joining. Apart from routine joining, laser is useful in joining dissimilar materials such as steel to Al, steel to alumina, polymer to metal, etc. Even, joining of inorganic thin films, microelectronic devices, human/animal tissues and organic substances is now commonplace using laser. The main issues for future research concern joining materials with dissimilar physical (melting point, density and diffusivity) and

mechanical (hardness and strength) properties and bonding (e.g. metallic to covalent). While several attempts have been made to develop mathematical models to predict the microstructure and width of the joint with similar materials, similar approaches now need to be extended for dissimilar materials. In this regard, capability to predict the microstructure, integrity and strength of the joints would be the core issue. The most useful contribution could be a multiscale model that can connect the atomic transport events to microstructural evolution of phases/phase aggregate and composition profile, and eventually to the mechanical properties in macroscopic length scale. In particular, investigations on the changes in microstructural and mechanical properties across the weldment as a function of the selected process/laser parameters, macroscopic length scales and material dimension/properties are needed. Perhaps, using filler rods can be dispensed with if mechanism for brittleness of the joints in certain combinations is well understood. On the other hand, using an intermediate barrier layer/metal that wets well with two dissimilar metals or components on either side but can prevent intermixing of these two components in the melt pool so that the formation of harmful brittle phases could be avoided would be a useful strategy. Apart from welding, attempts must be made to develop useful localised joining methods such as sintering and brazing to fabricate finished products of greater variety and challenges. Laser assisted welding could be extended to derive additional advantages of cladding, reclamation and refurbishing not only old and worn components but also to develop compositionally and functionally graded surfaces with tailored composition and microstructure. For further details about the technology and mechanism of laser joining, several recently published text books may be consulted.^{2,4,166}

Laser machining

Laser machining refers to controlled removal of material at a given rate by laser induced concomitant melting and evaporation from the designated area/spot on the surface or bulk of the workpiece. If dimension or volume adds in laser forming or joining, laser machining leads to a reduction in size or dimension of a component. Laser machining can achieve almost all types of conventional machining operations such as drilling, cutting, cleaning, marking and scribing up to a limited dimension or depth.^{1,8} Metals, ceramics, polymers and composites are all amenable to laser machining irrespective of the hardness and physical/chemical properties of the material. The entire process can be fully automated with high speed and precision retaining clean and sharp edges with minimum HAZ. Laser assisted cutting, drilling, cleaning (or paint stripping), marking and scribing can be applicable to any material including very soft polymers to extremely hard ceramics (through normal or oblique/inclined irradiation), ultrathin semiconductor chips to multilayer metallic films/sheets and simple/flat surfaces to complex/curved geometry. The discussion in this section will highlight the scope/versatility of laser machining and emphasise the recent advances and unresolved issues in using laser as a non-contact and non-contaminating tool for machining of engineering solids.

Laser cutting

Laser cutting does have a diversified application starting from thin sheet metal cutting for general purpose equipment (such as household appliances, electrical cabinets and automotive components) to thick section metal cutting (for trucks, buildings, stoves, construction equipment, shipbuilding, etc.). Research efforts that have been successfully conducted on laser assisted machining of engineering materials (both metallic and non-metallic) have been reported in the literatures.^{280–323} Titanium alloys cut in an inert atmosphere are used in airframe manufacture. Aluminium alloys have similar advantages in cutting by using the laser. Cutting of radioactive material is another important area of application of laser cutting. Cutting of wood up to 1 in thickness may be useful in the die board industry, furniture industry, puzzle and gift industry, crafts and trophies, etc. Plastics, rubbers, composites, cloth, ceramics and such diverse materials can be cut by laser in textile industry. Hard and brittle ceramics such as SiN can be cut 10 times faster by laser than by diamond saw. In the following section, laser cutting of different materials is discussed in details.

Laser assisted cutting of ferrous metals and alloy

A simple empirical relation to describe the parameters for laser cutting of iron and steel with a high quality of kerf was established by Belic and Stanic.²⁸⁰ This empirical relationship between laser power P (W), sheet thickness d (mm), cut width s (mm) and cutting speed V (mm min⁻¹) was established as: $P = 390d^{0.21}s^{0.01}V^{0.16}$ (ignoring the variation in thermal properties of material with temperature). It agreed well with the experimental results obtained from laser cutting of iron, steel and stainless steel with a 500 W CW CO₂ laser. A systematic investigation of the effect of gas composition and pressure on CO₂ laser assisted cutting of mild steel was reported by Chen.²⁸¹ The gas mixtures used were oxygen, argon, nitrogen and helium. It was found that high purity oxygen was conducive for better cutting performance. Low pressure inert gas could not produce a good quality cut. The influence of high pressure oxygen jet on the cutting efficiency was studied by O'Neill and Gabzdyl²⁸² for thick sections of mild steel plates (20, 30 and 50 mm in thickness) with a CW CO₂ laser at an incident power of 1 kW. Striation formation at the cut edge is often encountered during oxygen cutting of mild steel, which was attributed to: cyclic variation in the driving force of the oxidation reaction owing to the changes in the oxygen partial pressure in the melt zone melt; and viscosity and surface tension effects associated with melt removal.²⁸³

Danishman *et al.*²⁸⁴ measured the electron temperature and density of the surface plasma generated during CO₂ laser cutting of mild steel. Electron temperature and density were found to oscillate with cutting time. Electron number density decreased with increasing electron temperature. Electron number density also diminished when an argon+oxygen gas mixture was used rather than oxygen. It was concluded that the cut quality improved at high electron temperature and low electron number density.

Wang and Wong²⁸⁵ carried out a detailed investigation on the machinability of galvabond (galvanised commercial steel sheet with spangled surface) steel sheet of 0.55, 0.8 and 1 mm in thickness using a CW CO₂

laser. A relation among the cutting speed, laser power and workpiece thickness was developed by statistical analysis for selecting the cutting parameters for process control and optimisation. Similarly, Prasad *et al.*²⁸⁶ discussed the laser beam machining of metal coated steel sheet (mainly galvalume, galvaneal and galvabond) of 1 mm in thickness with a 500 W CW CO₂ laser and achieved an improved cut quality in terms of good surface finish, reduced kerf width and dross. An analytical model was developed to establish the parametric relation of the cutting process. The model suggested that an efficient choice of the process parameters was a prerequisite for minimum thermal damage of the coatings as the former had a significant influence on the topographical characteristics of the uncut through kerf, surface roughness and micro- and macromechanics of the cutting process. The comparison of magnetic properties of mechanically and laser cut uncoated non-oriented electrical steel suggests that the best magnetic property was obtained after mechanical cutting.²⁸⁷ King and Powell²⁸⁸ cut mild steel using continuous and pulsed CO₂ laser and concluded that pulsed laser irradiation is better than continuous one. Tabata *et al.*²⁸⁹ carried out a detailed study of laser cutting of steel using CO₂, Nd:YAG and diode laser and compared the effect of different lasers on cutting quality. Zheng *et al.*²⁹⁰ undertook laser cutting of stainless steel, aluminium and mild steel with CO₂ laser and showed that laser cuts had narrower kerf width with better accuracy than that obtained by mechanical cutting. Skvarenina and Shin²⁹¹ studied the machinability of compacted graphite iron by varying the depth of cut, feed and material removal temperature and then evaluating resultant cutting forces, specific cutting energy, surface roughness and tool wear. At a material removal temperature of 400°C, laser machining was successful to increase the tool life to 60% greater than conventional conditions at a feedrate of 0.150 mm rev⁻¹ without affecting the microstructure. Surface roughness was improved by 5% as compared with conventional machining at a feedrate of 0.150 mm rev⁻¹. A 20% reduction in the cost of machining was also achieved by laser assisted machining.

Laser assisted cutting of non-ferrous metals

Among the common aeronautical materials, Al alloys are one of the most suitable materials for laser machining. Araujo *et al.*²⁹² did a detailed investigation of the microstructures generated from laser machining of 2024 Al alloy using CO₂ laser. Owing to the large α -liquid region at 4%Cu in the Al–Cu phase diagram and low laser absorption factor of solid Al (at 10.6 μ m wavelength), the material partially melts before evaporation/sublimation of the liquid. On the other hand, the high pressure gas causes dripping of the high viscous melt and degrades the surface quality of the machined material. This promotes the generation of high stress concentration regions that might enhance its fatigue resistance. Indeed, the fatigue resistance of laser machined Al 2024-T3 aluminium alloy was found higher than that in the same alloy in as received condition.²⁹³ Laser assisted cutting of Al based metal matrix composites was extensively studied to understand the microstructure, residual stress, mechanical properties (wear and fatigue) and process optimisation.^{294–296} Yue and Lau²⁹⁷ have studied the pulsed laser cutting of Al–Li/SiC metal matrix composite to examine the influences

of laser cutting parameters on the quality of the machined surface. Proper process control seems effective to minimise the HAZ, improve the quality of the machined surface and predict the maximum depth of cut for the composite. The effects of femtosecond laser machining on surface characteristics and subsurface microstructure of a Nitinol alloy were studied by Huang *et al.*²⁹⁸ The femtosecond laser machining produced an average surface roughness of 0.2 μ m similar to the surfaces machined by precision milling.

Laser assisted machining also offers the ability to machine superalloys more efficiently and economically by providing the local heating of the workpiece before material removal by a single point cutting tool. The machinability of Inconel 718 under varying conditions was evaluated by examining tool wear, forces, surface roughness and specific cutting energy by Anderson *et al.*²⁹⁹ With increasing material preheating temperature from room temperature to 620°C (by laser heating), the benefit of laser assisted machining was demonstrated by a 25% decrease in specific cutting energy, two- to threefold improvement in surface roughness and 200–300% increase in ceramic tool life over that in conventional machining. Moreover, an economic analysis shows significant benefits of laser assisted machining of Inconel 718 over conventional machining with carbide and ceramic inserts.

Laser assisted cutting of ceramics

The high hardness and brittleness of ceramics make them difficult and expensive to machine using conventional cutting methods or tools. Laser cutting provides a good means for cutting of ceramics in non-contact mode with excellent precision. In laser machining of ceramics, the machining process consists of three inherent stages, i.e. vapourisation cutting, melting and ejection and controlled fracture.³⁰⁰ Lu *et al.*³⁰¹ studied the problem of cracking of ceramic plates during laser assisted cutting and identified that the most relevant parameters for laser cutting were laser power P and scan speed v , specific heat c , conductivity k and thermal expansion coefficient α . Tsai and Chen³⁰² attempted synchronous application of Nd:YAG and CO₂ lasers for laser cutting of thick ceramic substrates by controlled fracture. Tsai and Chen³⁰³ proposed a new image processing system to determine the laser movement path by iterative path revision algorithm useful for monitoring laser assisted cutting.

Cutting of engineering ceramics such as silicon nitride (Si₃N₄) was studied by several groups including Hong *et al.*,³⁰⁴ Hong and Li³⁰⁵ and Rozzi *et al.*³⁰⁶ with Q switched pulsed YAG laser. Considering the influence of the cut front shape on the absorption of the laser beam, a simplified two-dimensional mathematical model was developed based on the conservation of energy during pulsed laser vapourisation assisted cutting process. The experimental results showed that crack free cutting could be obtained by using high speed and multipass feed cutting routine. A detailed parametric investigation of laser assisted machining of aluminium nitride ceramic was undertaken with an UV and near infrared Nd:YAG laser by Gilbert *et al.*³⁰⁷ The effect of pulse overlap and pulse frequency on material removal rate was studied. The effect of laser power, scan speed, mode of cutting and shield gas composition on cutting of thick ceramic tiles using a CW CO₂ laser was studied by Black and

Chua³⁰⁸ and Black *et al.*³⁰⁹ Crack formation was one of the major problems associated with the cutting process. Large thermal gradient between the cut zone and associated glazed zone was responsible for the crack formation. Proper choice of laser parameters was essential to allow sufficient time to cool and minimise the tendency for crack formation. In this regard, Lit and Sheng³¹⁰ opined that fracture initiation during laser cutting of alumina was primarily related to laser parameters.

Pascual-Cosp *et al.*³¹¹ studied the cutting behaviour of highly vitrified ceramic materials and fine porcelain stoneware tile using Nd:YAG laser to determine the optimum processing zone for laser cutting. Preheating of the workpiece (to 1000°C) was beneficial to avoid catastrophic breakdown of the material under the action of laser. Besides cutting thick substrates, Park *et al.*³¹² investigated the scope of precision micromachining of chemical vapour deposited diamond film using excimer laser and concluded that the principle and mechanism were identical to those for laser cutting of thick substrates.

Pfefferkorn *et al.*³¹³ studied the machining of magnesia doped partially stabilised zirconia using a CW CO₂ laser and concluded that the mechanism of material removal is a combination of brittle fracture and ductile deformation with the latter becoming dominant as the uncut chip thickness decreases near the penultimate point of tool contact.

Rebro *et al.*³¹⁴ showed the optimum processing conditions for a defect free laser assisted machining of pressure less sintered mullite ceramics using a CW CO₂ laser which otherwise is difficult due to its low thermal diffusivity and inferior tensile strength.

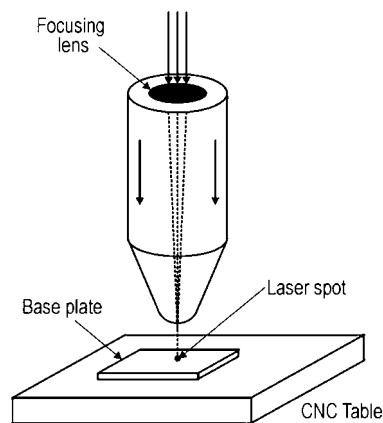
Laser assisted cutting of polymers

Laser cutting is equally useful in manufacturing special materials including circuit boards, resistance and functional trimming of circuits and microlithography. The growing use of the excimer laser in above operations may enable new and precision machining. Lum *et al.*³¹⁵ studied laser cutting of medium density fibreboard to determine the influence of laser parameters on the cut quality and optimum laser parameters (in terms of cutting speeds, material thickness and pulse ratio) for effective cutting of these fibreboards by CO₂ laser both under CW and pulsed modes of cutting.

Laser cutting of composites of aramid, graphite and glass cloth reinforced polyester were successfully implemented through studies on morphology of the cut surfaces by scanning electron microscopy and by numerical simulation of laser induced thermal changes.³¹⁶ The thermal properties of the fibres and matrix were identified as the principal factors that affected cutting performance. Laser machining has also been reported on polymer matrix composite.^{317–323}

A theoretical model describing the temperature distribution during CO₂ laser cutting of thin thermoplastic material (used in the packaging industry) was developed by the Green function method.³²⁴ It was shown that suitable control of laser beam diameter allowed two types of cutting operations: a simple cut, when the beam focus coincided with the surface and a cut with welding, when the beam was defocused.

Graff and Meijer³²⁵ investigated CO₂ laser cutting of newly developed materials (synthetic aluminium laminates) for the aeronautic and automotive industry and found that



10 Schematic of laser assisted drilling of thin sheets of metal, semiconductor or polymers with provisions for changing focus and rate of shroud gas flow

these materials could be cut at the same speed as homogeneous aluminium alloys, although some damage on the synthetic layers was unavoidable. Based on these experimental results, a dedicated computer simulation model was developed using appropriate mass, power and force balance equations that considered splitting of the laminate in several horizontal layers according to the concerned thermophysical and optical properties.

Laser drilling

Laser drilling is a machining operation by melting and evaporation of the workpiece using a high power laser beam. Metals/alloys, polymers and ceramics may be drilled by laser assisted drilling.^{326–337} High power CO₂ laser or Nd:YAG laser is generally used to drill holes. Figure 10 shows the basic set-up used in laser drilling, which is not too different from that used for other laser machining processes except that the workpiece remains stationary with regard to the laser spot. Laser drilling can be done both in pulsed and CW modes with suitable laser parameters. The advantage of using laser is that it can drill holes not only vertically but also at an angle inclined to the surface (e.g. fine lock pinholes in Monel metal bolts). Mechanical drilling is slow and causes extrusions at both ends of the hole that have to be cleaned. Mechanical punching is fast but is limited to shallow holes with greater than 3 mm diameter. Electrochemical machining is quite a slow process though produces a neat and precise hole. Electrodischarge machining is expensive and slow with a typical rate such as 58 s/hole. Electron beam drilling is fast at 0.125 s/hole but needs a vacuum chamber and is more expensive than a Nd:YAG laser processing. In comparison, a Nd:YAG laser can produce a hole in 4 s to outsmart all other methods.¹

Low *et al.*³²⁶ studied the rate of sputter deposition concomitant with laser drilling in Nimonic 263 alloy for various laser processing parameters using a fibre optic delivered 400 W Nd:YAG laser. It was observed that shorter pulse widths, lower peak powers and higher pulse frequencies generated smaller sputter deposition areas.

Laser micromachining

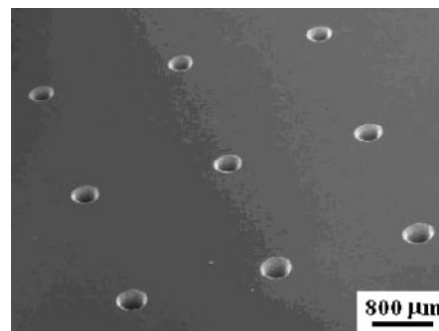
The ability to machine very small features such as holes into a metal, ceramic, semiconductor or polymer sheet/film by laser ablation with an unmatched precision, accuracy and speed has opened a very useful scope of

application of laser material processing in microelectronic industry. For instance, holes with a diameter of 300 nm and a depth of 52 nm could be drilled in metal films with minimum distortion and HAZ using 200 fs and 800 nm pulses from a Ti:sapphire laser focused to a spot size of 3000 nm. Lehan and Kwok³²⁸ have developed a novel method for improving the efficiency of laser drilling using two synchronised free running laser pulses from a tandem head Nd:YAG laser capable of drilling through 1/8 in thick stainless steel targets at a standoff distance of 1 m without gas assist. The combination of a high energy laser pulse for melting coupled with a properly tailored high intensity laser pulse for liquid expulsion results in efficient drilling of metal targets. The improvement in drilling is attributed to the recoil pressure generated by rapid evaporation of the molten material by the second laser pulse. Similarly, Zhu *et al.*³²⁹ have carried out a detailed experimental study of drilling submicrometre holes in thin aluminium foils with thickness ranging from 1.5 to 50 μm , and W, Mo, Ti, Cu, Fe, Ag, Au and Pb foils of 25 μm in thicknesses with femtosecond Ti:sapphire laser pulses of 800 nm in width. The influence of laser parameters and material properties on hole drilling processes at submicrometre scale has been examined and a simple model to predict the ablation rate for a range of metals has been developed.

Laser precision machining has been applied to fabricate metallic photonic bandgap crystals operating in the microwave frequency.³³⁸ Transmission measurements showed that the periodic crystals exhibited a cutoff frequency in the 8–18 GHz range allowing no propagation below this level. Furthermore, the cutoff frequency could easily be tuned by varying the interlayer distance or the filling fraction of the metal. The combination of plates with different hole diameters creates defect modes with relatively sharp and tunable peaks. Illyeaaalvi-Vitez³³⁹ has attempted utilisation of CO₂ and frequency multiplied Nd:YAG lasers (using five wavelengths, i.e. 10 600, 1064, 532, 355 and 266 nm) for drilling and direct patterning of copper clad glass fibre reinforced epoxy laminates, polyester foils and similar packaging structures. Laser processing was combined with through contacting of the generated vias by screen printing with polymer thick films, by wet chemical direct plating and by evaporation of thin metal layers. The results show promising opportunities for laser processing of metal layers and polymeric materials in microelectronic packaging.

Pronko *et al.*³⁴⁰ demonstrated that for micromachining, femtosecond pulses offer an advantage over current technology of employing nanosecond pulses because of the reduced effects of thermal diffusion. Using the femtosecond pulses, holes with a diameter of 300 nm, roughly 10% of the spot size was produced, whereas the smallest diameter made using the nanosecond pulses was roughly 60% of the spot size. The depth of the femtosecond hole was 52 nm.

Different types of glass materials were machined by Nikumb *et al.*³⁴¹ using a short pulse solid state laser with pulse duration in the nanosecond to femtosecond range. The effect of pulse duration and other process parameters on the machined features was analysed to reveal the underlying thermal effects and non-linear processes. Edge quality, circularity, aspect ratio, formation of the redeposit material and machining rate were also studied



11 Array of through vias in 3C-SiC wafer with thickness of 400 μm drilled with solid state laser providing pulses with duration of 300 fs at wavelength of 1040 μm (Ref. 342)

with respect to the process variables such as focusing optics, laser power, wavelength and repetition rate. Arrays of drilled microhole patterns and fabricated microfeatures are demonstrated with discussions on their potential applications.

Zoppel *et al.*³⁴² studied laser ablation of silicon carbide with emphasis on the influence of pulse duration and laser wavelength on ablation efficiency. A KrF laser of 248 nm wavelength (34 ns pulse duration), ArF laser of 193 nm wavelength (25 ns pulse duration) as well as fs-UV experiments with a distributed feedback dye laser system amplified by an excimer laser, delivering 500 fs pulses at 248 nm were employed for laser processing. Additional experiments were performed with solid state lasers providing radiation at 1040 and 1026 nm at pulse durations of 300 and 400 fs respectively. Figure 11 shows an array of vias drilled in 3C-SiC at a wavelength of 1040 nm with a pulse duration of 300 fs. Despite the relatively low fluence of 10.6 J cm^{-2} , through via drilling was possible by employing trepan drilling, where the beam moves along a circular path.

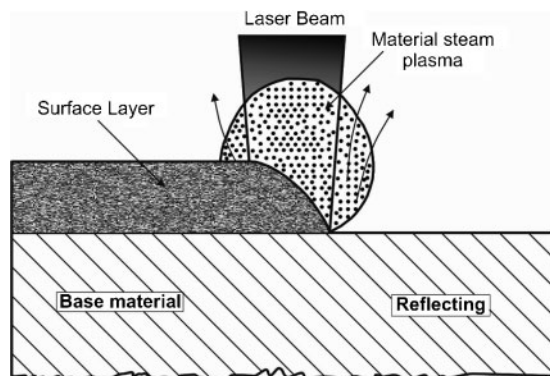
Laser cleaning

Removal of small particles or continuous layers from a metal surface can be carried out by laser beam using a selected area irradiation at an optimum combination of incident power, interaction/pulse time and gas flowrate (that sweeps the dislodged atoms from the surface).^{343–354}

Figure 12 schematically shows the mechanism of laser cleaning. At the initial stage, a plasma plume is formed due to ionisation of the atoms vapourised from the surface and blocks the beam/surface contact. As the irradiation stops, the temporary compression on the surface changes into tension and causes spallation of the oxidised layer. A dramatic improvement in cleaning efficiency in terms of area and energy is possible when laser beam irradiates the workpiece at an oblique or glancing angle of incidence rather than direct or perpendicular irradiation of the surface.

The influence of pulsed laser irradiation on the oxide scale removal from the surface of stainless steels oxidised using a pulsed Nd:YAG laser was investigated by Psyllaki and Oltra.³⁴³ At an appropriate angle of incidence, laser irradiation with an energy density of $\sim 10^4 \text{ J m}^{-2}$ resulted in expulsion of the oxide layer without any material removal from the underlying metal.

The side wall and bottom polymers resulting from reactive ion etching of via holes can be removed by a non-contact dry laser cleaning technique using pulsed



12 Schematic of laser assisted cleaning

excimer laser irradiation.³⁴⁴ Similarly, laser cleaning is capable of removing the polymers by subthreshold ablation, even at fluences limited by the damage threshold ($2500\text{--}2800\text{ J m}^{-2}$) of the underlying Al–Cu metal film with titanium nitride (TiN) antireflective coating. Comparing ablation results obtained using Nd:YAG laser and excimer laser shows that although the shorter 7 ns Nd:YAG laser pulse gives a greater etch thickness than the 23 ns excimer laser pulse, it also tends to damage the metal films and the silicon substrates of the via wafers more easily.

Tsunemi *et al.*³⁴⁵ have demonstrated that pulsed laser irradiation of oxidised metallic surfaces in an electrolytic cell under proper voltage conditions could be a promising new approach for effective removal of oxide films.

Schmidt *et al.*³⁴⁶ studied the process characteristics for removing white chlorinated rubber coatings, pigmented with titanium white (TiO_2), from concrete surfaces utilising a 120 W CW diode laser. About 0.3 mm thick double coating layers were removed utilising oxygen process gas flow.

Laser removal of small copper particles from silicon wafer surfaces was carried out using appropriate laser radiation from near infrared (1064 nm) through visible (532 nm) to UV (266 nm) wavelength range.³⁴⁷ It has been found that both 266 and 532 nm are successful in removing the particles from the surface, whereas 1064 nm was shown to be ineffective for removal of particles. The cleaning efficiency increased with a shorter wavelength. The mechanism of laser cleaning was explained by the evolution of tensile forces exceeding the adhesion force of the particle to the surface at given irradiation wavelengths.

Meja *et al.*³⁴⁸ studied the cleaning of oxidised iron samples using a Nd:YAG laser to model the influence of wavelength in air and the role of environment on the cleaning behaviour. This study showed that cleaning occurs in a small energy density range irrespective of the wavelength of the laser. The adhesion force of the oxide layer influences the optimal laser parameters, and the repetition rate determines the minimal spot size for the desired cleaning. Furthermore, it was also concluded that the application of pulsed laser irradiation in liquid confinement under controlled electrochemical potential enhanced the removal efficiency of oxide films from metal surfaces.

The influence of excimer laser irradiation (of 193 nm in wavelength and 20 ns pulses and of 248 nm in wavelength and 20 ns and 500 fs pulses and fluencies

between 8 and 19 J cm^{-2}) on the structure and morphology of plain and gold film coated mica surfaces was investigated.³⁴⁹ Surfaces treated with laser fluences below the ablation threshold were well suited for controlled growth of metallic films, whereas above threshold treated samples form roughened surfaces with close packed arrays of cones. It was seen that the first step of the ablation process included reorientation of dipole domains on the surface and the ablated products possessed large kinetic energy and high directionality.

Single crystal SiC substrate was successfully photo etched at a remarkably high etch rate of 35 nm s^{-1} by (133, 141, 150, 160, 171 and 184–266 nm) multi-wavelength laser ablation, combined with a chemical post-treatment in a solution of $\text{HCl} + \text{H}_2\text{O}_2 + \text{H}_2\text{O}$ and $\text{HF} + \text{H}_2\text{O}$.³⁵⁰ The analysis of the etched samples by scanning electron microscopy and scanning probe microscopy indicated that an array of square holes having well defined patterned structures and clean substrate surfaces were obtained. X-ray photoelectron spectroscopy analysis indicated that the SiC samples etched by multiwavelength laser have a similar stoichiometry after chemical post-treatment as the virgin SiC. The mechanism of high quality ablation using multi-wavelength laser was discussed and compared with that for ablation using 266 nm single wavelength. The chemical post-treatment contributed to removing the residues from the laser photolysis of SiC.

Laser marking

Laser, as a source of heat, may be employed in marking of metallic, polymeric and glassy materials.^{355–362} Laser ablation can be applied as a means for direct marking on plastics or polymer compounds as it is fast, economical and ecofriendly compared with the conventional marking processes.³⁵⁵ Besides embossing a mark/design on a surface for identification or aesthetics, laser marking is also useful in non-contact and dry etching and printing of polypropylene or polyethylene plastics using a 532 nm laser pulse.³⁵⁶ Similarly, laser irradiation can produce controlled microcracking or fracture of soda lime and borosilicate glasses.³⁵⁷

Beu-Zion *et al.*³⁵⁸ have designed and constructed a CO_2 laser marking system based on a scanned flexible wave guide (silver halide infrared optical fibre or a hollow glass wave guide) capable of cutting, heating or marking on various surfaces. Recently, permanent markings of various colours on different ceramic and plastic substrates (with orasols dispersed palladium acetate organometallic films) have been made by a photo thermal deposition process using an argon ion laser (514 nm).³⁵⁹

Recently, attempts have been made to develop a dual laser writing scheme in which an unmodulated short wavelength red laser augments the writing process effectively by a longer- wavelength laser. Apart from increasing the thermal efficiency of the laser marking process itself, the dual beam writing scheme may decrease the mark width and the recorded mark length variability. Coupled with the increased resolution of the short wavelength read spot, these enhancements could improve the overall writing or marking performance.

Laser scribing

Scribing involves laser ablation of a groove or row of holes that form perforation lines to separate a large

substrate into individual circuits. Laser scribing is extensively used for machining of ceramics in the micro-electronics industry. The quality of scribing, particularly for silicon chips and alumina substrates is judged by the amount of debris produced and size of HAZ (the smaller the better for both). Lump and Allen³⁶³ have drilled high aspect ratio, straight walled 60 and 300 μm diameter via holes in 635 μm thick aluminium nitride with or without a metallisation layer deposited on the via walls using an excimer laser. Attachment of a second substrate or metal sheet prevents damage to the back surface.

In fabrication of core laminations for motors and transformers, laser scribing may induce a reduction in hysteresis and eddy current losses. Laser scribing needs a very low power so as not to melt the surface. For example, pulsed excimer laser (248 nm and 23 ns) irradiation to scribe the grain oriented M-4 electrical steel has resulted in a 26% core loss reduction due to the beneficial thermal stresses that refined the magnetic domains and reduced the eddy current losses.³⁶⁴ Similarly, 50% reduction in core loss in 50 μm thick amorphous alloy ribbon has been achieved by laser scribing at 2–5 mm line spacing at the same excitation power.³⁶⁵ The mechanisms responsible for improvement include magnetic domain refinement, stress relaxation and inhibition of domain wall movement. Laser scribing also relieves the stresses that are induced in the material during manufacture. The scribe lines increase the surface resistivity of the material, resulting in reduced eddy current loss. Laser scribing of polycrystalline thin films used for solar cells can eliminate the frequently observed problem of ridge formation along the edges of scribe lines in the semiconductor films.³⁶⁶

Laser micromachining is a precision manufacturing technique, where pulses from ultrashort laser are used to induce micrometre sized structures on the surface or in the bulk of solid materials and an attractive technique for the high quality micromachining of many materials with minimum damage and maximum precision.^{367,368} The technique has been established since over a decade with a potential scope of application in thin film deposition, microelectromechanical systems fabrication, surface micromachining, three-dimensional internal waveguide fabrication, nanotechnology and surgical applications.^{369,370} A detailed basic understanding on micromachining with ultrashort laser pulses and technological development in this field for achieving an unprecedented level of accuracy at acceptable expenses has been presented by Dausinger *et al.*³⁷¹

The mechanism of laser ablation is different for different materials. Hence, the optimisation of process parameters is essential for efficient and precision machining.³⁷² Cheng *et al.*³⁷³ investigated the ultrafast laser microstructuring of Ti alloy, Al and Cu alloys using a Clark-MXR CPA2010 femtosecond laser system and a HighQ picosecond laser system. Ablation rate was found to increase with increasing repetition rate due to the combined effects of plasma, residual thermal energy and phase transition.

Summary and future scope

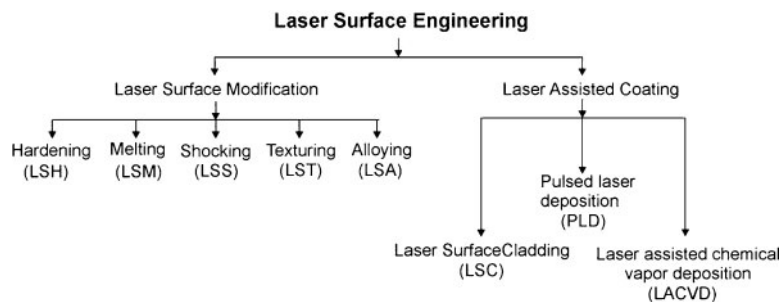
Laser machining requires controlled removal of material by vapourising within a narrow dimension incurring least heating and damaging of the surrounding region. The process can have many variations while removing

controlled amount of materials: cutting, drilling, scribing, marking or cleaning. In all these processes, material removal without damaging the surrounding and maintaining precision and accuracy is a challenge. The precise spatial and temporal resolution of laser is certainly an advantage here. The recent advances are now based on selecting appropriate wavelength, using multiple beams, allowing inclined/oblique incidence and removing material in stages. Multiple beams can preheat, reduce thermal shock, remove debris or vapour and increase thickness of cut in sequence or stages. A large variety of materials starting from human/animal tissues to diamond can be laser machined. The mechanism of material removal obviously varies while slicing animal tissues or soft organics than when ultrahard diamond tools are reshaped or a silicon wafer is machined. However, an appropriate choice of laser power and wavelength is crucial for the success of the operation. Owing to the diversity of materials, geometry and conditions that can be handled, laser machining now often relies on more than one laser for the same operation. Use of advanced optics and better control of the sample stage are the other concerns with regard to improving the quality of the cut. Future challenges include increasing the capability to machine thicker sections, curved surfaces and dissimilar/heterogeneous materials. Similarly, assessment of material damage needs more close control and monitoring of the microstructural change/damage across the cut. Development of intelligent machines for machining diverse materials would require interfacing a vast database with the hardware. Both speed and precision in fully automated operation are the biggest advantage of laser machining. Thus, continued efforts are needed to model laser machining processes with suitable experimental validation of the predicted results.

Laser surface engineering

Failure of engineering components or materials due to chemical (corrosion and oxidation) or mechanical (wear and erosion) interaction is most likely to initiate from the surface because both external and internal surfaces/interfaces are more prone to environmental degradation, and intensity of externally applied load is usually the highest at the surface.³⁷⁴ The engineering solution to minimise or eliminate such surface initiated failure lies in tailoring the surface composition and/or microstructure of the near surface region of a component without affecting the bulk.^{2,3,374–376} In this regard, the more commonly practiced or conventional surface engineering techniques such as galvanising, diffusion coating, carburising and nitriding possess several limitations such as high time/energy/material consumption, poor precision and flexibility, lack in scope of automation/improvisation and requirement of complex heat treatment schedule. Furthermore, thermodynamic constraint of restricted solid solubility and kinetic limitation of thermally activated solute transport to solid state diffusion impose further limits to the level of improvement possible through these conventional or near equilibrium processes.^{375–378}

Among the notable characteristic features, laser surface engineering enables delivery of a controlled quantum of energy (10^4 – $3 \times 10^5 \text{ J m}^{-2}$) or power (10^8 – 10^{11} W m^{-2}) with precise temporal and spatial distribution either in short pulses (10^{-3} – 10^{-12} s) or CW. The process is characterised by an extremely fast heating/



13 General classification of laser surface engineering

cooling rate (10^4 – 10^{11} K s⁻¹), very high thermal gradient (10^6 – 10^8 K m⁻¹) and ultrarapid resolidification velocity (1 – 30 m s⁻¹).^{374–376} These extreme processing conditions often develop a novel microstructure and/or composition in the near surface region with large extension of solid solubility and formation of metastable including nanocrystalline and amorphous phases.

Figure 13 presents a brief classification of different laser surface engineering methods that involve mainly two types of processes. The first type is meant only for only microstructural modification of the surface without any change in composition (hardening, melting, remelting, shocking, texturing and annealing), while the second one involves both microstructural as well as compositional modification of the near surface region (alloying, cladding, etc.).

Laser surface engineering of ferrous alloys

As already mentioned, laser surface engineering is mostly utilised to enhance resistance to corrosion, oxidation, wear and similar surface degradation of engineering materials. In this section, we review the scope and current status of understanding about laser surface engineering of Fe based or ferrous alloys that involves surface hardening, melting, alloying or cladding using a high power laser beam to improve the surface dependent engineering properties of interest. Detailed studies under each category are discussed in the following subsections.^{379–475}

Laser transformation hardening

Transformation hardening is a standard heat treatment for ferrous alloys (steel, cast iron, etc.) that involves heating to austenite (face centred cubic) phase and subsequent quenching to ambient or subambient temperature to enforce a shear induced austenite to martensite (body centred tetragonal) transformation instead of the equilibrium product/phase, i.e. ferrite (body centred cubic). In laser transformation hardening, only a thin surface layer of the substrate (instead of the entire bulk) is rapidly heated to the austenite phase field followed by rapid self-quenching (by conduction through the cooler bulk) to produce the desired martensitic microstructure.³⁷⁹ Heating and cooling rates of 10^4 K s⁻¹ or greater are typical of the laser surface hardening process and the entire thermal cycle in the irradiated volume may be accomplished in <0.1 s. Kennedy *et al.*³⁸⁰ have presented a detailed review on the fundamentals of laser hardening, associated benefits compared with the conventional hardening techniques and notable experimental studies particularly using high power diode lasers.

Prior microstructure of the substrate influences the final microstructure of the laser surface treated steel to a

large extent. Grum and Slabe³⁸¹ observed the formation of nanometric precipitates with a diverse microstructure up to varying depth following laser surface hardening of maraging steel. Heidkamp *et al.*³⁸² studied the effect of laser beam hardening of TiN coated [by chemical vapour deposition (CVD)] AISI 4140 (EN42CrMo4), AISI A2 (ENX100CrMoV5-1) and AISI D2 (ENX153CrMoV12) substrates. A composite microstructure with strength and hardness gradients from surface to core together with a high surface hardness and wear resistance was achieved. Kulka and Pertek³⁸³ heat treated the boride layers produced on 41Cr4 medium carbon steel (0.43%C and 1.02%Cr) using a CW CO₂ laser to obtain a laser surface modified surface with a reduction in hardness and brittleness. Similarly, laser heat treatment of carburised steel showed an improvement in microhardness and wear resistance.

Surface condition (roughness, residual stress and phase aggregate) is important in determining the characteristics and performance of the hardened layer. The effect of different types of absorption coating on laser surface hardening of C45E steel was studied by Grum and Kek³⁸⁴ by using a photodiode with infrared radiation. The results confirmed that there was a strong correlation between the voltage signal of the infrared radiation and the size of the hardened layer on the steel surface depending on the type of absorbent chosen.

Heitkemper *et al.*³⁸⁵ showed that fatigue and fracture behaviour of laser heat treated high nitrogen martensitic tool steel (X30CrMoN15) with a 3 kW Nd:YAG laser was significantly improved as compared with that of the untreated one. The residual stresses generated during laser treatment had a significant influence on crack initiation, while that produced during the transformation of retained austenite to martensite had a minor influence on mechanical behaviour. The impact resistance of laser surface hardened (using a 3 kW CW CO₂ laser) hypoeutectoid 2Cr13 martensite stainless steel was improved in laser treated sample as compared with that of the same processed under conventional heat treatment conditions.³⁸⁶ The superior impact wear resistance of laser surface processed steel was attributed both to grain refinement and higher martensite content near the surface that resulted from rapid cooling. Iordanova *et al.*³⁸⁷ carried out laser surface melting of 0.46 mm thick cold rolled low carbon steel by pulsed Nd:glass laser up to a thickness of 80 μ m. Prior cold rolling developed residual tensile stress both along rolling and particularly along the transverse directions. Subsequent laser melting did affect the residual stress along the rolling direction, but significantly increased that along the transverse direction with a weaker and scattered texture.

Ashby and Easterling³⁸⁸ developed a heat transfer model to predict the surface temperature during laser transformation hardening. The residual stress developed in the vicinity of the workpiece surface following repetitive pulse heating was predicted from the two-dimensional heat transfer and fluid flow model using a controlled volume approach by Shuja *et al.*³⁸⁹ Laser annealing of aged 17-4 PH (AISI 630) steel (aged at 482°C) was carried out using a 5 kW Rofin-Sinar CO₂ laser.³⁹⁰ Immersion test in the H₂S saturated solution indicated that laser annealed and H900 (peak aged) specimens had the similar ability to resist hydrogen induced cracking. Laser annealing led to an improved fatigue property predominantly because of the presence of residual compressive stress ahead of the laser annealed zone in the base metal. Amulevicius *et al.*³⁹¹ studied the influence of simultaneous laser irradiation and ultrasound on the phase evolution in the Fe-Si-C system (grey cast iron) by means of Mössbauer spectroscopy. The application of ultrasound reduced the amount of retained austenite in the samples by 2–3 times due to activated dissolution and redistribution of carbon between austenite and iron-carbon clusters. Chen and Shen³⁹² optimised the process parameters for laser transformation hardening of SNCM 439 steel using Taguchi methodology and fuzzy evaluation technique. Under optimal conditions, the hardness of laser hardened steel using a long pulsed Nd:YAG laser was improved from 52.5 to 63.9 HRC, the hardening width was increased from 0.43 to 0.89 mm and the erosion loss was reduced from 69.55 to 40.94 mg. Tobar *et al.*³⁹³ optimised the process parameters for laser transformation hardening of a tool steel and simulated it experimentally. The method comprised an analytical three-dimensional transient model and a numerical finite element based approach, both concerned with solution of a transient heat conduction equation.

Molian³⁹⁴ studied the effect of laser surface heat treatment on the microstructure of AISI 4340 steel and compared it with a conventional furnace heat treated one. The microstructure exhibited a mixture of dislocated and twinned martensites with an enhancement in volume fraction of retained austenite as compared with the conventional furnace heat treated (870°C treatment) specimens, though, prior austenite grain size and martensitic substructure remained essentially the same. Roy and Manna³⁹⁵ have demonstrated that laser surface hardening, instead of laser surface alloying or melting, is more effective in enhancing hardness and wear resistance of unalloyed austempered ductile iron. The improvement in wear resistance following laser surface hardening is attributed to the martensitic surface with residual compressive stress. Lakhkar *et al.*³⁹⁶ developed a numerical model to predict the back tempering in multitrack laser hardening of AISI 4140 steel. The model was capable of predicting the volume fraction of martensite that has been tempered and the corresponding hardness values. Temperature field in the laser hardening process was numerically simulated by Wang *et al.*³⁹⁷ using a dedicate software to study the influence of energy density on laser hardening process. Simulation result was verified by a thermocouple temperature transducer measuring the specimen surface temperature following laser irradiation. Experimental curves of temperature versus time are in agreement with simulation

results. The simulation results are useful in designing laser surface engineering experiments. Several other examples of laser surface hardening and melting of steel are listed in Table 3.^{383–475}

Laser surface melting

Laser surface melting is another important and effective technique primarily to modify the microstructure of the near surface region. Lo *et al.*^{398,399} showed that laser surface melting of 440C martensitic stainless steel was more effective in improving the corrosion resistance (in NaCl solution) by carbide refinement, while laser transformation hardening was useful in enhancing the cavitation erosion resistance by formation of an appropriate amount of martensite and austenite. Pan *et al.*⁴¹⁵ eliminated the carbides formed during sensitising treatment and homogenised the microstructure leading to a significant improvement in corrosion resistance by laser surface melting of austenitic stainless steel. Similar attempts to eliminate the susceptibility to intergranular corrosion of the cold worked and sensitised AISI 316 stainless steel and nitrogen bearing AISI 316L stainless steel were reported by Parvathavarthini *et al.*⁴⁰² and Mudali *et al.*⁴¹⁹ respectively. The improvement by laser surface melting is attributed to the complete dissolution of M₂₃C₆ type of carbides and suppression of reprecipitation due to rapid quenching. Laser surface melting of DIN X42Cr13 samples showed a higher abrasive wear resistance than the same after tempering treatments due to strain induced transformation of austenite into martensite and the work hardening of austenite during the wear tests.⁴¹¹ Wilde *et al.*⁴²⁷ showed that laser surface melting of AISI 4135 low alloy steel resulted in reduction in the hydrogen absorption kinetics as measured by the permeation test, and an improved resistance to hydrogen induced fracture in the slow strain rate tensile test under galvanostatic charging conditions.

Laser glazing is a process of rapid melting and solidification of materials using a high power laser beam as a source of heat.⁴⁴⁴ Laser glazing was successfully applied to improve the surface properties of the VPS NARloy-Z coatings developed on steel.⁴⁴⁵

Laser shock processing

Laser shock processing involves rapid irradiation of the component surface with a laser (at a power density of $\sim 10^{12}$ W m⁻²) resulting in generation of shock wave (due to a volume expansion of the plasma plume formed on the surface) and subsequent alteration of microstructure/state of stress. This processing has the capability to improve material hardness and fatigue strength in a number of alloys.⁴¹³ The effects of laser shock processing (with a Nd:glass phosphate laser with a 600 ps pulse width and an up to 120 J pulse energy at power densities $>10^{16}$ W m⁻²) on the microstructure, microhardness and residual stress of low carbon steel were studied by Chu *et al.*⁴¹³ Surface hardness increased up to 80% after the laser shock processing up to 100 μ m in depth and introduced residual compressive stress on the surface. The strengthening effect was attributed to the presence of a high dislocation density. Peyre *et al.*⁴⁰⁸ showed that laser shock processing offered a better pitting corrosion behaviour of AISI 316L stainless steel than that obtained by shot peening in saline environment.

Dumitru *et al.*⁴⁰⁶ used a Q switched Nd:YAG laser to develop microholed microstructure on AISI 440C

Table 3 Summary of selected studies on laser surface hardening and melting of steel

System	Results	Reference	Year
AISI 304 stainless steel	The fatigue behaviour and stability of the residual stress state and new surface microstructure of laser shock peened and deep rolled austenitic stainless steel AISI 304 was investigated	Nikitin and Altenberger ⁴¹²	2007
General	Review on the application of high power diode laser in surface hardening	Kennedy <i>et al.</i> ³⁸⁰	2004
AISI H13 steel plate	The method comprises an analytical three-dimensional transient model and a numerical finite element method one, based both on the solution of a transient heat conduction equation	Tobar <i>et al.</i> ³⁹³	2006
Carburised and borocarbured 15CrNi6 low carbon steel	Laser heat treatment improves the hardness and wear resistance of carburised and borocarbured steel	Kulka and Pertek ³⁸³	2007
High nitrogen martensitic steel	Laser heat treatment improved the tribological and fatigue properties	Heitkemper <i>et al.</i> ³⁸⁵	2003
2Cr13 steel	Impact property was improved by CW CO ₂ laser treatment attributed to grain refinement and higher martensite content near the surface	Shao <i>et al.</i> ³⁸⁶	2003
AISI 440C martensitic steel	Cavitation erosion resistance was significantly increased by laser transformation hardening	Lo <i>et al.</i> ^{398,399}	2003
Nodular cast iron	Nodular cast iron was modified by laser microprecision treatment. The substantial increase in the wear resistance was found	Xia <i>et al.</i> ⁴⁰⁰	2002
Low C steel (0.05%C steel)	Pulsed Nd:YAG laser treatment caused scattering of crystallographic texture and introduction of significant tensile residual macrostresses transverse to the rolling direction	Iordanova <i>et al.</i> ³⁸⁷	2002
Steel	The effect of repetitive pulse heating on the residual stress developed and its distribution was calculated using a mathematical model	Shuja <i>et al.</i> ³⁸⁹	2002
Stainless steel	A combination of analytical and numerical model to modify and predict the process parameters for laser surface hardening of stainless steel	Yanez <i>et al.</i> ⁴⁰¹	2002
AISI 316 stainless steel	Laser surface melting is effective in desensitisation	Parvathavarthini <i>et al.</i> ⁴⁰²	2001
AISI 304 stainless steel	Tendency for pitting decreases due to dissolution of nitrogen and formation of nitrides	Conde <i>et al.</i> ⁴⁰³	2001
Austempered ductile iron	Laser surface hardening is superior to laser surface melting in terms of microhardness, residual stress and adhesive wear resistance	Roy and Manna ³⁹⁵	2001
Austenitic 316 stainless steel and high speed tool steel	Morphology of surface microstructure and topographical features were studied as a function of laser parameters	Trtica <i>et al.</i> ⁴⁰⁴	2001
17-4 PH steel	Stress corrosion cracking and fatigue crack growth behaviour were determined in 17-4 PH stainless steel by laser surface annealing treatments	Tsay <i>et al.</i> ³⁹⁰	2001
Borided steel	Laser surface treatment of borided steels results in the deterioration of the case microstructure and properties	Gopalakrishnan <i>et al.</i> ⁴⁰⁵	2001
AISI 440C steel	Microstructural modification on the steel was done by a Q switched Nd:YAG laser. Tribological test showed a significant improvement in wear behaviour	Dumitru <i>et al.</i> ⁴⁰⁶	2000
Fe–Si–C system (grey cast iron)	Simultaneous laser irradiation and ultrasound had a great influence on dissolving carbon and disintegrating oversaturated γ -Fe(C) solution	Amulevicius <i>et al.</i> ⁴⁰⁷	2000
AISI 316L stainless steel	Laser peening (LP) improved the pitting corrosion resistance of AISI 316L stainless steel	Peyre <i>et al.</i> ⁴⁰⁸	2000
SNM 439 steel	Optimum process parameters for laser surface hardening were derived by Taguchi methodology and fuzzy evaluation approach	Chen and Shen ⁴⁰⁹	1999
Mild steel (EN8)	Interaction of CO ₂ , Nd:YAG, excimer and high power diode lasers with surface of mild steel (EN8) changed the wettability characteristics	Lawrence and Li ⁴¹⁰	1999
DIN X42Cr13 tool steel (Fe–0.4 wt-%C–13 wt-%Cr)	Abrasive wear resistance of laser surface melted sample decreased with increasing volume fraction of retained austenite	Colação <i>et al.</i> ⁴¹¹	1999
Low C steel	Laser shock induced plastic deformation resulted in extensive formation of dislocations with increased surface hardness	Chu <i>et al.</i> ⁴¹³	1999
Carbon steel	Conversion electron Mossbauer spectroscopy study was conducted to observe the effect of CO ₂ laser surface irradiation on carbon steel	Agudelo <i>et al.</i> ⁴¹⁴	1999
321 austenitic stainless steel	Laser surface melting eliminated the carbide formed by sensitisation and homogenised the microstructure leading to improvement of localised corrosion resistance	Pan <i>et al.</i> ⁴¹⁵	1998

Table 3 Continued

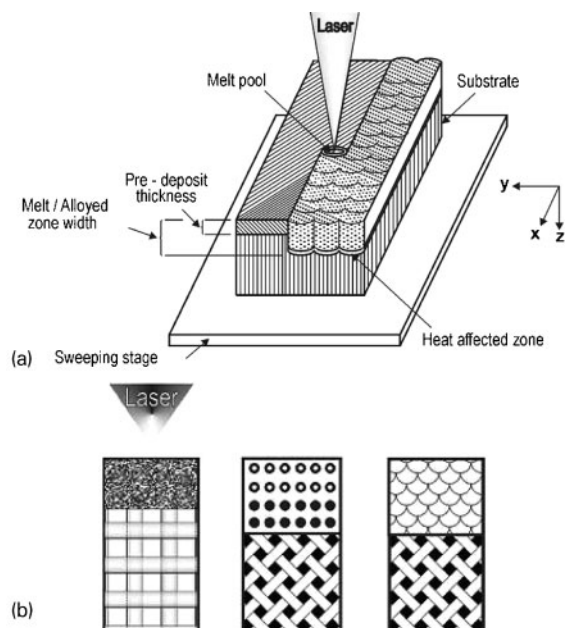
System	Results	Reference	Year
AISI 4150 steel	Resistance to fatigue crack propagation rate is significantly improved in laser surface heated specimen as compared with that of as received AISI 4150 steel (0.47C, 0.16Ni, 1.04Cr, 0.21Mo, 0.25Si, 0.80Mn, 0.015P and 0.02S)	Tsay and Lin ⁴¹⁶	1998
AISI 420 stainless steel	Amount of retained austenite in AISI 420 stainless steel is reduced by reducing laser scanning speed and increasing the laser power density	Colação and Vilar ⁴¹⁷	1998
DIN X42Cr13 martensitic tool steel	Laser melted steel was secondary hardened at 600°C which is 100°C higher than conventionally heat treated steel (composition: C: 0.47, Cr: 12.82, Mo: 26, Ni: 0.19, Si: 0.40, Mn: 0.78, P: 0.024 and S: 0.011)	Colação and Vilar ⁴¹⁸	1998
Nitrogen bearing 316L stainless steel	Laser surface melting is helpful in desensitisation of stainless steel by complete dissolution of M ₂₃ C ₆ type of carbides and suppression of reprecipitation due to rapid quenching	Mudali <i>et al.</i> ⁴¹⁹	1998
Pearlitic rail steel	Laser surface melting produces a thin glazed layer with better hardness and lower friction rate	Di-Melfi <i>et al.</i> ⁴²⁰	1998
EN31 bearing steel	Single or double glazed laser surface hardening tracks have better wear and friction property	Zhang <i>et al.</i> ⁴²¹	1997
Pure iron and low alloy steel	The effect of prior austenite grain size on the hardness of the hardened layer is established	Das ⁴²²	1997
1Cr17Ni2 stainless steel	A detailed microstructural investigation of laser melted steel was conducted	Song and Shen ⁴²³	1997
304L stainless steel	Laser surface melting of 304L stainless steel (using a CO ₂ laser)	Akgun <i>et al.</i> ⁴²⁴	1995
Ductile cast iron	Laser surface melting led to sevenfold enhancement in cavitation erosion resistance of ductile iron	Gadag and Srinivasan ⁴²⁵	1995
Ductile iron	The novel Fe based material containing rapidly solidified eutectic cementite and the ultrafine graphite particles was fabricated by laser melting of ductile iron	Wang and Bergmann ⁴²⁶	1995
AISI 4135 low alloy steel	The effect of laser heat treatment of steel on stress corrosion and hydrogen embrittlement was studied	Wilde <i>et al.</i> ⁴²⁷	1995
Ductile iron	Laser surface melting led to a significant improvement in wear resistance attributed to the ultrafine microstructure and high hardness	Gadag and Srinivasan ⁴²⁸	1994
2Cr13 stainless steel	Effect of laser surface treatment on the fretting wear resistance of 2Cr13 stainless steel was studied	Yang <i>et al.</i> ⁴²⁹	1994
Fe–26Mn–7Al–0.9C	A microstructural study and aging behaviour of laser surface melted austenitic steel was undertaken	Tjong and Tsang ⁴³⁰	1993
Fe–Mn–Al austenitic steel	Laser surface treatment results in significant improvement in fatigue crack growth resistance of a Fe–Mn–Al austenitic steel	Saxena <i>et al.</i> ⁴³¹	1993
0.4%C steel	Laser surface melting of 0.4%C led to a marginal improvement in hardness without any change in erosion behaviour	Rao <i>et al.</i> ⁴³²	1993
304L stainless steel	Laser surface melting decreased the tensile strength and hardness, which was attributed to an overall decrease in dislocation density	Akgun <i>et al.</i> ⁴³³	1992
AISI 420 steel	Laser surface melting was proved to be beneficial to improve the corrosion resistance	Escudero <i>et al.</i> ⁴³⁴	1992
EN40-B substrate	The metallurgical and mechanical properties of laser nitrided surface was reported	Karamis and Yilbas ⁴³⁵	1991
16 wt-%Cr white cast iron	Laser surface melting and hardening reduced the erosion rate in distilled water considerably	Tomilson and Talks ⁴³⁶	1990
Pearlitic grey and ductile cast iron	The erosion rate of laser treated grey and ductile iron were 5 and 25 times enhanced	Molian and Baldwin ⁴³⁷	1987
Pure iron and plain C steel with up to 0.2%C	Laser surface melting and resolidification developed a hard homogeneous martensitic structure	Hegge and de Hosson ⁴³⁸	1987
AISI 4340 steel	The effect of laser heat treatment on the microstructures of the hardened layer was studied and compared with the same by conventional processing	Molian ³⁹⁴	1981

stainless steel for improving the friction and wear properties. The hole diameters of about 5–10 μm and structures with spatial frequencies $>60\text{ mm}^{-1}$ were achieved. The depth of the laser induced microholes was up to 5–8 μm (one pulse per hole) and could be increased if more pulses were applied for each micro-hole. Tribological tests, performed on samples structured with arrays of microholes having a diameter of

10 μm and different spatial frequencies (from 30 to 60 mm^{-1}) showed significant increase in the lifetime of the laser processed samples, in comparison with that in unstructured ones.

Laser surface alloying

Among the various laser surface engineering methods, laser surface alloying involves melting of predeposited or



14 **a** schematic of experimental set-up for laser surface alloying and **b** stages of irradiation, melting, intermixing and solidification involved in laser surface alloying

concomitantly added materials (powders, wire, film and layer) along with a part of the underlying substrate to form an alloyed zone for improvements in wear, corrosion and oxidation resistance.^{440–443} Figure 14a illustrates the scheme of laser surface alloying with a CW laser. It includes three major parts: a laser source with a beam focusing and delivery system, a lasing chamber with controlled atmosphere and a microprocessor controlled sweeping stage where the specimen is mounted for lasing. The process includes melting, intermixing and rapid solidification of a thin surface layer with a thin coating layer comprising the desired alloying elements (Fig. 14b). The coating layer may be predeposited by any of the conventional means such as electrodeposition, plasma spray and physical/chemical vapour deposition. As an alternative, alloying powder or powder mixture may be injected into the melt pool during laser irradiation through a process called codeposition. In either strategy of pre- or codeposition, a 20–30% overlap of the successive molten/alloyed track is intended to ensure microstructural/compositional homogeneity of the laser treated surface in laser alloyed zone that has a composition distinctly different than that of the underlying base substrate. The sweeping stage (x – y or x – y – z – θ) allows laser irradiation of the intended area of the sample surface at an appropriate rate and interaction time/frequency. The irradiation results in transient melting of the deposit with a part of the underlying substrate, rapid mass transfer by diffusion/convection in the melt pool and ultrafast solidification to form the alloyed zone. The depth, chemistry, microstructure and associated properties of the alloyed zone depend on the suitable choice of laser/process parameters, i.e. incident power/energy, beam diameter/profile, interaction time/pulse width, pre- or codeposition thickness/composition and concerned physical properties such as reflectivity, absorption coefficient, thermal conductivity, melting point and density. Laser surface alloying has been successfully applied

to develop alloyed and clad zone on the surface of ferrous based materials.^{476–527}

Laser surface alloying of AISI 1020 carbon steel was carried out using a 4.5 kW CW CO₂ laser by predeposition of alloying mixture of Fe–Cr–Ni Si₃N₄ and Fe–Cr–Mn–Si₃N₄ powders and subsequently melting it by laser irradiation.⁴⁶⁵ Laser treated Fe–Cr–Ni and Fe–Cr–Ni–Si–N layers exhibited a higher resistance to pitting corrosion in chloride ion containing solution as compared with that of AISI 316L and P900N stainless steels. However, the pitting corrosion resistance of both Fe–Cr–Mn and Fe–Cr–Mn–Si–N were inferior to those of the AISI 316L and P900N stainless steels. Nickel coated cast iron was laser treated using pulsed excimer radiation.⁴⁶⁰ The amount of Ni after the processing was varied by changing the laser power density, number of pulses per step and degree of overlap between consecutive steps. The surface roughness was a non-linear function of the laser pulse number. Surface layers presented higher microhardness as compared with that of untreated material. The corrosion resistance of the laser treated material was found to be a function of the number of laser pulses.

Laser surface alloying of A7 tool steel with Ti using an excimer laser was found to reduce dry, sliding friction coefficient due to the formation of low friction transfer film.⁴⁶² Two Ni–P coatings of different thicknesses were deposited on mild steel and laser melted at different scanning rates.⁴⁶⁴ The corrosion resistance of laser surface alloyed Ni–P coatings was determined by the degree of compositional changes or dilution by mixing with the steel substrate and by the degree of crystallisation that took place in the dendritic solidification process after laser melting. The presence of Fe in Ni–P coatings considerably reduced its corrosion resistance. Khanna *et al.*⁴⁷¹ coated stainless steel powder on mild steel substrate using a plasma coating technique and melted it with a CW CO₂ laser. The porous coating thus obtained was modified using laser surface alloying. The oxidation behaviour of the plasma coated as well as the laser treated materials was studied at 1073 K in air; while plasma coating decreased the oxidation rate considerably, the improvement in laser treated condition was significantly higher than that of the base stainless steel. The adherence of the scale, tested through acoustic emission technique and thermal cycling, was remarkably strong.

Dutta Majumdar and Manna⁴⁵⁸ investigated the effect of laser surface melting and alloying of plasma spray deposited Mo on AISI 304 stainless steel. The optimum conditions for the formation of a homogeneous microstructure and composition for improvement in pitting corrosion resistance and mechanical property were established.⁴⁵⁸ Potentiodynamic anodic polarisation tests of the substrate and laser surface alloyed samples in 3.56 wt-%NaCl solution (both in forward and reverse potential) showed that the critical potentials for pit formation E_{PP1} and growth E_{PP2} were significantly (2–3 times) improved from 75 mV(SCE) in the substrate to 550 mV(SCE) after laser surface alloying.⁴⁵⁸ E_{PP2} has also been found to be nobler in as lased specimens than that in the stainless steel substrate.

The poor pitting corrosion resistance of the plasma sprayed stainless steel samples (without laser remelting) was due to the presence of surface defects present in

plasma deposited layer. Standard immersion test was conducted in a 3.56 wt-%NaCl solution to compare the effect of laser surface alloying on the pitting corrosion resistance. A detailed study of the kinetics of erosion corrosion (in terms of weight loss per unit area followed by continuous circulation of the samples at 750 rev min⁻¹ for 10–75 h in a medium containing 20 wt-% sand in 3.56 wt-%NaCl solution) in the stainless steel substrate before and after laser surface alloying (with 1210 MW m⁻² power density and 31.7 mm s⁻¹ scan speed for deposit thickness of 250 µm) showed a significant decrease in the kinetics of erosive corrosion loss due to laser surface alloying.⁴⁵⁸ It was thus concluded that laser surface alloying was capable of imparting an excellent superficial microhardness and resistance to corrosion and erosion–corrosion properties to austenitic stainless steel due to Mo both in solid solution and as precipitates.

Oxidation is another serious mode of surface degradation that gets aggravated under unabated counter ionic transport of cations and anions at elevated temperature. Unlike electrochemical corrosion, oxidation occurs in dry condition (without aqueous medium) and solid state ionic transport through the oxide scale. In the past, several attempts have been made to enhance the resistance to oxidation by laser surface alloying, cladding and similar laser surface engineering techniques.^{475,483,489,461} Pillai *et al.*⁴⁴⁷ have obtained enhanced oxidation resistance (at 873 K for up to 200 h) of plain carbon steel following laser surface alloying with Al due to several intermetallic phases such as Al₁₃Fe₄ and Al₂Fe₂. Thermal barrier coatings could also be effective for improving oxidation resistance of stainless steel.⁴⁸⁵ However, such coatings are brittle and hence are applied with a favourable composition gradient such as applying a bond coat (Ni–22Cr–14Al–1Y) between the topcoat (ZrO₂ + 7.5 wt-%Y₂O₃) and stainless steel substrate before single or multipass laser melting treatments. Oxidation tests at 1200°C indicated that double pass laser surface melting was more effective to impart a better oxidation resistance.⁴⁸⁵ Similar attempts of improving oxidation resistance have been made with Incoloy 800H by laser surface alloying with Al^{461,489} and by laser surface cladding of AISI 316L stainless steel with Fe–Cr–Al–Y alloy coatings.⁴⁸⁹ In all these cases, it appears that the degree of enhancement in oxidation resistance by laser surface engineering primarily depends on stability, adherence, imperviousness and strength of the complex oxide scale comprising multiple oxide on top of the base metal.

The effect of laser melting on the oxidation resistance of plasma spray deposited stainless steel on mild steel substrates was studied by Khanna *et al.*⁴⁹¹ The oxidation resistance was improved significantly by plasma spray deposition which was reported to improve further improved by laser melting.⁴⁹¹ The adherence of the scale, tested using an acoustic emission technique and thermal cycling, showed significant improvement. The comparative tribological behaviour of plasma sprayed and laser alloyed coatings was also studied. Wear resistance of laser melted surface followed by plasma spraying was improved as compared with plasma sprayed and as received substrate.⁴⁹²

Manna *et al.*⁴⁵⁰ attempted enhancing the high temperature oxidation resistance (>873 K) of 2.25Cr–1Mo ferritic steel by laser surface alloying with codeposited

Cr using a 6 kW CW CO₂ laser. The main process variables chosen for optimising the laser surface alloying routine were laser power (from 2 to 4 kW), scan speed of the sample stage (150–400 mm min⁻¹) and powder feedrate (16–20 mg s⁻¹). Isothermal oxidation studies in air at 973 and 1073 K for up to 150 h revealed that laser surface alloying had significantly enhanced the oxidation resistance attributed to the presence of an adherent and continuous Cr₂O₃ layer.⁴⁵⁰ Metal matrix composites generally possess an enhanced wear resistance than that of the base substrate or the matrix. Development of composite layer on the surface by conventional means is extremely difficult. Laser melting of substrate and subsequent feeding of ceramic particles into the molten matrix is an effective means of developing composite layer on the surface through a process called laser composite surfacing.

Several attempts have been made to develop a composite layer on metallic matrix by this technique. Cheng *et al.*⁴⁴⁶ have added a mixture of WC–Cr₃C₂–SiC–TiC–CrB₂ and Cr₂O₃ to produce a metal matrix composite surface on stainless steel UNS S31603. Following laser surface melting, cavitation erosion resistance improved in all cases except for Cr₂O₃. Wu and Hong⁴⁵⁴ have attributed the improvement in hardness and corrosion/wear resistance of stainless steel to the presence of amorphous phase in the Zr rich surface (with 7.8–14.5 at-%Zr) following laser surface alloying of stainless steel with Zr nanoparticles. Agarwal and Dahotre⁴⁵⁵ have reported a substantial improvement in resistance to adhesive/abrasive wear of steel following laser surface alloying with TiB₂. The surface composite layer containing ~69 vol.-%TiB₂ particles recorded an elastic modulus of 477.3 GPa. Similar improvement in wear resistance of mild steel was reported by Tondur *et al.*⁴⁵⁷ due to the formation of an FeCr+TiC composite coating formed by laser assisted self-propagating high temperature synthesis. Earlier, Zhukov *et al.*⁴⁹⁰ achieved a similar carbide dispersed composite coating on steel by laser assisted self-propagating high temperature synthesis.

Pelletier *et al.*⁴⁸³ applied an Fe–Mn–C Hadfield steel coating by laser surface cladding on steel and achieved significant improvement in hardness (>800 VHN), elastic modulus (210 GPa) and yield strength (1200 MPa) mostly due to the deformation induced twinning transformation. Laser surface alloying of A7 tool steel with Ti was reported to reduce the dry sliding friction coefficient due to the formation of a low friction transfer film.⁴⁶² Similarly, stellite coating by laser surface cladding is a standard method of reducing wear loss of ferrous substrate.⁴⁸⁸ Besides providing with a harder and tougher surface, stellite coating develops a tougher oxide scale and reduces wear under severe condition. Improvement in hardness and wear resistance has been achieved by laser surface cladding of martensitic steel with Ni alloy⁴⁸⁴ and laser surface alloying of ferritic steel with Cr₃C₂ + SiC.⁴⁶⁷

Recently, an attempt has been made to develop a hard *in situ* boride dispersed composite layer on the surface of AISI 304 stainless steel substrate to improve its wear resistance property.⁴⁷⁸ Laser processing was carried out by melting the surface of sand blasted AISI 304 stainless steel substrate using a CW CO₂ laser and simultaneous deposition of a mixture of K₂TiF₆ (potassium titanium hexafluoride) and KBF₆ (potassium hexafluoroborate)

(in the weight ratio of 2:1) using Ar as shrouding environment. Powder feedrate was maintained constant at 4 g min^{-1} . Irradiation results in dissociation of a predeposited mixture along with a part of the stainless steel substrate, intermixing and rapid solidification to form the composite layer on the surface. The microstructure of composite layer consisted of dispersion of titanium boride particles in AISI 304 stainless steel matrix with an uniform volume fraction of particles and an improved microhardness (250–350 VHN as compared with 220 VHN of the AISI 304 stainless steel substrate) and wear resistance.⁵⁷⁸ The mechanism of improvement in microhardness in the composite layer is due to both grain refinement and dispersion strengthening. It may be noted that both laser power and scan speed should be carefully chosen in order to achieve an improved microhardness, microstructural homogeneity and uniformity of dispersion in the composite layer. Several other examples of laser surface alloying and cladding of steels are enlisted in Table 4.

Laser surface engineering of non-ferrous alloys

Surface dependent degradation by wear, oxidation and corrosion is as much a problem in non-ferrous metals and alloys as that in ferrous alloys. Thus, laser surface engineering is widely applied to Al, Ti, Cu, Mg and other important non-ferrous metals and alloys to extend the service life of components. In this section, the scope and current status of understanding regarding the application of laser assisted surface engineering to enhance surface dependent properties of non-ferrous alloys has been briefly reviewed. For a ready reference, Table 5 summarises the major work done in the recent past in this area.^{445–585}

Laser surface engineering of Al alloys

Several attempts have been made in the past to improve corrosion resistance of Al based alloys by laser surface melting, alloying, cladding or remelting. Table 5a summarises the notable outcome on laser assisted surface engineering of Al based alloys.^{493–556} Grain refinement and microstructural homogeneity are responsible for better corrosion resistance in laser surface melted alloy systems with dispersion of intermetallic and/or amorphous phases.

Yue *et al.*⁵¹² surface melted 7075-T651 Al alloy using an excimer laser with pulse energy of 800 J mm^{-2} . Within the resolidified laser melted layer, the grain boundaries were found to be free from precipitates. The results of the stress corrosion cracking tests showed that the untreated specimens after the 30 days immersion test was severely attacked with intergranular cracks that penetrated deep inside the specimen. In contrast, no intergranular cracks were found in the laser treated specimen. The electrochemical impedance measurements conducted during the stress corrosion cracking test showed that the film resistance of the laser treated specimen was always higher than that of the untreated specimen.

Chan *et al.*⁴⁹³ modified the surface microstructure of 7075-T651 aluminium alloys using an excimer laser (KrF) with a wavelength of 248 nm. A detailed transmission electron microscopy study revealed that laser surface melting caused reductions both in number and size of constituent particles and refinement of the grain size. As a result, the corrosion resistance of the

aluminium alloy was improved. Under dry fatigue condition, the total fatigue life of the laser treated specimens, in which the crack initiation period is of considerable significance, was lower than that of the untreated specimens. However, after shot peening, the fatigue life of the laser treated specimens was recovered. The fatigue resistance of the shot peened laser treated specimens, tested in 3.5 wt-%NaCl solution with 48 h prior immersion, was greater than the untreated specimens with an increase of two orders of magnitude in fatigue life. This was primarily due to the elimination of surface defects and the reduction in corrosion pits. Li *et al.*⁵¹³ surface melted 2024-T351 aluminium alloy using a CW CO₂ laser. The laser treatment changed both the anodic polarisation behaviour and the form of localised corrosion in deaerated 3%NaCl solution. Immersion tests in the same solution under the condition of natural aeration showed that both intergranular and pitting corrosion occurred with pits distributed mainly along the rolling direction in the as received alloy while, only pitting corrosion was present with pits distributed uniformly for the laser surface melted material. This difference in corrosion behaviour as a result of laser surface melting is attributed to changes in the uniformity of distribution and composition of the second phase particles present in the alloy.

Badekas *et al.*⁴⁹⁶ irradiated an Al-4.5%Cu alloy with an excimer laser. The residual stress of the treated surface changed from tensile to compressive with an improved microhardness. Gremaud *et al.*⁵¹⁰ laser surface melted Al-Fe alloy (with Fe content varying from 0.25 to 8 wt-%) using a 1.5 kW CW CO₂ laser and observed significant influence of process parameters on the morphology and degree of fineness of the precipitates/grains.

Hari Prasad and Balasubramaniam⁴⁹⁵ laser irradiated an Al-Li-Cu alloy (of nominal composition 2.5%Li, 1%Cu and 0.12%Zr) with an Nd:YAG laser for the purpose of surface modification and studied its oxidation behaviour at 450°C in air by thermogravimetry and compared with the untreated specimen. Parabolic oxidation kinetics was studied for both the laser treated and untreated specimens. The laser treated specimen exhibited improved oxidation resistance. X-ray diffraction (XRD) analysis indicated that the major component of the scale was Li₂CO₃. The improved oxidation behaviour of the laser treated specimen was related to the compact oxide structure produced on its surface during oxidation. The nature of scale formation on the surface was also addressed. Wang *et al.*⁶² melted a series of Al-Si alloys with various Si contents using a 2 kW CW CO₂ laser. A desirable kind of microstructure in laser melted zone of Al-Si alloys was obtained within the power density range of $156\text{--}208 \text{ W mm}^{-2}$. The degree of hardening of the laser melted surface increased with silicon content. It was found that laser treatment refined the grains and altered the morphology of eutectic silicon from lath-like to coralline-like with a higher Si content in Al matrix as compared with equilibrium Si content in Al. X-ray diffraction analysis revealed that the lattice parameter of Al reduced after laser melting. Wong *et al.*⁵¹¹ reported that the corrosion behaviour of laser surface melted Al-Si alloy (using a CW CO₂ laser) was significantly improved in 10%H₂SO₄ and 10%HNO₃ solutions, although similar level of improvement was not observed in 10%HCl+5%NaCl solution.

Table 4 Summary of selected studies on laser surface engineering of ferrous metals and alloys

a. Laser surface alloying			
System	Results	Reference	Year
31603 stainless steel (CrB ₂ , Cr ₃ C ₂ , SiC, TiC, WC and Cr ₂ O ₃)	Erosion resistance improves considerably for all carbides and borides except Cr ₂ O ₃	Cheng <i>et al.</i> ⁴⁴⁶	2001
Steel (Al)	Laser surface alloying with Al forms several aluminides that impart good oxidation resistance at 600°C up to 200 h	Pillai <i>et al.</i> ⁴⁴⁷	2001
Electroless nickel coated mild steel	Laser melting exhibited a better corrosion resistance in deaerated Na ₂ SO ₄ solution	Renaud <i>et al.</i> ⁴⁴⁸	1990
AISI 1040 and AISI 316L	Following laser surface alloying NiCoCrB with corrosion resistance is improved in mild steel but deteriorated in stainless steel	Kwok <i>et al.</i> ⁴⁴⁹	2001
Cr–Mo steel (Cr)	Oxidation resistance improved at 800–1000°C due to Cr ₂ O ₃ rich scale	Manna <i>et al.</i> ⁴⁵⁰	2000
UNS S31603 stainless steel	Laser surface alloying with Co, Ni, Mn, C, Cr, Mo, Si, SiFe, Si ₃ N ₄ and NiCrSiB was carried out. A significant improvement in corrosion resistance was achieved with Si and Si ₃ N ₄	Kwok <i>et al.</i> ⁴⁵¹	2000
UNS S31603 austenitic stainless steel (Co, Ni, Mn, Cr and Mo)	Resistance to erosion is improved due to dispersed ceramic/intermetallic phases but pitting resistance is decreased	Kwok <i>et al.</i> ⁴⁵²	2000
AISI 304 stainless steel (Si)	Si turns the matrix ferritic. Post-laser surface alloying homogenisation improved corrosion resistance significantly	Isshiki <i>et al.</i> ⁴⁵³	2000
Austenitic stainless steel (nanoZr)	Hardness and wear/erosion resistance improves significantly due to dispersion of Zr rich amorphous phase	Wu and Hong ⁴⁵⁴	2000
AISI 1040 (TiB ₂)	Laser surface alloying improves resistance to adhesive/ abrasive wear and reduces friction	Agarwal and Dahotre ⁴⁵⁵	2000
Plain carbon steel (TiB ₂)	Complex oxide layers develop on steel surface exposed to 600–1000°C following parabolic growth rate	Agarwal <i>et al.</i> ⁴⁵⁶	2000
Steel (FeCr–TiC)	Laser surface remelting homogenises the microstructure and improves wear resistance	Tondu <i>et al.</i> ⁴⁵⁷	2000
AISI 304 stainless steel (Mo)	Significant improvement in microhardness, pitting and erosion corrosion resistance by laser surface alloying of AISI 304 stainless steel with Mo	Dutta Majumdar and Manna ⁴⁵⁸	1999
Fe (Ni)	A theoretical model was proposed to calculate the mixture ratio between the alloy and substrate D_j and the transport coefficient m_i of the i_{th} element from the added material and the substrate	Xie ⁴⁵⁹	1999
Cast iron (Ni)	Pulsed excimer laser irradiation of nickel coated cast iron improved the microhardness and corrosion resistance	Panagopoulos <i>et al.</i> ⁴⁶⁰	1998
Inconel 800H (Al)	Laser surface alloying with Al develops an Al rich surface that considerably improves resistance to oxidation up to 1000°C	Gutierrez and Damborenea ⁴⁶¹	1997
A7 tool steel	Excimer laser surface treatment and surface alloying with Ti had no discernible effect on mechanical properties but the friction coefficient was significantly reduced	Jervis <i>et al.</i> ⁴⁶²	1997
304 stainless steel and 2-25Cr–1Mo steel	Laser surface melting improved the tribological properties of plasma sprayed coated 304 stainless steel and 2-25Cr–1Mo steels with WC, Co, Mo and Cr	Khedkar <i>et al.</i> ⁴⁶³	1997
Mild steel (Ni–P)	Laser surface alloying of Ni–P on mild steel deteriorated the corrosion resistance because of the presence of Fe in the alloyed layer	Garcia-Alson <i>et al.</i> ⁴⁶⁴	1996
316L and P900N stainless steel	Laser surface melting of Fe–Cr–Ni Si ₃ N ₄ and Fe–Cr–Mn–Si ₃ N ₄ powders introduced nitrogen and exhibited higher resistance to pitting corrosion	Huang <i>et al.</i> ⁴⁶⁵	1996
Mild steel (0-21C–1-46Cr–3-52Ni–0-80W)	Laser surface cobaltising of mild steel achieved a Co content of 9-96 wt-%. The performance of the surface at elevated temperature was improved	Ding <i>et al.</i> ⁴⁶⁶	1996
35 NCD 16 ferritic steel (Cr ₃ C ₂ /SiC)	Hardness and oxidation resistance were increased due to Si containing oxides	Gemelli <i>et al.</i> ⁴⁶⁷	1996
AISI 304 stainless steel (Mo and Ta)	Laser surface melting produces δ -ferrite and deteriorates corrosion. Laser surface alloying with Mo is useful	Akgun and Inal ⁴⁶⁸	1995
Mild steel (Fe, Cr, Si and N)	Laser surface alloying with Fe–Cr–Si–N produces a fine duplex microstructure and greatly increases corrosion resistance	Chong <i>et al.</i> ⁴⁶⁹	1995
AISI 316L (Si ₃ N ₄)	Laser surface alloying of AISI 316L stainless steel with silicon nitride improves the hardness to 300–1200 VHN and varied with Si content	Tsai <i>et al.</i> ⁴⁷⁰	1994

Table 4 Continued

a. Laser surface alloying			
System	Results	Reference	Year
Mild steel (stainless steel)	Stainless steel powder was coated on mild steel substrates using a plasma coating technique and laser melting. The oxidation rate in air (at 1073 K) decreased after plasma coating and was further improved by laser melting	Khanna <i>et al.</i> ⁴⁷¹	1992
Roping wires of grade BS 2763 and 1770	Laser surface alloying proved durable and friction/wear resistant as compared with thin coatings by physical vapour deposition. The rope compositions were about 0.65–0.75%Cr, 0.65–0.85%Mn and 0.15–0.35%Si. The coating material was Cr and Zr	Batchelor <i>et al.</i> ⁴⁷²	1992
Low C steel	Laser surface alloying of low carbon steel with chromium resulted in an increase in surface hardness and resistance to abrasive wear compared with the bulk material	Tan and Doong ⁴⁷³	1991
b. Laser surface cladding			
System	Results	Reference	Year
Plain carbon steel	Laser surface cladding with 25%Cr+10%Ni formed a refined microstructure with a minimum dilution at the interface	Barnes <i>et al.</i> ⁴⁷⁴	2003
Mild steel	A process for producing a dense, hard, iron–aluminide coating on a mild steel substrate has been successfully developed using pulsed laser assisted powder deposition	Corbin <i>et al.</i> ⁴⁷⁵	2003
Carbon steel	In the present study, Ni–Cr–B–Si cladding on carbon steel has been achieved by high power diode laser. The microstructure is composed of a matrix of α -Ni with the presence of nickel (Ni ₃ Fe and Ni ₃ Si), boron (B ₄ C and CrB) and chromium (Cr ₃ Si) intermetallics	Conde <i>et al.</i> ⁴⁷⁶	2002
AISI 304 stainless steel	A comparison between plasma transferred arc welded and laser hardfaced Co based superalloy on AISI 304 stainless steel showed that, for normal temperature application, laser hardfacing shows the best results; however, for high temperature application, plasma transferred arc deposits exhibited superior microstructure	d'Oliveira <i>et al.</i> ⁴⁷⁷	2002

Liu *et al.*⁵¹⁶ laser surface melted Al–Si–graphite particulate composite. The wear resistance was improved because of microstructural refinement and improved microhardness. Watkins *et al.*⁵⁰³ melted the surface of a series of Al transition metal alloys (Al–Cu, Al–Si, Al–Zn and Al–Fe) and obtained unique microstructural and compositional characteristics including improved hardness, wear and pitting corrosion resistance using a selected set of laser parameters. SiC dispersed Al alloy composite is the most promising class of metal matrix composite for applications in automobile and aircraft industries. However, these composites undergo severe degradation in chloride containing environment.

Sicard *et al.*⁴⁹⁴ reported enhancement of mechanical and chemical properties of aluminium alloys (AlSi7Mg0.3) by excimer laser assisted nitriding treatment. The special nitriding treatment requires an excimer laser to irradiate the sample placed inside the cell with 1 bar nitrogen pressure allowing expansion of the vapour plasma, dissociation/ionisation of nitrogen by laser generated shock wave and subsequent penetration (adsorption and diffusion) of nitrogen from plasma up to some depth. While adequate laser fluence is required to create the plasma, this fluence must be limited below a threshold to prevent laser induced surface roughness. The resultant polycrystalline nitride layer is several micrometres thick and comprises pure AlN columns (200–500 nm thick) on top of equiaxed AlN grains in the diffusion layer. Heat conduction calculations show that a 308 nm laser is better suited for greater nitride thickness, as it corresponds to a weaker reflectance value for aluminium. Laser induced

plasma formation was utilised to develop AlN coating on Al surface using XeCl laser (wavelength=308 nm) operated at 20 Hz repetition rate.⁴⁹⁷ The vacuum chamber was filled with N₂ gas to a pressure of 760 torr. The interaction between the laser beam with metal surface led to melting of the substrate and reaction with the confined plasma causing the formation of AlN on the surface of Al. Wear resistance was improved by surface nitriding with a significant reduction in friction coefficient. However, it was observed that marginal presence of oxygen and greater degree of overlapping are beneficial for further improvement in wear resistance.

Laser surface alloying of 2014-T6 [Al–4.65Cu–0.5Mg–0.8Si (wt-%)] Al alloy with 25Cr–75Al, 75Cr–25Al, 25W–75Al, 75W–25Al, 8Al–72Zr–20Ni and 35Al–50Ti–15Ni powder mixtures was carried out using an Electro 2 kW CW CO₂ laser.⁵⁰³ The effects of alloying and overlapping on the microstructure and corrosion behaviour were studied. Similar attempt of laser surface alloying of Al with Mo (by injecting Mo powder into melt pool) was undertaken by laser surface alloying with a composition ranging from 14.8 to 19.1 wt-%Mo and resulted in 85–100 VHN in hardness and 84–92 GPa in Young's modulus (determined by nanoindentation).⁵⁰² Tang *et al.*⁵¹⁸ observed that laser surface alloying of manganese–nickel–aluminium bronze with Al was more effective in enhancing corrosion and cavitation erosion resistance in a 3.56 wt-%NaCl solution than that after laser surface melting. By employing polarisation resistance measurements under quiescent and cavitating conditions, the contributions

Table 5 Summary of selected studies on laser surface engineering of non-ferrous materials**a. Aluminium and its alloys**

System	Results	Reference	Year
7075-T651 Al alloy	Laser surface melting with excimer (KrF) laser caused reduction in number and size of constituent particles and refinement of grain structure with an improved corrosion property	Chan <i>et al.</i> ⁴⁹³	2003
AlSi7Mg0.3	The excimer laser nitriding process was developed to enhance the mechanical and chemical properties of aluminium alloys. The polycrystalline nitride layer composed of a pure AlN (200–500 nm thick) on an AlN (grains) in alloy diffusion layer was developed	Sicard <i>et al.</i> ⁴⁹⁴	2001
Al–Li–Cu alloy (2.5Li, 1Cu and 0.12Zr)	The laser treated (using Nd:YAG laser) specimen exhibited improved oxidation resistance attributed to the compact oxide structure produced on its surface during oxidation	Hari Prasad and Balasubramaniam ⁴⁹⁵	1997
Al–4%Cu	Residual stress changes from tensile to compressive due to laser surface treatment (using excimer laser). The microhardness is increased at a depth <10 mm from the surface	Badekas <i>et al.</i> ⁴⁹⁶	1988
Al (AlN)	AlN coating was developed on Al substrate (using excimer laser) by laser induced plasma surface engineering with a significant improvement in wear resistance	Meneau <i>et al.</i> ⁴⁹⁷	1998
6061 Al (TiC)	The four point bend test of Nd:YAG laser treated sample showed propagation of cracks from the surface and delamination of coating	Katipelli <i>et al.</i> ⁴⁹⁸	2000
Al–Si (Al ₂ O ₃ –TiO ₂)	A graded coating consisting of an Ni clad Al bond layer, a 50 wt-%Ni clad Al + 50 wt-%(Al ₂ O ₃ –13 wt-%TiO ₂) intermediate layer and an Al ₂ O ₃ –13 wt-%TiO ₂ overlayer (or ceramic layer), was produced on an Al–Si substrate by plasma spraying and then was remelted by laser irradiation	Wang <i>et al.</i> ⁴⁹⁹	1993
Al ₃ Ti or intermetallic compound matrix composite	Intermetallic compound Al ₃ Ti or intermetallic compound matrix composite (IMC) layers were formed on Al surface by CW CO ₂ laser treatment. Intermetallic compound matrix composite surface layer improved the wear property of Al substrate	Uenishi and Kobayashi ⁵⁰⁰	1999
Al (TiC)	TiC was deposited on 2024 and 6061 Al using Nd:YAG laser. The microstructure was found to vary with laser parameters	Kadolkar and Dahotre ⁵⁰¹	2003
Al (Cr)	Microstructure of laser surface alloyed Al with Cr contains relatively large amounts of intermetallic compounds dispersed in a matrix of α -Al	Almeida <i>et al.</i> ⁵⁰²	1995
Laser surface alloying and laser surface melting of Al Al7175–7351 (Cr)	A review of literature on laser surface melting and alloying of Al is summarised	Watkins <i>et al.</i> ⁵⁰³	1998
	Laser surface alloying of Al with Cr improved crevice corrosion resistance	Ferriera <i>et al.</i> ⁵⁰⁴	1996
Al6013–SiC	Laser surface treatment of Al6013–SiC composite using a CW Nd:YAG laser resulted in selective vapourisation of the aluminium matrix with improved pitting resistance	Yue <i>et al.</i> ⁵⁰⁵	1997
Al (Cr)	Vacuum deposited chromium films were irradiated in air using a Nd:glass laser. For a thin Cr deposition, Cr was evaporated; however, for a thick coating, it was uniformly distributed	Jain <i>et al.</i> ⁵⁰⁶	1981
Al–4.3Cu	Al–4.3%Cu alloy was developed by laser surface alloying of Cr on Al substrate and subsequent remelting. The microstructure was investigated in details	Almeida <i>et al.</i> ⁵⁰⁷	2001
Al–Cu	Mechanical and microstructural properties were found to vary with laser scan speed	van Otterloo and de Hosson ⁵⁰⁸	1994
2014 Al	This corrosion behaviour of the overlapped areas in the laser melting and alloyed (with Cr, W, Zr–Ni or Ti–Ni) alloy was undertaken. Electrochemical testing showed microsegregation within the overlapped areas leading to the initiation of pitting corrosion	Watkins <i>et al.</i> ⁵⁰³	1998
Al (Cu)	The microstructure and hardness variations throughout samples of an aluminum–copper alloy (Al–15 wt-%Cu) were analysed	Pinto <i>et al.</i> ⁵⁰⁹	2003
Al–Fe alloy	Laser melting of Al–Fe alloy containing 0.25–8.0 wt-%Fe was carried out to study the effect of solidification rate on the microstructure	Gremaud <i>et al.</i> ⁵¹⁰	1990
Al–Si	Microstructure of the laser surface melted Al–Si alloy using a CO ₂ laser was refined with extension of solid solubility of Si in Al. The corrosion resistance after laser treatment was improved in 10%H ₂ SO ₄ and 10%HNO ₃ solutions. However, laser treatment did not produce significant improvement in studies with 10%HCl and 5%NaCl solutions	Wong <i>et al.</i> ⁵¹¹	1997

Table 5 Continued

a. Aluminium and its alloys			
System	Results	Reference	Year
SiC reinforced Al AA 2009 (Al-3.8Cu-1.3Mg)	The corrosion resistance was significantly improved by laser surface melting attributed to redistribution of CuAl ₂ and extension in solid solubility of Cu in Al	Yue <i>et al.</i> ⁵¹²	1999
2024-T351 aluminium alloy	The improved corrosion resistance due to laser surface melting (using a CO ₂ laser) is attributed to changes in the distribution and composition of the second phase particles present in the alloy	Li <i>et al.</i> ⁵¹³	1996
7075 Al alloy	Laser shock processing of A356, Al2Si and 7075 aluminium alloys (using excimer laser) improved fatigue performance attributed to compressive residual stress levels	Peyre <i>et al.</i> ⁴⁰⁸	2000
Al (SiC)	Al/SiC metal matrix composit layers were prepared by laser injection of SiC particles into Al substrate. An interface reaction layer consisted of Al ₄ C ₃ plates forming a good mechanical bonding between the SiC particles and Al matrix	Vreeling <i>et al.</i> ⁵¹⁴	2000
Al	Al was irradiated in nitrogen atmosphere in pulsed nanosecond excimer laser and there was formation of AlN phase. The effect number of pulses and beam fluence on nitrogen incorporation was studied	Carpene and Schaaf ⁵¹⁵	2002
Al-Si-C composite	Laser surface melting of Al-Si-C composite was found to refine the grains with a significant improvement in hardness and wear resistance	Liu <i>et al.</i> ⁵¹⁶	1997
Al-6061	ZrO ₂ -7 wt-%Y ₂ O ₃ powder mixture was plasma sprayed on Al followed by laser surface melting with an Nd:YAG laser. The wear resistance improved significantly	Fu <i>et al.</i> ⁵¹⁷	1997
Al (TiB ₂)	Uniform <i>in situ</i> dispersion of TiB ₂ was obtained by laser melting of a mixture of K ₂ TiF ₆ and KBF ₆ (2:1 weight ratio) on Al substrate, which improved hardness and wear resistance	Dutta Majumdar <i>et al.</i> ⁵¹⁹	2006
b. Copper and its alloys			
System	Results	Reference	Year
Cu (Al)	Laser surface alloying of Al on Cu substrate using a ruby laser formed Al ₂ Cu phase. A theoretical model predicted cooling rate	Manna <i>et al.</i> ⁵²²	1994
Zn coated Cu	Zinc coated copper specimens were irradiated with a CW CO ₂ laser created surface alloys between zinc and copper on the surface of copper substrate; the morphology and its distribution were found to vary with laser intensity	Michaelides and Panagopoulos ⁵²³	1993
Al bronzes (-5 at-%Fe, 13-28 at-%Al)	Laser melting of commercial Al bronze was conducted. A detailed microstructural and phase analysis was conducted	Vandenberg and Draper ⁵²⁴	1984
Cu (Cr)	Laser surface alloying enhances resistance to adhesive/abrasive wear and erosion due to solid solution and dispersion hardening	Dutta Majumdar and Manna ⁵²⁵	1999
c. Titanium and its alloys			
System	Results	Reference	Year
Ti-6Al-4V (TiN)	TiN forms by laser surface alloying and improves wear resistance that occurs in three stages	Yilbas <i>et al.</i> ⁵³⁴	2001
Ti (Si, Al and Si+Al)	Laser surface alloying with Si is more effective than Al/Si+Al to improve wear and corrosion resistance	Dutta Majumdar <i>et al.</i> ⁵²⁸⁻⁵³⁰	2000 2002 1999
Ti-6Al-4V (WC and TiC)	Laser surface alloying with TiC improves, but that with WC deteriorates corrosion resistance	Cooper <i>et al.</i> ⁵³⁵	1998
Ti (Ti+TiC)	Laser surface alloying with TiC significantly enhances the hardness of substrate Ti	Fouillard-Paille <i>et al.</i> ⁵³⁶	1997
Ti and Ti-IMI829 alloy (N ₂)	Laser surface alloying enhances hardness and corrosion resistance with a smooth surface finish	Weerasinghe <i>et al.</i> ⁵³⁷	1996
Ti alloy (WC and TiC)	Laser surface alloying significantly enhances hardness	Masse and Mathieu ⁵³⁸	1996

Table 5 Continued

d. Magnesium and its alloys

System	Results	Reference	Year
Stainless steel on Mg-17 vol.-%SiC composite	Laser cladding of stainless steel on ZK60/SiC composite (using a CW Nd:YAG laser) was achieved by thermal spray and subsequent laser remelting. The corrosion resistance was improved	Yue <i>et al.</i> ⁵⁴⁴	2001
Al-12.5 wt.-%Si	Laser cladding of Al-12.5 wt.-%Si alloy onto a Mg/SiC composite (using a Nd:YAG laser) led to a significant improvement in corrosion resistance	Wang and Yue ⁵⁴⁵	2001
17 vol.-%SiC/ZK60	Excimer laser surface treatment of Mg metal matrix composite caused dissociation of SiC	Wang and Yue ⁵⁴⁸	2001
17 vol.-%SiC/ZK60	Excimer laser surface treatment of Mg-SiC metal matrix composite improved the corrosion resistance property significantly attributed to grain refinement. The improvement is more in nitrogen atmosphere	Yue <i>et al.</i> ^{541,544}	1997 2001
MEZ	Laser surface melting improves microhardness and corrosion resistance significantly by refinement of grains and supersaturation of rare earth element	Dutta Majumdar <i>et al.</i> ⁵⁴³	2003
AZ91D and AM60B Mg alloys	Uniform and refined microstructure improves corrosion resistance	Dube <i>et al.</i> ⁵³⁹	2001
AZ91 Mg alloy	Pulsed excimer laser irradiation improves corrosion resistance	Schippman <i>et al.</i> ⁵⁴⁰	1999
Mg-ZK60+SiC composite	Uniform and refined microstructure produced by excimer laser improves corrosion resistance	Yue <i>et al.</i> ⁵⁴¹	1997

of erosion induced corrosion and corrosion induced erosion were determined.

Jain *et al.*⁵⁰⁶ vacuum deposited Cr film (thickness varying from 650 to 1350 nm) on Al substrate and irradiated with single pulses (with a full width at half maximum of 7 ns) from a Nd:glass laser operating in a TEM00 mode at peak power densities of 6000–15 000 GW m⁻² in air. Rutherford backscattering of 3.05 and 4.8 MeV He⁺ ions was employed to determine depth profiles of chromium in aluminium. Almost complete removal of chromium at 8000 J m⁻² is followed by the rapid diffusion of chromium (in molten Al) with an accompanying reduction in evaporation loss at higher energy densities. Almeida *et al.*⁵⁰⁷ carried out laser surface alloying of Al with Cr using powder injection method with a laser beam power of 2000 W and a scan speed of 5 mm s⁻¹. Cr content in the alloyed zone was 4.3 wt.-%. The microstructure of the alloyed zone consists of dispersion of long faceted intermetallics of Al-Cr organised radially along primary α -solid solution matrix. The microhardness of the alloyed layer is significantly improved and increased with Cr content in the alloyed zone. Similarly, Ferreira *et al.*⁵⁰⁴ also reported the improvement in crevice corrosion in Al 7175-T7351 alloy by laser surface alloying with Cr.

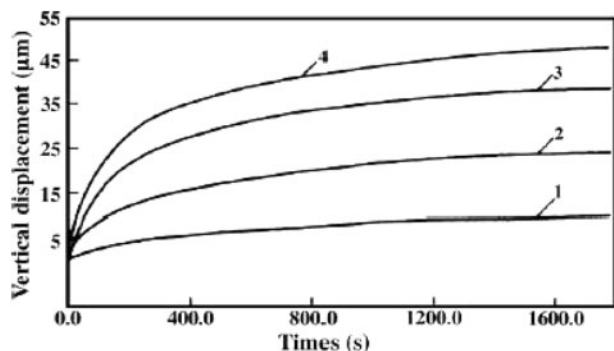
Dutta Majumdar *et al.*⁵¹⁹ developed a titanium boride dispersed Al based metal matrix composite on the surface of pure Al by melting the surface of as received substrate using a CW CO₂ laser and simultaneous deposition of a mixture of K₂TiF₆ and KBF₆ (in the weight ratio of 2:1) through an external feeder (at a feedrate of 4 g min⁻¹). The microstructure of the surface layer consisted of uniformly dispersed titanium boride (TiB) and titanium diboride (TiB₂) particles in grain refined Al matrix. Figure 15 shows the variation in wear loss (in terms of depth of wear) as a function of time for as received and laser composite surfaced Al with TiB₂ (lased with a power of 1.2 kW and a scan speed of

700 mm min⁻¹) using a friction and wear monitor unit with applied loads of 500 and 900 g respectively. It is evident that the rate of wear increases both with time and applied load. In pure Al (curves 3 and 4), the wear rate is very high in the initial stage (up to 5 min) following which the same decreases.

On the other hand, the magnitude and rate of wear are significantly lower in laser composite surfaced Al than those of the as received Al, although the same increases with increasing load for both as received and laser composite surfaced Al with TiB₂. The kinetics of wear was, however, found to vary with laser parameters. Maximum wear resistance was achieved in laser composite surfaced Al lased at a power of 1.2 kW and a scan speed of 700 mm min⁻¹. The improved wear resistance of laser composite surfaced Al was attributed to improved microhardness of the composite layer because of both grain refinement and precipitation of fine and hard TiB₂ and TiB particles in the matrix.

An aluminium oxide layer of 100 μ m in thickness has been successfully coated on aluminium alloy 6061 and pure aluminium using a powder mixture of silicon oxide and aluminium by laser cladding using a CW CO₂ laser. A strong Al/Al₂O₃ interface was formed possibly due to an exothermic chemical reaction between SiO₂ and Al.⁵²⁰ Laser cladding on pure Al substrate was carried out using a CW Nd:YAG laser and a coaxial powder injection system to develop composite coatings of 0–40 wt.-%Si and 0–30 wt.-%TiC particles in Al matrix.⁵²¹

The microstructure of the coatings was homogeneous and crack/pore free with uniform distribution of carbides. The addition of Si and TiC reinforcement particles increased the hardness and wear resistance of the coatings. Katipelli *et al.*⁴⁹⁸ carried out a four point bend test on laser surface engineered TiC coating on 6061 Al alloy to study the interfacial strength or integrity of the coating including strain energy release rate and stress intensity factor. It was inferred that the



15 Cumulative wear loss (in terms of vertical displacement) as function of time for as received Al with applied loads of 500 g (plot 3) and 900 g (plot 4) and laser composite surfaced Al with TiB_2 (with power of 1.2 kW and scan speed of 700 mm min^{-1}) with applied loads of 500 g (plot 1) and 900 g (plot 2) using friction and wear monitor unit⁵¹⁹

interface plays a key role in determining the mechanical behaviour of ceramic coating system. Similar investigations on the laser surface alloying of TiC on aluminium (2024 and 6061) alloys by Nd:YAG laser showed the effect of process parameters on the microstructures of coating.⁵⁰¹ Fu *et al.*⁵¹⁷ studied the effect of laser surface melting on the wear behaviour of plasma sprayed ZrO_2 coating on Al. Porosity and roughness of the plasma sprayed coating were significantly reduced and the bond strength between the coating and substrate was increased by laser melting. However, the presence of cracks was observed despite the fact that wear resistance was significantly improved after laser remelting.

Wang *et al.*⁴⁹⁹ developed a graded coating consisting of a Ni clad Al bond layer, a 50 wt-%Ni clad Al + 50 wt-% (Al_2O_3 –13 wt-% TiO_2) intermediate layer and an Al_2O_3 –13 wt-% TiO_2 overlayer (or ceramic layer) on an Al–Si alloy substrate by plasma spraying followed by laser surface remelting. While the plasma sprayed coatings seem prone to spallation at the different depths originating from macrocracks in the ceramic layer during thermal cycling, the laser remelted coatings, containing a network of microcracks in the ceramic layer reveal a significantly reduced tendency of spallation at the interface between the intermediate and bond layer. Thus, laser remelting improved the spalling resistance of plasma sprayed coating to thermal shock, although cracking due to thermal shock remains a problem to be solved. Uenishi and Kobayashi⁵⁰⁰ obtained Al_3Ti dispersed intermetallic matrix composite on Al by laser surface cladding.

Laser surface alloying of copper alloys

Laser surface alloying of copper and its alloys has been extensively studied by several researchers and are summarised in Table 5b. Manna *et al.*⁵²² made an attempt of laser surface alloying of Cu with Al (predeposited by physical vapour deposition, and subsequently laser melted) using ruby laser (irradiated in air for varying numbers of 30 ns pulses with 0.2 J mm^{-2} energy density) and identified the phases through careful XRD and photoelectron spectroscopy. Michaelides and Panagopoulos⁵²³ irradiated Zn coated Cu with a CW CO_2 laser to produce a thin Zn–Cu surface alloy on copper substrate. The amount and distribution of zinc in the alloyed zone and the intensity of the plasma created on the surface during laser irradiation

were found to be a function of the laser power density and interaction time. Vanderberg and Draper⁵²⁴ laser surface melted commercial Al bronzes (containing 5 at-%Fe and 13–28 at-%Al) and found a variety of phases formed in near surface region. The microstructure consisted of α -phase (face centred cubic solid solution), β_1 phase (product of ordering transformation of the high temperature disordered body centred cubic phase β) and even β phase (martensite formed by shear transformation on quenching). Rapid quenching in laser treatment also resulted in β_1 retention (by suppressing of martensite change) at Al concentration that was lower than what was usually observed with splat quenching experiments.

Dutta Majumdar and Manna⁵²⁵ attempted to enhance the wear and erosion resistance of Cu by laser surface alloying with Cr (electrodeposited with 10 and $20 \mu\text{m}$ thickness t_z). Total Cr content X_{Cr} , Cr in the form of precipitates f_{Cr} and Cr in solid solution with Cu Cu_{Cr} in the alloyed zone were determined by energy dispersive spectrometry, optical microscope and XRD technique respectively. Laser surface alloying extended the solid solubility of Cr in Cu as high as 4.5 at-% as compared with the equilibrium solid solubility limit of 1 at-%.

The microhardness of the alloyed zone was found to improve significantly (as high as 225 VHN) following laser surface alloying as compared with 85 VHN of the base metal.⁵²⁵ Since hardness is related to Cr present in solid solution and dispersed as precipitates in the matrix, the variation in average microhardness of the alloyed zone as a function of the total Cr content, Cr dissolved in solid solution or volume fraction of Cr precipitated shows that hardness increases with all these microstructural factors, especially with the degree of solid solubility extension of Cr in Cu.⁵²⁵ A significant enhancement on wear resistance and erosion resistance (both at room temperature and at elevated temperature) properties was also achieved by laser surface alloying.⁵²⁵ Zhao *et al.*⁵²⁶ studied the effect of laser surface treatment on the surface energy of copper plate in terms of the surface microstructure analysis and theoretical computation. The surfaces of the copper plates were treated by Nd:YAG pulsed laser with different powers. The microstructures of the treated copper plates were analysed by optical microscopy and XRD, and the wetting experiment was performed to evaluate the variation in surface energy. The surface energy of the treated surface was found to vary with the applied laser power.

Laser surface alloying of titanium and its alloys

Tian *et al.*⁵²⁷ laser surface alloyed pure Ti with TiN–B–Si–Ni powders and detected the formation of different kinds on intermetallics in the alloyed zone with a very high microhardness (1500–1600 HV0.1), a low friction coefficient (~ 0.4) and a higher oxidation resistance than the untreated substrate.

Laser surface alloying of Ti with Si, Al and Si+Al (with ratios of 3:1 and 1:3 respectively) was conducted to improve the wear and high temperature oxidation resistance of Ti.^{528–530} Isothermal oxidation studies conducted at 873–1023 K showed that laser surface alloyed Ti with Si and Si+Al significantly improved the isothermal oxidation resistance. The effect of laser surface alloying of Ti with Si or Si+Al on wear resistance was also studied. Under comparable conditions of scratching,

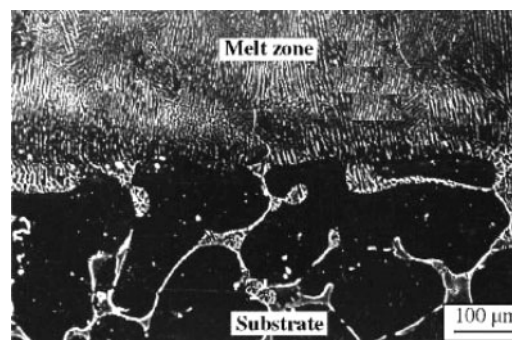
Ti undergoes the most rapid wear loss followed by that in laser surface alloyed specimens. The laser alloyed sample with Si undergoes the minimum wear loss. The improved wear resistance of laser surface alloyed Ti with Si was attributed to the formation of hard Ti_3Si_2 precipitates in the alloyed zone.⁵²⁸ Manna *et al.*^{531,532} made an attempt to develop an alternative electrode material for neural stimulation electrode by laser surface alloying of Ti with Ir that can mimic the normal spatiotemporal pattern of neuronal activation by reversible charge transfer. The usual electrode made of iridium is expensive, brittle and not amenable to miniaturisation by plastic deformation. In comparison, titanium is cheaper, biocompatible and amenable to drawing/etching. A few relevant literatures on the laser surface alloying of titanium and its alloys are summarised in Table 5c.^{528–538}

Laser surface alloying of magnesium and its alloys

Magnesium and its alloys are widely used in automotive and aerospace applications because of its lightweight. However, poor wear and corrosion properties are of serious concern for its application as structural components. Laser surface melting of magnesium and its alloys was reported to be effective to refine the microstructure and improve corrosion resistance in AZ91D/AM60B,⁵³⁹ AZ91,⁵⁴⁰ Mg-ZK60⁵⁴¹ and other commercial Mg based alloys.⁵⁴² Detailed studies on kinetics and mechanism of laser surface melted MEZ (a Mg–Zn alloy) were reported by Dutta Majumdar *et al.*⁵⁴³ Figure 16 shows the cross-section of the laser surface melted MEZ alloy, lased with a power of 2 kW and a scan speed of 200 mm min^{-1} , consisting of fine columnar grains growing epitaxially from the liquid/solid interface. The underlying substrate grains are significantly coarser with grain boundaries decorated with thick films of Mg/Zr/Ce rich compound (confirmed by energy dispersive spectroscopy). Furthermore, the melted zone/substrate interface is crack/defect free and well compatible with practically no noticeable amount of heat affected zone.

A detailed energy dispersive spectroscopy analysis of the melt zone suggests that laser melted zone has a more homogeneous elemental distribution than that in as received MEZ. Figure 17 presents the microhardness profiles of the laser surface melted MEZ specimens as a function of vertical distance from the surface measured on the cross-sectional plane and shows that the microhardness of the melted zone has significantly increased by 2–4 times (85–100 VHN) than that of the substrate (35 VHN) primarily due to grain refinement and solid solution hardening. While average microhardness varies with the concerned laser parameters, the maximum microhardness is achieved following laser surface melting with a power of 1.5 kW and a scan speed of 200 mm min^{-1} .

Figure 18 shows the kinetics of pit formation in the as received and laser surface melted (with a power of 2 kW and a scan speed of 200 mm min^{-1}) MEZ samples subjected to standard immersion tests in a 3.56 wt-% NaCl solution. It is evident that both the extent and rate of pitting are significantly reduced following laser surface melting. Furthermore, the rate of pitting is initially faster and decreases after 30 h of immersion in as received MEZ, while visible pits are observed in laser surface melted MEZ only after 12 h of exposure revealing a much slower kinetics of the process. In fact, the pits in the laser surface melted samples are both

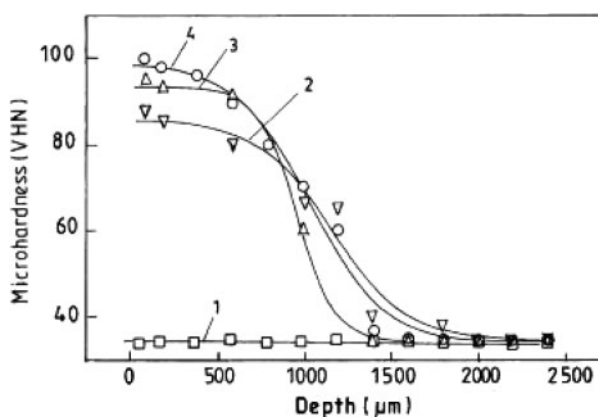


16 Scanning electron micrograph of cross-section of laser surface melted MEZ sample (lased with power of 2 kW and scan speed of 200 mm min^{-1}): defect free interface with significant grain refinement in laser melted zone may be noted

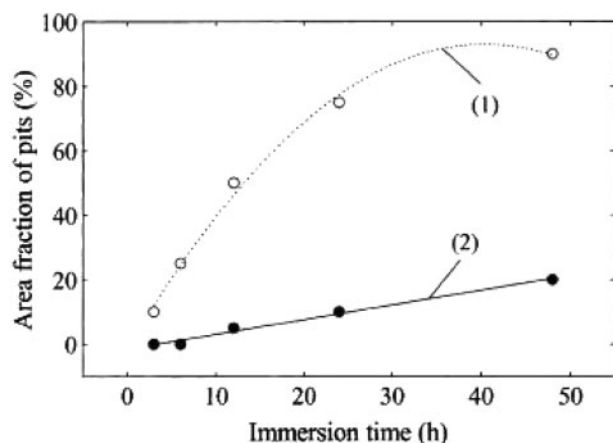
smaller in dimension and remain isolated compared with larger and interconnected pits in as received MEZ. Yue *et al.*⁵⁴⁴ surface melted Mg–SiC (17 vol.-%) composite using KrF excimer laser with Ar and N_2 as shrouding environment and achieved considerable improvement in corrosion resistance in 3.56%NaCl solution primarily due to microstructural refinement. Similar surface melting in N_2 atmosphere improved the corrosion resistance further because of the formation of magnesium nitrides.

Laser surface alloying was also attempted on magnesium and its alloys to improve the wear and corrosion resistance properties. Wang and Yue⁵⁴⁵ have achieved significant improvement in corrosion resistance of SiC dispersed Mg by LSC with Al–12Si alloy layer.

Ignat *et al.*⁵⁴⁶ studied the effect of laser surface alloying on surface alloying of WE43 and ZE41 Mg alloys with Al using a 3 kW CW Nd:YAG laser and found that the alloyed zone microstructure consisted of Al_3Mg_2 and $\text{Al}_{12}\text{Mg}_{17}$ intermetallics with an improved microhardness and corrosion property. Similar investigations of laser surface alloying of MEZ with Al + Mn (in 3 : 1 and 1 : 1 by weight) using a CW CO_2 laser significantly improved the wear and corrosion properties of the base alloy.⁵⁴⁷ Several other works related to the surface engineering of magnesium and its alloys are summarised in Table 5d.^{539–548}



17 Microhardness profile as function of depth from surface of (1) as received MEZ and laser surface remelted MEZ lased with (2) power of 2 kW and scan speed of 200 mm min^{-1} , (3) 2.0 kW and 300 mm min^{-1} and (4) 1.5 kW and 200 mm min^{-1} respectively⁵⁴³



18 Kinetics of pit formation (in terms of area fraction of pits as function of immersion time) in (1) as received and (2) laser surface melted (with power of 2 kW and scan speed of 200 mm min⁻¹) MEZ samples followed by immersion test in 3.56 wt-%NaCl solution⁵⁴³

Laser assisted CVD

Chemical vapour deposition encompasses a group of processes that involve depositing a solid material from a gaseous phase. Laser chemical vapour deposition (LCVD) involves techniques in generating solid deposits on the surface of a substrate by dissociation of reactants followed by heating using laser beam. Laser chemical vapour deposition deposits materials at a faster rate with a high accuracy, a low porosity and a high degree of crystallinity and hence, a coating with an excellent mechanical properties and thermal stability. Pyrolytic LCVD is a thermally driven process in which the energy of a laser beam is used to heat the surface of a substrate to the temperature required for chemical deposition. Pyrolytic LCVD has been applied to deposit a large varieties of materials including Cu, Au, Si, Al, W, C, WC, B, TiC, TiN, SiC and Si₃N₄.^{549–552} However, a careful control of process parameters is extremely important when depositing multiple layers of materials.⁵⁵³ Since the underlying deposit may exhibit irregularities, proper control is needed to insure that these irregularities are not amplified with each successive layer. Noel and Kovar deposited titanium carbide (TiC) coatings onto tantalum substrates using hydrogen gas, titanium tetrachloride (TiCl₄) and either methane (CH₄) or acetylene (C₂H₂) mixtures.^{554,555} The kinetics of deposition was found to vary with precursor composition. It was also observed that the phases (both quantity and area fraction) and orientation of the deposition varied with precursor composition and temperature. Laser chemical vapour deposition was also reported to develop oxide based ceramic coatings such as partially yttria stabilised zirconia and titania (TiO₂).⁵⁵⁶ The deposition rate was significantly higher than that of the conventional chemical vapour deposition which is attributed to the plasma formation around the deposition zone, above critical values of laser power and substrate preheating temperature. The activation energy for the deposition for both yttria stabilised zirconia and TiO₂ films was ~10 kJ mol⁻¹, suggesting a mass transfer limited process. A wide variety of film morphologies were obtained based on the applied parameters. The columnar structure contained a large amount of nanopores at columnar

boundary and inside grains. A detailed thermodynamic calculation was undertaken to understand the process sequences. On the other hand, a kinetic model is essential to determine the apparent activation energy and the apparent order of reaction.⁵⁵⁶

Laser surface engineering of ceramic materials

Laser surface engineering has also been extensively studied to tailor the surface of ceramic materials.^{557–569} Heras *et al.*⁵⁵⁷ attempted laser assisted removal of water repellent substances or coatings on stone and similar materials used in buildings or sculptures of artistic value. The effect of wavelength on laser assisted removal of two water repellent coatings or treatments applied on limestone, namely, Paraloid B-72, a copolymer of methyl acrylate and ethyl methacrylate, and Tegosivin HL-100, a modified polysiloxane resin, was investigated by using the four harmonics of a Q switched Nd:YAG laser (1064, 532, 355 and 266 nm). It was observed that the removal of the thin coatings depended on the laser irradiation conditions and on the characteristics of the coatings. The fundamental laser radiation was effective in removing both the coatings but thermal alteration processes were induced on the constituent calcite crystals. The best results were obtained by irradiation at the near UV wavelength of 355 nm.

Lawrence and Li⁵⁵⁸ glazed the surface of Al₂O₃ based refractory by means of high power diode laser and augmented the wear rate and wear life characteristics within both normal and corrosive (NaOH and HNO₃) environmental conditions. Life assessment testing revealed that diode laser generated glaze increased the wear life of the Al₂O₃ based refractory by 1.2–13.4 times depending upon the environmental conditions. Such improvements are attributed to the fact that the microstructure of the Al₂O₃ based refractory was altered from a porous, randomly ordered structure to a much more dense and consolidated structure after laser treatment that contained fewer cracks and porosities.

Sciti *et al.*⁵⁶¹ surface treated alumina and silicon carbide ceramics with a KrF excimer laser and studied the influence of laser fluence (1.8 and 7.5 J cm⁻²), number of laser pulses (1–500), frequency (1–120 Hz), pulse duration (25 ns) and shrouding environment on the characteristics of the treated surface. Microstructural analyses of the surface and cross-section of the laser processed samples at a low fluence (1.8 J cm⁻²) evidenced that the surface of both ceramics is covered by a scale due to melting/resolidification. At a high fluence (7.5 J cm⁻²), there are no continuous scales on the surfaces as material is removed by decomposition/vapourisation. On the alumina surface, a network of microcracks formed, while on the silicon carbide, products of different morphologies (flat and rugged areas, deposits of debris and discontinuous thin remelted scales) were detected.

Schmidt and Li⁵⁶³ studied the influence of laser glazing of sintered alumina as well as alumina–silica mixtures on the surface characteristics and service life using a high power diode laser and a high speed motion analysis system for use in a multitude of applications as insulators, crucibles, vacuum system windows, etc. The laser glazed material seems robust and exhibits good adherence to the bulk ceramic.

Stoltz and Poprawe⁵⁶⁴ modified the surface conductivity of dielectric ceramics by CW and pulsed laser irradiation. Conductive structures on Al₂O₃ were developed with a CO₂ laser and on AlN with an excimer laser

irradiation (pulse width=25 ns). Laser treated specimens were subjected to a detailed characterisation of electrical resistivity, chemical composition, structure and morphology.

Polycrystalline ceramics, notably alumina (polycrystalline aluminium oxide) are used widely in the electronics industry as substrates for metal film deposition, as well as in other industrial applications. Multishot pulsed XeCl excimer laser irradiation of commercial fine grained polycrystalline alumina substrates is found to significantly improve the bonding characteristics between the metal and film.⁵⁶⁷ Laser irradiation produces a smoother surface finish increasing the adhesion strength of subsequently deposited copper films to the laser treated alumina surface by a factor of 3–5 under optimum lasing conditions. X-ray photoelectron spectroscopy measurements suggest that the electrical activation of the near surface region may also contribute to the enhanced copper adhesion.

Buerhof *et al.*⁵⁶⁸ studied the effect of laser ablation on surface modification of glass using a 50 W CO₂ laser (10.6 μm) and a 40 W/2 J/20 Hz XeCl excimer laser (308 nm). The infrared radiation heated the sample surface locally while high power UV radiation caused ablation of glass with negligible thermal effects. Removal rates of various glasses during laser ablation have been determined.

Laser surface engineering of polymers

Laser induced surface modification and laser induced patterning of polymeric surfaces by deposition, etching and ablation have drawn considerable attention of late. Wong *et al.*⁵⁷⁰ irradiated the surface of polyethylene terephthalate with KrF excimer laser (having a wavelength of 248 nm) and studied the surface image using atomic force microscopy. The surface roughness and ripple spacing increased with laser energy. Merging of the ripples (due to surface tension forces) is the driving force for the increased ripple spacing. With appropriate laser treatment, the hydrophobicity of polyester can be greatly enhanced, making the treated polyester highly unwettable. This provides the textile industry a new method for making a high tech water repellent textile product. Adhi *et al.*⁵⁷¹ irradiated the surface of poly-tetra-fluoro-ethylene films by femtosecond UV radiation from an excimer laser (KrF: λ : 248 nm; t_p : 380 fs) in air and investigated the surface using X-ray photoelectron spectroscopy, and bulk by Fourier transform infrared spectroscopy. There was no signature of incorporation of hydrogen and/or oxygen, or the formation of a cross-linked network of carbon indicating chemically clean processing in contrast to nanosecond excimer laser processing which chemically degrades the surface. Huang *et al.*⁵⁷² discussed a method of improving adhesion to fluorocarbon resin by irradiation with an XeCl excimer laser. The respective adhesive force between the polymer and metal, and contact angles with water were measured by the shear test method and a Ne-He laser system. The adhesive force improved when the polymer surface was irradiated in the presence of water or solutions of boric acid, sodium hydroxide, copper sulphate and sodium aluminate. These results were interpreted in terms of a simple laser heating model. Csete *et al.*⁵⁷³ generated a new, two-dimensional grating-like structure on spin coated films made of polycarbonate which might be used as a special kind of mask.

Drinek *et al.*⁵⁷⁴ photolysed hexamethylcyclotrisilazane with an ArF excimer laser at 84 K. At room

temperature and in air, the film changed into a polymer having siloxane units. However, fragments generated by laser irradiation of this polymer in the frozen film made a similar polymer film having a siloxane structure on a polyvinyl alcohol film in air. The polymeric film with siloxane units prepared by the cryogenic laser ablation method displayed hydrophobic properties. Lu *et al.*⁵⁷⁵ surface modified polyimide film by pulsed UV laser. Irradiation caused a photochemical reaction in the hydrophilic groups. In addition, a ripple microstructure was formed on the surface when the angle of incidence of the laser beam was 20–50°. Csete and Bor⁵⁷⁶ generated laser induced periodic surface structures on oriented and amorphous thick, as well as on spin coated thin, polycarbonate films by a polarised ArF excimer laser. Studies on the influence of the film structure and thickness on the development of periodic surface structure revealed that the line shaped structure would transform into droplets below a critical thickness of the spin coated films. This droplet formation was explained by the laser induced melting across the film and subsequent dewetting on the substrate. The thickness of the layer melted by laser irradiation was computed by a heat conduction model, compared well with the critical thickness for the spin coated film. Hartwig⁵⁷⁷ studied the influence of excimer laser radiation (248 nm) on the surface properties of an epoxy resin. At low laser energy densities, the lap shear strength with the epoxy based adhesive was on an appropriate level with an improved reproducibility; however, at higher energy densities, the lap shear strength decreased due to the decomposition of the resin surface, as revealed by X-ray photoelectron spectroscopy results. Frerichs *et al.*⁵⁷⁸ conducted a detailed investigation on the laser ablation of different polymers such as polyamide, polystyrene, polycarbonate and polypropylene using a 248 nm excimer laser to study the effect of laser parameters on the surface structuring. Arenholz *et al.*⁵⁷⁹ showed that the adhesion of metal films on polymer foils could be improved by laser irradiation of the polymer surface before metal deposition. As a model system, metal films of different Co alloys such as CoNi, CoCr and CoNi oxide were chosen to form deposits on polytetraethylene foils. Laser irradiation was performed by means of a 248 nm KrF and a 308 nm XeCl laser lights in 1–20 pulses with typical fluences of 5 and 40 mJ cm⁻² respectively. Post-irradiation metal deposition was carried out by electron beam evaporation immediately after laser beam irradiation. Laser irradiation of untreated polytetraethylene foils resulted in the formation of wall type or map type structures, the mean distance of which was found to be related to laser parameters.

Different methods of laser treatment are suitable to generate grating-like periodic structures on polymer surfaces. One possibility is the projection of a mask and realisation of periodic ablation.⁵⁸⁰ The resulting periodicity is limited by the resolution of the applied optical system and by the complex response of the material to the laser illumination.⁵⁸¹ This method is suitable for generating structures in the micrometre sized region without difficulty. Smaller structures can be induced in the near field of the grating by the Talbot effect.^{582,583}

Bolle and Lazare⁵⁸⁴ produced a new class of sub-micrometre laser induced structures on various polymer surfaces by absorption of a polarised excimer laser beam of low fluence. Parallel ripples of <0.2 μm spacing in the

direction of the electric field are obtained with the ArF or KrF radiation when it is strongly absorbed by the polymer. Several polymers in thin or thick films of commercial or laboratory origin were textured in air or in vacuum or both depending on the chemical structure. Fluence should be below the ablation threshold and must be chosen in a narrow window, which depends on the polymer and the wavelength used. Surface characterisation by scanning and transmission electron microscopies and ellipsometry reveals a good insight into pattern generation.

In many applications, a controlled submicrometre level roughening of surfaces is desired, say, on thin film (0.2 μm) coated on silicon wafers. Owing to the simplicity of this irradiation procedure, large scale surface treatments are possible. Buchman *et al.*⁵⁸⁵ did preadhesion treatment of two thermoplastic substrates (Lexan polycarbonate and Ultem polyetherimide) by means of ArF excimer laser and evaluated the adhesion behaviour by comparing the lap shear strengths of polyurethane bonded joints with untreated, laser treated and SiC abraded adherends. Experimental results showed a significant increase in lap shear strength following ArF laser treatment. Surface analysis of the ArF laser treated adherends and the fractured polyurethane joints revealed morphological and chemical modifications due to the intense laser irradiation.

Summary and future scope

Surfaces of engineering components are prone to fail due to mechanical interaction or degrade under chemical/electrochemical attack. Surface engineering is based on the philosophy that performance and life of a component can be improved or extended by tailoring the surface microstructure and/or composition without altering that of the bulk. Laser is particularly suited to achieve this goal of surface engineering because of the exponential energy deposition profile of laser into any solid and associated advantages of non-contact, ultrafast, precise, fully automated and flexible mode of manipulating the level of operation encompassing material removal (machining and cleaning), addition (cladding and deposition) or simply modification (heating/cooling, hardening, melting and alloying). The irradiated zone is extremely small and hence the distortion is the minimum. In the process, laser surface engineering can vastly improve the surface dependent properties such as hardness, friction, fatigue and resistance to wear, corrosion, etc. There are several books^{1-4,8-10,20} and review articles^{11,165,166,374-377} exclusively dealing with this subject. The present section thus made an attempt mainly to review mainly the recent experimental studies concerning laser surface engineering with brief discussion on the mathematical modelling or simulation studies. For the latter, the readers may refer to the Refs. 5, 6 and 192.

Laser is now routinely used for surface engineering in automobile sector for hardening and cladding engine cam shafts or valve seats, hardening of cutting and bending edges on press tools; in power generation sectors for hard facing steam turbine blades, in oil and gas exploration fields for hardening and cladding of pumps, valves and tubular components; in aerospace sector for repair and cladding of turbine blades, hardening of bearing surfaces on rotating shafts, improving adhesion between metallic and composite joints, etc. The scope of laser surface engineering is also firmly extended to medical technology both for development and surface

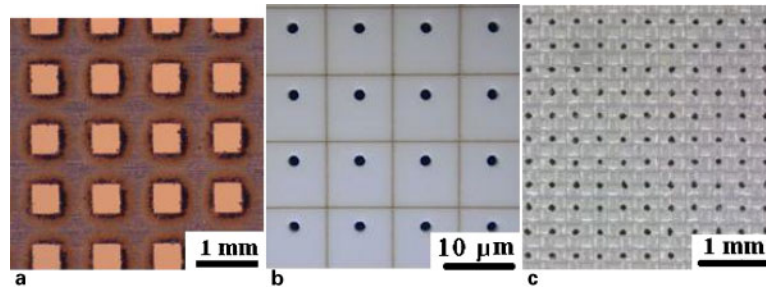
modification of prosthesis and implants and tissue engineering.

The current trend appears to suggest that the diode/fibre/disc lasers with shorter wavelength, fibre optic delivery system and multiple axis sample/component manipulator may soon substitute CO₂ even YAG lasers for laser surface engineering applications. Laser is such a useful tool for modifying the surface properties because of the great precision, inherent flexibility of delivery and manipulation of the laser light, and speed or productivity of the process. Thus many more new scopes of exploiting lasers for surface engineering are likely to emerge. However, extensive efforts are needed to establish the microstructural evolution and scope of improvement following laser surface engineering with these new lasers and processes.

Emerging trends and future challenges

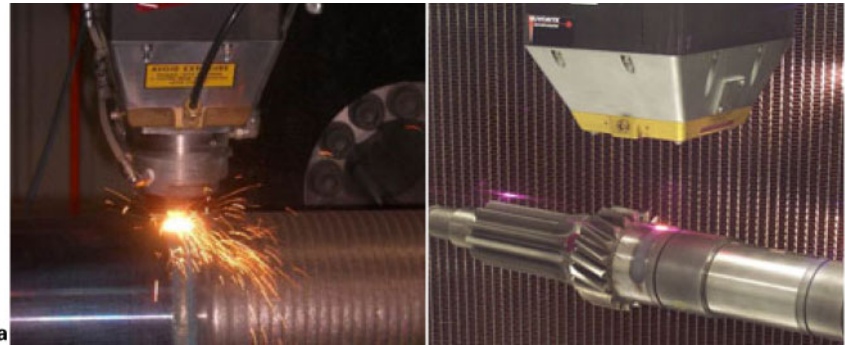
This article has presented an overview of the fundamentals of laser, laser-matter interaction and application of laser with regard to material processing divided into four major categories, namely, forming, joining, machining and surface engineering. Laser was invented in early twentieth century. Commercial high power lasers capable of delivering adequate power density for material processing involving heating, melting and vapourising from all engineering solids (metallic, ceramic, semiconducting, polymeric and composites) were developed nearly 40 years ago. Yet, exploiting laser for material processing became popular only in 1980s and onwards. However, the exploration and preliminary forays soon turned into a plethora of applications right up to large scale commercial production level. The success in engineering applications soon drew intense attention from the scientific community to understand the microstructure and phase evolution, influence of process parameters on material behaviour and correlation between microstructure and properties. The fact that laser not only produces intense localised heating but can also produce novel and non-equilibrium microstructure including amorphous and extended solid solutions by ultrarapid cooling due to self-quenching created immense scientific curiosity to develop metastable aggregate with enhanced properties, improved performance and extended component life. The unique spatial/temporal coherence and monochromaticity of laser, flexibility and speed of processing, and automation worthiness soon made laser an extremely useful tool from purely mundane (sheet metal cutting or drilling) to most sophisticated application and processing (joining of animal tissues or treating human tumours/stones, processing microelectronic devices, cleaning priceless paintings and sculptures, and precision machining of instruments and components).

The extensive amount of literature reviewed in this article clearly suggests that the field is far from being saturated and is rather rapidly growing into new and unexplored areas of application. The biggest advantage of laser is its ability to be a contactless heating tool that can deliver the desired quantum of thermal energy in all solids at a precise location irrespective of their chemical bonding/nature, physical dimension (small or big) and state of aggregate (pure, mixed and composite) with minimum damages to the surrounding. Although the laser induced heating or thermal profile decays exponentially with depth, the possibility of coupling laser with modern engineering practices such as computer aided design and



a 500 μm square holes with 500 μm gaps in between drilled into 200 μm thick copper foils by pulsed Nd:YAG laser; b 1 mm diameter holes with 10 μm wide lines scribed on alumina sheet by CO_2 laser; c high magnification view of array of 150 μm diameter holes drilled in fabric with a spacing of 625 μm by CO_2 laser

19 Examples of ultrafine drilling⁵⁸⁶



a laser cladding a shaft with Satellite 6; b laser transformation hardening a transmission shaft

20 Examples of CW CO_2 laser assisted⁵⁸⁶ forming/manufacturing of large engineering component

manufacturing using robot controlled stage and pneumatic/mechanically driven delivery system (of wire, powder and sheet) has enabled the creation of large components practically of any desired dimension and geometry. The versatility is further extended in the ability of laser to simultaneously heat, add, join or remove materials of dissimilar nature such as conducting and non-conducting, hard and soft, and dense and porous. Laser is now utilised to close surface porosities on coated and welded products and also for creating graded aggregate with controlled porosity, composition and microstructure. This microstructural or compositional gradation can be achieved within very narrow to wide length scales and along horizontal or vertical directions.

In order to showcase some of the recent exploits of laser that are unique to laser material processing, we pick up three examples from commercial products and practices. Figure 19a–c demonstrates the unique capability of laser to produce ultrafine holes in square, rectangular or round shapes in metals, ceramic or polymeric materials. Besides drilling, the same material can be marked, scribed or machined using the same tool and facility. The precision and dimensional accuracy of these holes are unmatched notwithstanding the advantage of fully automated processing at an extremely fast speed and productivity. The holes have completely vertical walls with no distortion of the bulk workpiece. Holes with tapered inclination, threads or steps are also possible. Similar perforations can be made on thin filter or cigarette papers, round rims or flat turbine blades with equal ease and precision.

Laser drilling can be used to produce microholes with very high accuracy and close tolerance in almost all solids (metals, ceramics, diamond, silicon and other semiconductors, polymers, glass and sapphire) in the

form of pinholes, nozzles, orifices, vias, photovoltaic cells, fuel injection nozzles, vertical probe cards, metered dose inhaler products, slits for scientific instrumentation, inkjet printer nozzles, detectors, sensors, high resolution circuitry, fuel cells, fibre optic interconnects and medical devices.

Figure 20a and b demonstrates how laser can be used for developing a clad or over layer on large shaft and heat treating (transformation hardening) transmission shaft, axle rods or gears. Similarly, laser cladding is exploited in refurbishing or reclaiming rolls, tools, blades, discs and turbine components. Transformation hardening can be applied mostly to steels and cast irons, but the ultrafine grain structure produced by surface melting and associated rapid cooling can also produce hard and wear resistant surfaces on non-ferrous metals and alloys. Controlled laser irradiation can produce localised heating, drilling vias and holes, annealing and even metallisation and doping in semiconductor wafers and films.

References

1. W. M. Steen and K. Watkins: 'Laser material processing', 1; 2003, New York, Springer-Verlag.
2. B. L. Mordike: in 'Materials science and technology', (ed. R. W. Cahn *et al.*), Vol. 15, 111; 1993, Weinheim, VCH.
3. J. Mazumdar: in 'Lasers for materials processing', (ed. M. Bass), 113; 1983, New York, North-Holland Pub. Co.
4. J. C. Ion: 'Laser processing of engineering materials principles, procedure and engineering application', 1; 2005, Oxford, Elsevier Butterworth-Heinemann.
5. J. Dowden: 'The theory of laser materials processing: heat and mass transfer in modern technology', 119; 2009, Bristol, Canopus Publishing Limited.
6. S. T. Picraux and D. M. Follstaedt: in 'Laser–solid interactions and transient thermal processing of materials', (ed. J. Narayan *et al.*), 751; 1983, New York, North-Holland Pub. Co.

7. A. Einstein: *Z. Phys.*, 1917, **18**, 121–128.
8. J. Dutta Majumdar and I. Manna: *Sadhana*, 2003, **28**, 495–562.
9. T. H. Maiman: *Nature*, 1960, **187**, 493–494.
10. O. Svelto: 'Principles of lasers', 1; 1986, New York, Plenum Press.
11. C. W. White and M. J. Aziz: in 'Surface alloying by ion, electron and laser beams', (ed. L. E. Rehn *et al.*), 19; 1987, Metals Park, OH, ASM.
12. D. Shealy: *Proc. SPIE*, 2002, **4770**, 28–47.
13. F. Dickey, L. Weichman and R. Shagam: *Proc. SPIE*, 2000, **4065**, 338–348.
14. Y. Liu and E. Kannatey-Asibu: *Manuf. Sci. Eng.*, 1993, **64**, 895–905.
15. Y. Liu and E. Kannatey-Asibu: *Weld. J.*, Sept.1997, 342–348.
16. A. Russ, W. Gref, M. Leimser, F. Dausinger and H. Hugel: *Proc. SPIE*, 2002, **5121**, 362–368.
17. J. Kell, J. Tyrer, R. Higginson, R. Thomson, J. Jones and S. Noden: Proc. ICALEO'06 Conf., Scottsdale, AZ, USA, October 2006, Laser Institute of America, 8–17.
18. J. Hammond, E. Johnson, C. Koehler, J. Stack and T. Suleski: *Proc. SPIE*, 1999, **3633**, 206–213.
19. M. H. Cho, D. Farson, Y. C. Lim and H. W. Choi: *Sci. Technol. Weld. Join.*, 2007, **12**, (8), 677–688.
20. M. H. Cho and D. Farson: *Weld. J.*, Sept.2007, 253–262.
21. J. Mombo-Charistan, R. Mueller, C. Sharp and M. McCay: Proc. ICALEO'97 Conf., San Deigo, CA, USA, November 1997, Laser Institute of America, Section G, 191–199.
22. J. Laeng, J. G. Stewart and F. W. Liou: *Int. J. Prod. Res.*, 2000, **38**, 3973–3996.
23. Y. Namba: Proc. ICALEO'85 Conf., Boston, MA, USA, September 1985, Laser Institute of America, 403–407.
24. Y. Namba: Proc. LAMP'87 Conf., Osaka, Japan, June 1987, High Temperature Society of Japan, 601–606.
25. F. Vollersten: 'Mechanisms and models for laser forming', in 'Laser assisted net shape engineering', (ed. M. Geiger and F. Vollersten), 345–360; 1994, Bamberg, Meisenbach.
26. F. Vollersten and M. Rödle: 'Model for the temperature gradient mechanism in laser bending', in 'Laser assisted net shape engineering', (ed. M. Geiger and F. Vollersten), 371–378; 1994, Bamberg, Meisenbach.
27. F. Vollersten and S. Holzer: *VDI-Z Integr. Prod.*, 1994, **136**, 35–38.
28. M. Geiger and F. Vollersten: *Ann. CIRP*, 1993, **42**, 301–304.
29. H. Arnet and F. Vollertsen: *J. Eng. Manuf.*, 1995, **209**, 433–442.
30. K. C. Lee and J. Lin: *Opt. Laser Technol.*, 2002, **34**, 639–648.
31. F. Vollertsen and M. Geiger: *Blech Rohre Profile*, 1993, **40**, 666–668.
32. T. Hennige, S. Holzer, F. Vollertsen and M. Geiger: *J. Mater. Process. Technol.*, 1997, **71**, 422–432.
33. M. Geiger, P. Hoffmann and J. Hutfless: *Prod. Eng. (Annals of the German Academic Society for Production Engineering)*, 1993, **1/1**, 37–40.
34. J. Magee and L. J. de Vin: *J. Mater. Process. Technol.*, 2002, **120**, 322–326.
35. Y. Li, N. R. Quick and A. Kar: *J. Mater. Process. Technol.*, 2002, **123**, 451–458.
36. D. Schuöcker: *J. Mater. Process. Technol.*, 2001, **115**, 104–107.
37. D. Chen, S. Wu and M. Li: *J. Mater. Process. Technol.*, 2004, **148**, 30–34.
38. S. C. Wu and Z. Ji: *J. Mater. Process. Technol.*, 2002, **121**, 269–272.
39. M. Marya and G. R. Edwards: *J. Mater. Process. Technol.*, 2001, **108**, 376–383.
40. K. Okuda, Y. Sugie and M. Nunobiki: *Key Eng. Mater.*, 2007, **329**, 625–630.
41. P. J. Cheng and S. C. Lin: *J. Mater. Process. Technol.*, 2001, **108**, 314–319.
42. R. Kovacevic, Z. Hu, M. Labudovic and H. Wang: *Int. J. Mach. Tools Manuf.*, 2001, **41**, 589–607.
43. S. Safdar, L. Li, M. A. Sheikh and Z. Liu: *Opt. Laser Technol.*, 2007, **39**, 1101–1110.
44. K. R. Edwards, G. Dearden, K. G. Watkins and S. P. Edwardson: Proc. Conf. 'Photon06', Manchester, UK, April 2006, Institute of Physics, 100.
45. Y. J. Guan, S. Sun, G. Q. Zhao and Y. G. Luan: *J. Mater. Process. Technol.*, 2003, **142**, 400–407.
46. L. W. Zhang, Q. Zhong, B. Lu, J. B. Pei and G. L. Zhang: *Met. Form. Technol. (China)*, 2003, **21**, 25–29.
47. Z. Hu, R. Kovacevic and M. Labudovic: *Int. J. Mach. Tools Manuf.*, 2002, **42**, 1427–1439.
48. J. Liu, S. Sun and Y. Guan: *J. Mater. Process. Technol.*, 2009, **209**, 1580–1587.
49. J. Bao and L. Yao: *J. Manuf. Sci. Eng. (Trans. ASME)*, 2001, **123**, 53–61.
50. T. Hennige: *J. Mater. Process. Technol.*, 2000, **103**, 102–108.
51. S. Silve, B. Podschies, W. M. Steen and K. G. Watkins: Proc. ICALEO'99 Conf., San Diego, CA, USA, November 2000, Laser Institute of America, F87–F96.
52. P. J. Cheng and S. C. Lin: *J. Mater. Process. Technol.*, 2000, **101**, 260–267.
53. T. Bergs and S. Heinemann: *Manuf. Eng.*, 1998, **121**, 50–51.
54. G. Thomson and M. Pridham: *Opt. Laser Technol.*, 1998, **30**, 141–146.
55. C. Peng and Y. Y. Lawrence: *J. Manuf. Sci. Eng. (Trans. ASME)*, 2005, **127**, 572–582.
56. S. C. Wu and J. S. Zheng: *J. Mater. Process. Technol.*, 2001, **110**, 160–163.
57. C. L. Yau, K. C. Chan and W. B. Lee: *J. Mater. Process. Technol.*, 1998, **82**, 117–121.
58. R. E. Holt: *Weld. Eng.*, 1965, **12**, 49–53.
59. J. Lawrence, M. J. J. Schmidt and L. Li: *Int. J. Mach. Tools Manuf.*, 2001, **41**, 967–977.
60. S. Yoshioka, T. Miyazaki, M. Imai, Y. Shirai and T. Misu: Proc. ICALEO'98 Conf., Orlando, FL, USA, November 1998, Laser Institute of America, Section E, 161–169.
61. J. Dutta Majumdar, A. K. Nath and I. Manna: *Mater. Sci. Eng.*, 2004, **385**, 113–122.
62. X. F. Wang, J. Takacs, G. Krallics, A. Szilagyi and T. Markovits: *J. Mater. Process. Technol.*, 2002, **127**, 388–391.
63. W. Li and Y. L. Yao: *Trans. ASME*, 2001, **123**, 674–681.
64. N. Hao and L. Li: *Appl. Surf. Sci.*, 2003, **208–209**, 432–436.
65. D. Schuöcker, F. Kilian, C. Zeinar and A. Kratky: Proc. ICALEO'01 Conf., Jacksonville, FL, USA, October 2001, Laser Institute of America, 10.
66. G. Thomson and M. Prisham: *J. Mater. Process. Technol.*, 2001, **118**, 40–44.
67. R. P. Martukanitz, E. W. Reutzel, R. W. Erskine and S. Nasia: Proc. ICALEO'00 Conf., Dearborn, MI, USA, October 2000, Laser Institute of America, D47–D52.
68. W. Li and Y. L. Yao: Conf. on 'Laser Materials Processing' Proc. ICALEO'00 Conf., Dearborn, MI, USA, October 2000, Laser Institute of America, D53–D62.
69. T. Hatayama and M. Osawa: *J. Jpn Soc. Heat Treat. (Japan)*, 2000, **40**, 116–121.
70. A. K. Kyrzanidi, Th. B. Kermanidis and Sp. G. Pantelakis: *J. Mater. Process. Technol.*, 1999, **87**, 281–290.
71. S. Silve, W. M. Steen and D. Podschies: Proc. ICALEO'98 Conf., Orlando, FL, USA, November 1998, Laser Institute of America, Section E, 151–160.
72. M. Geiger and A. Coreman: *Blech Rohre Profile*, 1995, **42**, 329–334.
73. M. Geiger, J. Kraus and F. Vollertsen: *Bander Bleche Rohre*, 1994, **35**, 26–37.
74. H. Frackiewicz: *Tube Pipe Q.*, 1992, **3**, 12–15.
75. F. R. Liu, K. C. Chan and C. Y. Tang: *Mater. Sci. Eng. A*, 2005, **A396**, 172–180.
76. J. A. Ramos, J. Magee, K. G. Watkins, W. M. Steen and F. Noble: Proc. ICALEO'98 Conf., Orlando, FL, USA, November 1998, Laser Institute of America, 178–185.
77. C. L. Yau, K. C. Chan and W. B. Lee: Proc. ICALEO'96 Conf., Detroit, MI, USA, October 1996, Laser Institute of America, E124–E131.
78. K. G. Chan and J. Liang: *J. Mater. Process. Technol.*, 2000, **100**, 214–218.
79. K. C. Chan and J. Liang: *Compos. Sci. Technol.*, 2001, **61**, 1265–1270.
80. J. Magee, K. G. Watkins, W. M. Steen, N. J. Calder, J. Sidhu and J. Kirby: Proc. ICALEO'97, San Diego, CA, USA, November 1997, Laser Institute of America, E156–E165.
81. K. Okuda, S. Shimoyama and M. Nunobiki: *Key Eng. Mater.*, 2004, **257–258**, 541–546.
82. K. C. Chan, Y. Harada, J. Liang and F. Yoshida: *J. Mater. Process. Technol.*, 2002, **122**, 272–277.
83. K. C. Chan and J. Liang: *Mater. Lett.*, 2001, **49**, 51–55.
84. J. Magee, K. G. Watkins and W. M. Steen: *J. Laser Appl.*, 1998, **10**, 235–246.
85. J. Magee, K. G. Watkins, W. M. Steen, R. L. Cooke and J. Sidhu: Proc. ICALEO'98 Conf., Orlando, FL, USA, November 1998, Laser Institute of America, E141–E150.
86. J. Magee, J. Sidhu and R. L. Cooke: *Opt. Lasers Eng.*, 2000, **34**, 339–353.
87. R. J. Blake: Proc. ICALEO'96 Conf., Detroit, MI, USA, October 1996, Laser Institute of America, E105–E114.
88. R. J. Blake, R. M. Pearson, A. B. Royell and W. E. Simon: *J. Laser Appl.*, 1998, **10**, 199–204.
89. W. Maher, K. O. Tong, C. Bampton, M. Bright, J. Wooten and C. Rhodes: Proc. ICALEO'98 Conf., Orlando, FL, USA, November 1998, Laser Institute of America, 141–150.

90. T. Hatayama and M. Osawa: *J. Jpn Soc. Heat Treat. (Japan)*, 2000, **40**, 116–121.
91. M. Marya and G. R. Edwards: Proc. 5th Int. Conf. on 'Trends in welding research', Pine Mountain, GA, USA, June 1998, ASM International, 982–987.
92. W. H. Wang, M. R. Holl and D. T. Schwartz: *J. Electrochem. Soc.*, 2001, **148**, 363–368.
93. A. Greco, A. Licciulli and A. Maffezzoli: *J. Mater. Sci.*, 2001, **36**, 99–105.
94. L. Lu, J. Y. H. Fuh, Z. D. Chen, C. C. Leong and Y. S. Wong: *Mater. Res. Bull.*, 2000, **35**, 1555–1561.
95. K. Daneshvar, M. Raissi and S. M. Bobbio: *J. Appl. Phys.*, 2000, **88**, 2205–2210.
96. S. Shoji and S. Kawata: *Appl. Phys. Lett.*, 2000, **76**, 2668–2670.
97. O. Lehmann and M. Stuke: *Science*, 1995, **270**, 1644–1646.
98. M. Burns: 'Automotive fabrication: improving productivity in manufacturing'; 1993, Englewood Cliffs, NJ, PTR Prentice-Hall Inc.
99. B. K. Paul and S. Baskaran: *J. Mater. Process. Technol.*, 1996, **61**, 168–172.
100. D. Kochan: 'Solid freeform manufacturing'; 1993, Amsterdam, Elsevier.
101. Y. Z. Zhang, L. K. Shi, J. Cheng, M. Z. Xi and J. Xu: *J. Adv. Mater.*, 2003, **35**, 36–40.
102. E. Goode: *Adv. Mater. Process.*, 2003, **161**, 66–67.
103. K. Dai and L. Shaw: *Acta Mater.*, 2004, **52**, 69–80.
104. K. McAlea, C. Nelson, U. Heimadi and S. Seitz: Proc. Conf. on 'Rapid prototyping/laser applications in the automotive industries', Florence, Italy, June 1997, Automotive Automation Limited, 159–167.
105. A. Bauer, J. Ganz, K. Hesse and E. Kohler: *Appl. Surf. Sci.*, 1990, **46**, 113–120.
106. F. Micheli and I. W. Boyd: *Opt. Laser Technol.*, 1987, **18**, 75–82.
107. M. L. Griffith, M. E. Schlienger, L. D. Harwell, M. S. Oliver, M. D. Baldwin, M. T. Ensz, M. Essien, J. Brooks, C. V. Robino, J. E. Smugeresky, V. P. Mishina, D. L. Wainstein, V. I. Titov, V. F. Moiseev and N. K. Tolochko: *J. Mater. Eng. Perform.*, 2002, **11**, 492–495.
108. J.-Y. Jeng and M.-C. Lin: *J. Mater. Process. Technol.*, 2001, **110**, 98–103.
109. C.-L. Kuo, J.-D. Huang and H.-Y. Liang: *Int. J. Adv. Manuf. Technol.*, 2003, **21**, 796–800.
110. D. S. Choi, S. H. Lee, B. S. Shin, K. H. Whang, Y. A. Song, S. H. Park and H. S. Jee: *J. Mater. Process. Technol.*, 2001, **113**, 273–279.
111. A. L. Kovalev, V. P. Mishina, D. L. Wainstein, V. I. Titov, V. F. Moiseev and N. K. Tolochko: *J. Mater. Eng. Perform.*, 2002, **11**, 492–495.
112. J.-Y. Jeng, S.-C. Peng and C.-J. Chou: *Int. J. Adv. Manuf. Technol.*, 2000, **16**, 681–687.
113. J. Pinkerton and L. Li: *Int. J. Mach. Tools Manuf.*, 2004, **44**, 573–584.
114. A. J. Pinkerton and L. Li: *Appl. Surf. Sci.*, 2003, **208–209**, 405–416.
115. J. Dutta Majumdar, A. Pinkerton, Z. Liu, I. Manna and L. Li: *Appl. Surf. Sci.*, 2005, **247**, 320–327.
116. J. Dutta Majumdar, A. Pinkerton, Z. Liu, I. Manna and L. Li: *Appl. Surf. Sci.*, 2005, **247**, 373–377.
117. Y. Li, H. Yang, X. Lin, W. Huang, J. Li and Y. Zhou: *Mater. Sci. Eng. A*, 2003, **A360**, 18–25.
118. A. Simchi and H. Pohl: *Mater. Sci. Eng. A*, 2003, **A359**, 119–128.
119. P. J. Maziasz, E. A. Payzant, M. E. Schlienger and K. M. McHugh: *Scr. Mater.*, 1998, **39**, 1471–1476.
120. J. D. Hu, Z. X. Guo, Q. F. Guan and Y. L. Li: *Opt. Laser Technol.*, 1997, **29**, 75–78.
121. G. J. Davies and S. Zhen: *J. Mater. Sci.*, 1983, **18**, 1899–1902.
122. Y. P. Kathuria: *J. Mater. Process. Technol.*, 2003, **142**, 466–470.
123. Y. Tang, H. T. Loh, Y. S. Wong, J. Y. H. Fuh, L. Lu and X. Wang: *J. Mater. Process. Technol.*, 2003, **140**, 368–372.
124. X. H. Wang, J. Y. H. Fun, Y. S. Wong, L. Lu, H. T. Loh, Y. X. Tang and H. H. Zhu: *Mater. Sci. Forum*, 2003, **437–438**, 273–276.
125. M. W. Khaing, J. Y. H. Fuh and L. Lu: *J. Mater. Process. Technol.*, 2001, **113**, 269–272.
126. Y. P. Kathuria: Proc. Conf. on '60 years of scientific co-operation in welding', Timisoara, Romania, November 1997, Romanian Welding Society, 199–205.
127. H. H. Zhu, L. Lu and J. Y. H. Fuh: *J. Mater. Process. Technol.*, 2003, **140**, 314–317.
128. J. D. Kim and B. C. Jun: Proc. Conf. on 'Rapid prototyping/laser applications in the automotive industries', Florence, Italy, June 1997, Automotive Automation Limited 201–207.
129. W. Bradley, B. Stucker, P. J. Eubank, S. Norasetthekul, E. Kim and B. Bozkurt: Proc. Conf. on 'Rapid prototyping/laser applications in the automotive industries', Florence, Italy, June 1997, Automotive Automation Limited, 181–184.
130. S. Hayano: *Titan. Jpn*, 2002, **50**, 43–47.
131. N. K. Tolochko, V. V. Savich, T. Laoui, L. Froyen, G. Onofrio, E. Signorelli and V. Totov: *J. Mater.: Des. Appl.*, 2002, **216**, (L4), 267–270.
132. S. Y. Gao, Y. Z. Zhang, L. K. Shi and D. W. Wang: *Mater. Sci. Forum*, 2003, **426–432**, 695–700.
133. J. Mei, J. Liang, R. Sharman, W. Voice and X. Xu: Proc. Conf. on 'Rapid prototyping of materials', Columbus, OH, USA, October 2002, TMS, 133–138.
134. F. G. Arcella and F. H. Froes: *J. Met.*, 2000, **52**, 28–30.
135. D. H. Abbott and F. G. Arcella: *Adv. Mater. Process.*, 1998, **153**, 29–30.
136. D. Srivastava, I. T. H. Chang and M. H. Loretto: *Mater. Des.*, 2000, **21**, 425–433.
137. D. Srivastava, I. T. H. Chang and M. H. Loretto: *Intermetallics*, 2001, **9**, 1003–1013.
138. R. Banerjee, P. C. Collins, A. Gene and H. L. Fraser: *Mater. Sci. Eng. A*, 2003, **A358**, 343–349.
139. R. Banerjee, P. C. Collins, D. C. Bhattacharyya, S. Banerjee and H. L. Fraser: *Acta Mater.*, 2003, **51**, 3277–3292.
140. R. Banerjee, P. C. Collins and H. L. Fraser: *Adv. Eng. Mater. (Germany)*, 2002, **4**, 847–851.
141. H. Tomochika, H. Kikuchi, T. Araki and M. Nishida: *Mater. Sci. Eng. A*, 2003, **A356**, 122–129.
142. X. D. Zhang, C. Brice, D. W. Mahaffey, H. Zhang, K. Schwendner, D. J. Evans and H. L. Fraser: *Scr. Mater.*, 2001, **44**, 2419–2424.
143. X. Wu and J. Mei: *J. Mater. Process. Technol.*, 2003, **135**, 266–270.
144. Y. X. Tang, L. Lu, Y. H. J. Euh, H. T. Loh, Y. S. Wong and C. H. Ng: *Mater. Sci. Forum*, 2003, **437–438**, 301–304.
145. C. A. Brice and H. L. Fraser: *J. Mater. Sci.*, 2003, **38**, 1517–1521.
146. W. L. Karlson, J. Kotila and J. E. Lind: *Int. J. Powder Metall.*, 2004, **40**, 29–41.
147. L. Xue, J. Y. Chen and M. U. Islam: Proc. ICALEO'99 Conf., San Diego, CA, USA, November 1999, Laser Institute of America, Section E, F172–F181.
148. S. Das, T. P. Fuesting, G. Danyo, L. E. Brown, J. J. Beaman and D. L. Bourell: *Mater. Des.*, 2000, **21**, 63–73.
149. K. Cai, D. Guo, Y. Huang and J. L. Yang: *J. Eur. Ceram. Soc.*, 2003, **23**, 921–925.
150. A. Larrea, G. F. de la Fuente, R. I. Merino and V. M. Orera: *J. Eur. Ceram. Soc.*, 2002, **22**, 191–198.
151. L. Lu, J. Y. H. Fuh, Z. D. Chen, C. C. Leong and Y. S. Wong: *Mater. Res. Bull.*, 2000, **35**, 1555–1561.
152. W. Liu and J. N. DuPont: *Scr. Mater.*, 2003, **48**, 1337–1342.
153. Y. P. Kathuria: *Surf. Coat. Technol.*, 1999, **116–119**, 643–647.
154. H. M. Wang, D. Y. Luan and L. Y. Zhang: *Scr. Mater.*, 2003, **48**, 1179–1184.
155. G. Duan and H. M. Wang: *J. Mater. Sci.*, 2002, **37**, 1981–1985.
156. J. Cheng, Y. Z. Zhang, P. Z. Zhang, L. K. Shi and M. Z. Xi: *Heat Treat. Met. (China)*, 2002, **3**, 32–35.
157. J. Bosbach, D. Martin, F. Stietz, T. Wenzel and F. Trager: *Appl. Phys. Lett.*, 1999, **74**, 2605–2608.
158. K. Daneshvar, M. Raissi and S. M. Bobbio: *J. Appl. Phys.*, 2000, **88**, 2205–2210.
159. M. C. Wanke, O. Lehmann, K. Muller, Q. Z. Wen and M. Stuke: *Science*, 1997, **275**, 1284–1286.
160. Y. F. Lu and H. Qiu: *J. Appl. Phys.*, 2000, **88**, 1082–1087.
161. S. Fukumoto, A. Hirose and K. F. Kobayashi: *Mater. Sci. Technol.*, 1993, **9**, 264–271.
162. C. A. Forbis-Parrott: *Weld. J.*, 1991, **70**, 37–42.
163. S. T. Riches: *Weld. Met. Fabr.*, 1993, 79–83.
164. R. S. Parmar: 'Welding engineering and technology'; 1999, New Delhi, Khanna Publishers.
165. J. F. Lancaster: 'Metallurgy of welding'; 1980, London, George Allen and Unwin.
166. W. W. Duley (ed.): 'Laser welding', 1; 1999, New York, John Wiley & Sons, Inc.
167. K. H. Leong, P. A. Kirkham and K. C. Meinert, Jr: *J. Laser Appl.*, 2000, **12**, 181–184.
168. A. Hirose, H. Todaka, K. Yamaoka, N. Kurosawa and K. F. Kobayashi: *Metall. Mater. Trans. A*, 1999, **30A**, 2115–2120.
169. D. C. Weckman, H. W. Kerr and J. T. Liu: *Metall. Mater. Trans. B*, 1997, **28B**, 687–700.
170. M. F. Lee, J. C. Huang and N. J. Ho: *J. Mater. Sci.*, 1996, **31**, 1455–1468.
171. I. R. Whitaker and D. G. McCartney: *Mater. Sci. Eng. A*, 1995, **A196**, 155–163.
172. E. Biro, Y. Zhou, D. C. Weckman and K. J. Ely: *J. Laser Appl.*, 2001, **13**, 96–104.

173. T. Fuhrich, P. Berger and H. Hugel: *J. Laser Appl.*, 2001, **13**, 178–186.
174. M. Farid and P. A. Molian: *J. Mater. Sci.*, 2000, **15**, 3817–3826.
175. E. S. Ng and I. A. Watson: *J. Laser Appl.*, 1999, **11**, 273–278.
176. A. M. El-Batahy: *Mater. Lett.*, 1997, **32**, 155–163.
177. Z. Szymanski, J. Kurzyina and W. Kalita: *J. Phys. D: Appl. Phys.*, 1997, **30**, 3153–3162.
178. W.-H. Cheng, W.-H. Wang and J.-C. Chen: *IEEE Trans. Compon. Packag. Manuf. Technol.*, 1996, **19**, 764–769.
179. G. Wang and K. N. Tandon: *Microgravity Sci. Technol.*, 1995, **8**, 131–133.
180. T. T. Hsu, Y. R. Wang, S. K. Wu and C. Chen: *Metall. Mater. Trans. A*, 2001, **32A**, 569–576.
181. J. M. Jouvard, K. Girard and O. Perret: *J. Phys. D: Appl. Phys.*, 2001, **34(18)**, 2894–2901.
182. L. W. Tsay and C. Y. Tsay: *Int. J. Fatigue*, 1997, **19**, 713–720.
183. A. Hirose, S. Fukumoto and K. F. Kobayashi: *Key Eng. Mater.*, 1995, **104–107**, 853–872.
184. M. Marya and G. R. Edwards: *J. Mater. Eng. Perform.*, 2001, **4**, 435–443.
185. T. Shida, M. Hirokawa and S. Sato: *Weld. Res. Abroad*, 1997, **43**, 36.
186. M. Kern, P. Berger and H. H. Ugel: *Weld. J.*, 2000, **3**, 72–75.
187. R. S. Xiao, G. Ambrosy, T. C. Zuo and H. Hugel: *J. Mater. Sci. Lett.*, 2001, **20**, 2163–2165.
188. J. W. Hirsch, L. G. Olson, Z. Nazir and D. R. Alexander: *Opt. Lasers Eng.*, 1998, **29**, 465–484.
189. X. H. Ye and X. Chen: *J. Phys. D: Appl. Phys.*, 2002, **35**, 1049–1056.
190. P. Solana and G. Negro: *J. Phys. D: Appl. Phys.*, 1997, **30**, 3216–3222.
191. T. M. Yue, C. Y. Jiang, J. H. Xu and W. S. Lau: *J. Mater. Process. Technol.*, 1996, **57**, 316–319.
192. A. Kar and J. Mazumdar: *J. Appl. Phys.*, 1995, **78**, 6353–6360.
193. B. S. Yilbas, R. Davies and Z. Yilbas: *J. Mater. Process. Technol.*, 1991, **25**, 139–151.
194. T. Chande and J. Majumdar: *J. Appl. Phys.*, 1984, **56**, 1981–1986.
195. V. V. Semak, R. J. Steele, P. W. Fuerschbach and B. K. Damkroger: *J. Phys. D: Appl. Phys.*, 2000, **33**, 1179–1185.
196. Y. Peng, W. Chen, C. Wang, G. Bao and Z. Tian: *J. Phys. D: Appl. Phys.*, 2001, **34**, 3145–3149.
197. A. Kar and J. Mazumdar: *J. Appl. Phys.*, 1995, **78**, 6353–6360.
198. R. Rai, G. G. Roy and T. DebRoy: *J. Appl. Phys.*, 2007, **101**, 1–10.
199. T. DebRoy and S. A. David: *Rev. Mod. Phys.*, 1995, **67**, 85–112.
200. N. Chakraborty and S. Chakraborty: *Int. J. Heat Mass Transf.*, 2007, **50**, 1805–1822.
201. G. Phanikumar, P. Dutta and K. Chattopadhyay: *Metall. Trans. B*, 2004, **35B**, 339–350.
202. C. O. Brown and C. M. Banas: ‘Deep penetration laser welding’, Proc. AWS Annual Meet., San Francisco, CA, USA, April 1971, AWS, 50.
203. W. W. Duley: ‘CO₂ laser: effects and applications’, 1976, New York, Academic Press.
204. J. F. Ready: ‘Effects of high power laser radiation’, 1971, New York, Academic Press.
205. J. R. Berretta, W. de Rossi, M. D. M. das Neves, I. A. de Almeida and N. D. Vieira, Jr: *Opt. Lasers Eng.*, 2007, **45**, 960–966.
206. M. M. Schwartz: ‘Metallurgical joining manual’, 1979, New York, McGraw Hill.
207. Z. Li and S. L. Gobbi: *J. Mater. Process. Technol.*, 1997, **70**, 137–144.
208. E. Schubert, M. Classen, I. Zerner, C. Walz and G. Sepold: *J. Mater. Process. Technol.*, 2001, **115**, 2–8.
209. N. Postacioglu, P. Kapadia and J. M. Dowden: *J. Phys. D: Appl. Phys.*, 1997, **30**, 2304–2312.
210. N. Postacioglu, P. Kapadia and J. M. Dowden: *J. Phys. D: Appl. Phys.*, 2000, **33**, 1739–1746.
211. C. F. Niordson: *Int. J. Fract.*, 2001, **111**, 53–69.
212. E. L. Baarden, D. J. Schmatz and R. E. Bisaro: *Weld. J.*, 1973, **52**, 227.
213. R. A. Willgoss, J. H. P. C. Megaw and J. N. Clark: *Opt. Laser Technol.*, 1979, **11**, 73.
214. R. C. Crafer: Proc. 4th Int. Conf. on ‘Advances in welding processes’, Harrogate, UK, May 1978, The Welding Institute, 267–278.
215. J. Mazumdar and W. M. Steen: *Metall. Trans. A*, 1982, **13A**, 865.
216. Y. S. Yang and S. H. Lee: *J. Mater. Process. Technol.*, 1999, **94**, 151–156.
217. C. M. Banas: ‘Laser welding developments’, Proc. GECB Int. Conf. on ‘Welding research related to power plants’, September 1972, Southampton, UK, GECB, B52–B57.
218. B. G. Chung, S. Rhee and C. H. Lee: *Mater. Sci. Eng. A*, 1999, **A272**, 357–362.
219. F. Malek Ghaini, M. J. Hamed, M. J. Torkamany and J. Sabbaghzadeh: *Scr. Mater.*, 2007, **56**, 955–958.
220. C. T. Kwok, S. L. Fong, F. T. Cheng and H. C. Man: *J. Mater. Process. Technol.*, 2006, **176**, 168–178.
221. N. A. McPherson, K. Chi and T. N. Baker: *J. Mater. Process. Technol.*, 2003, **134**, 174–179.
222. V. A. Ventrella, J. R. Berretta and W. De Rossi: *J. Mater. Proc. Technol.*, 2010, **210**, 1838–1843.
223. S. M. Chan, L. C. Chan and T. C. Lee: *J. Mater. Process. Technol.*, 2003, **132**, 95–101.
224. Z. Sun, M. Kuo and T. Moiso: *J. Mater. Sci. Lett.*, 1998, **17**, 355–357.
225. A. P. Costa, L. Quintino and M. Greitmann: *J. Mater. Process. Technol.*, 2003, **141**, 163–173.
226. R. Akhter, M. Davis, J. Dowden, P. Kapadia, M. Ley and W. M. Steen: *J. Phys. D*, 1989, **21**, 23–28.
227. S. Iqbal, M. M. S. Gualini and F. Grassi: *J. Mater. Process. Technol.*, 2007, **184**, 12–18.
228. L. W. Tsay, M. C. Young and C. Chen: *Corros. Sci.*, 2003, **45**, 1985–1997.
229. E. Capello, P. Chiarello, B. Previtali and M. Vedani: *Mater. Sci. Eng. A*, 2003, **A351**, 334–343.
230. W. S. Chang and S. J. Na: *J. Mater. Process. Technol.*, 2002, **120**, 208–214.
231. Y.-F. Tzeng: *J. Mater. Process. Technol.*, 2000, **102**, 40–47.
232. Z. Li and G. Fontana: *J. Mater. Process. Technol.*, 1998, **74**, 174–182.
233. T. Narikiyo, H. Miura, S. Fujinaga, A. Ohmori and K. Inoue: *J. Phys. D: Appl. Phys.*, 1998, **31**, 2331–2337.
234. J. Onoro and C. Ranninger: *J. Mater. Process. Technol.*, 1997, **68**, 68–70.
235. H. Kusuda, T. Takasago and F. Natsumi: *J. Mater. Process. Technol.*, 1997, **71**, 136–140.
236. A. Roustila, N. Kuromoto, A. M. Brass and J. Chene: *J. Nucl. Mater.*, 1994, **211**, 156–167.
237. R. Akhter, K. G. Watkins and W. M. Steen: *Mater. Lett.*, 1990, **9**, 550–556.
238. C. Bagger and F. O. Olsen: *J. Laser Appl.*, 2005, **17**, 2–14.
239. T. Graf and H. Staufer: *Weld. J.*, 2003, **82**, 42–48.
240. J. Yan, M. Gao and X. Zeng: *Opt. Lasers Eng.*, 2010, **48**, 512–517.
241. M. Gao, X. Zeng, J. Yan and Q. Hu: *Appl. Surf. Sci.*, 2008, **254**, 5715–5721.
242. J. Zhang, D. C. Weckman and Y. Zhou: *Weld. Res.*, 2008, **87**, 18–30.
243. H. Zhao and T. DebRoy: *J. Appl. Phys.*, 2003, **93**, 10089–10096.
244. F. Dausinger, P. Berger and H. Hugel: Proc. ICALEO’02 Conf., Phoenix, AZ, USA, October 2002, Laser Institute of America, 2651–2658.
245. M. F. Lee, J. C. Huang and N. J. Ho: *J. Mater. Sci.*, 1996, **31**, 1455–1468.
246. I. R. Whitaker, D. G. McCartney, N. Calder and W. M. Steen: *J. Mater. Sci.*, 1993, **28**, 5469–5478.
247. A. Haboudou, P. Peyre, A. B. Vannes and G. Peix: *Mater. Sci. Eng. A*, 2003, **A363**, 40–52.
248. R. Braun: *Mater. Sci. Eng. A*, 2006, **A426**, 250–262.
249. T. M. Yue, J. H. Xu and H. C. Man: *Appl. Compos. Mater.*, 1997, **4**, 53–64.
250. H. M. Wang, Y. L. Chen and L. G. Yu: *Mater. Sci. Eng. A*, 2000, **A293**, 1–6.
251. Y. L. Qi, J. Deng, Q. Hong and L. Y. Zeng: *Mater. Sci. Eng. A*, 2000, **A280**, 177–181.
252. O. Perret, M. Bizouard, Ph. Naudy, G. Pascal, D. Nore, Y. Horde and Y. Delaisse: *J. Appl. Phys.*, 2001, **90**, 27–30.
253. A. P. Wu, G. S. Zou, J. L. Ren, H. J. Zhang, G. Q. Wang, X. Liu and M. R. Xie: *Intermetallics*, 2002, **10**, 647–652.
254. K. Uenishi and K. F. Kobayashi: *Intermetallics*, 1996, **4**, 95–101.
255. Z. Li, S. L. Gobbi, I. Norris, S. Zolotovskiy and K. H. Richter: *J. Mater. Process. Technol.*, 1997, **65**, 203–208.
256. A. Falvo, F. M. Furgiuele and C. Maletta: *Mater. Sci. Eng. A*, 2005, **A412**, 235–240.
257. K. Girard, J. M. Jouvard and Ph. Naudy: *J. Phys. D: Appl. Phys.*, 2000, **33**, 2815–2824.
258. Z. Xu, K. Natesan, C. B. Reed and D. L. Smith: *Int. J. Refract. Met. Hard Mater.*, 2000, **18**, 231–236.
259. L. Zhang, S. L. Gobbi and K. H. Richter: *J. Mater. Process. Technol.*, 1997, **70**, 285–292.
260. M. L. Santos, H. A. Acciari, L. C. O. Vercik and A. C. Guastaldi: *Mater. Lett.*, 2003, **57**, 1888–1893.
261. G. Song, L. Liu and P. Wang: *Mater. Sci. Eng. A*, 2006, **A429**, 312–319.

262. L. M. Liu, J. F. Wang and G. Song: *Mater. Sci. Eng. A*, 2004, **A381**, 129–133.
263. J. H. Kim, C. Lee, D. M. Lee, J. H. Sun, S. Y. Shin and J. C. Bae: *Mater. Sci. Eng. A*, 2007, **A449–A451**, 872–875.
264. M. Rohde, C. Markert and W. Pfleger: *Int. J. Adv. Manuf. Technol.*, 2010, **50**, 207–215.
265. G. Sepold, E. Schubert and E. Zerner: 'Laser beam joining of dissimilar materials', IIW IV-739-99, Lissabon, Portugal, 1999.
266. Z. Sun and J. C. Ion: *J. Mater. Sci.*, 1995, **30**, 4205–4214.
267. G. Sierra, P. Peyre, F. Deschaux-Beaume, D. Stuart and G. Fras: *Mater. Sci. Eng. A*, 2007, **A447**, 197–208.
268. B. Majumdar, R. Galun, A. Weisheit and B. L. Mordike: *J. Mater. Sci.*, 1997, **32**, 6191–6200.
269. L.-M. Liu, H.-Y. Wang and Z.-D. Zhang: *Scr. Mater.*, 2007, **56**, 473–476.
270. T. A. Mai and A. C. Spowage: *Mater. Sci. Eng. A*, 2004, **A374**, 224–233.
271. P. Kolodziejczak and W. Kalita: *J. Mater. Process. Technol.*, 2009, **209**, 1122–1128.
272. X. D. Qi and G. Song: *Mater. Des.*, 2010, **31**, 605–609.
273. L. M. Liu and X. Zhao: *Mater. Charact.*, 2008, **59**, 1279–1284.
274. E. Schubert, T. Seefeld, I. Zerner, M. Grupp and G. Sepold: *Laser Opt.*, 1999, **31**, 75–77.
275. F. Guo, J. P. Lucas and K. N. Subramanian: *J. Mater. Sci.: Mater. Electron.*, 2001, **12**, 27–35.
276. M. Brandner, G. Seibold, C. Chang, F. Dausinger and H. Hugel: *J. Laser Appl.*, 2000, **12**, 194–199.
277. H. Berkowitz and J. Walvoord: *Solder. Surf. Mt. Technol.*, 1997, **9**, 41–43.
278. K. M. McNally, B. S. Sorg, A. J. Welch, J. M. Dawes and E. R. Owen: *Phys. Med. Biol.*, 1999, **44**, 983–1002.
279. J. Tang, J. M. Evans, V. Petricevic, P. Ho and R. R. Alfano: *IEEE J. Sel. Top. Quantum Electron.*, 1999, **5**, 1103–1106.
280. I. Belic and J. Stanic: *Opt. Laser Technol.*, 1987, **19**, 309–311.
281. S. L. Chen: *J. Mater. Process. Technol.*, 1998, **73**, 147–159.
282. W. O'Neill and J. T. Gabzdyl: *Opt. Lasers Eng.*, 2000, **34**, 355–367.
283. A. Ivarson, J. Powel, J. Kamalu and C. Magnusson: *J. Mater. Process. Technol.*, 1994, **40**, 359–374.
284. K. Danisman, B. S. Yilbas, A. Gorur, R. Davies, C. Ciftlikli, Z. Yilbas and F. Begh: *Opt. Laser Technol.*, 1992, **24**, 33–38.
285. J. Wang and W. C. K. Wong: *J. Mater. Process. Technol.*, 1999, **95**, 164–168.
286. G. V. S. Prasad, E. Siores and W. C. K. Wong: *J. Mater. Process. Technol.*, 1998, **74**, 234–242.
287. A. Belhadji, P. Baudouin, F. Breaban, A. Deffontaine, M. Dewulf and Y. Houbart: *J. Magn. Magn. Mater.*, 2003, **256**, 20–31.
288. T. G. King and J. Powell: *Wear*, 1986, **109**, 135–144.
289. N. Tabata, S. Yagi and M. Hishii: *J. Mater. Process. Technol.*, 1996, **62**, 309–314.
290. H. Y. Zheng, Z. Z. Han, Z. D. Chen, W. L. Chen and S. Yeo: *J. Mater. Process. Technol.*, 1996, **62**, 294–298.
291. S. Skvarenina and Y. C. Shin: *Int. J. Mach. Tools Manuf.*, 2006, **46**, 7–17.
292. O. D. Araujo, F. J. Pacheco, D. Mendez, A. J. Garcia, M. P. Villar, R. Garcia, D. Jimenez and L. Rubio: *Appl. Surf. Sci.*, 2003, **208–209**, 194–198.
293. F. J. Carpio, D. Mendez, A. J. Garcia, M. P. Villar, R. Garcia, D. Jimenez and L. Rubio: *Appl. Surf. Sci.*, 2003, **208–209**, 210–217.
294. F. Muller and J. Molaghan: *J. Mater. Process. Technol.*, 2001, **118**, 278–285.
295. L. M. Wee and Lin Li: *Appl. Surf. Sci.*, 2005, **247**, 277–284.
296. L. Hong, R. M. Vilar and W. Youming: *J. Mater. Sci.*, 1997, **32**, 5545–5550.
297. T. M. Yue and W. S. Lau: *Mater. Manuf. Process.*, 1996, **11**, 17–29.
298. H. Huang, H. Y. Zheng and G. C. Lim: *Appl. Surf. Sci.*, 2004, **228**, 201–206.
299. M. Anderson, R. Patwa and Y. C. Shin: *Int. J. Mach. Tools Manuf.*, 2006, **46**, 1879–1891.
300. J. H. Zhang, T. C. Lee, X. Ai and W. S. Lau: *J. Mater. Process. Technol.*, 1996, **57**, 304–310.
301. G. Lu, E. Siores and B. Wang: *J. Mater. Process. Technol.*, 1999, **88**, 154–158.
302. C. H. Tsai and H.-W. Chen: *J. Mater. Process. Technol.*, 2003, **136**, 166–173.
303. C. H. H. Tsai and C. J. Chen: *Opt. Lasers Eng.*, 2004, **41**, 189–204.
304. L. Hong, L. Li and C. Ju: *Opt. Lasers Eng.*, 2002, **38**, 279–289.
305. L. Hong and L. J. Li: *Opt. Laser Technol.*, 1999, **31**, 531–538.
306. J. C. Rozzi, F. P. Incropera and Y. C. Shin: *Int. J. Heat Mass Transf.*, 2000, **43**, 1425–1437.
307. T. Gilbert, V. D. Krstic and G. Zak: *J. Mater. Process. Technol.*, 2007, **189**, 409–417.
308. I. Black and K. L. Chua: *Opt. Laser Technol.*, 1997, **29**, 193–205.
309. I. Black, S. A. J. Livingstone and K. L. Chua: *J. Mater. Process. Technol.*, 1998, **84**, 47–55.
310. K. Lit and P. Sheng: *Int. J. Mach. Tools Manuf.*, 1995, **35**, 1493–1506.
311. J. Pascual-Cosp, A. J. Ramirez del Valle, J. Garcia-Forte and P. J. Sanchez-Soto: *Mater. Lett.*, 2002, **55**, 274–280.
312. J. K. Park, V. M. Ayres, J. Asmussen and K. Mukherjee: *Diam. Relat. Mater.*, 2000, **9**, 1154–1158.
313. F. E. Pfeifferkorn, Y. C. Shin, Y. Tian and F. P. Incropera: *Trans. ASME*, 2004, **126**, 42–51.
314. P. A. Rebro, Y. C. Shin and F. P. Incropera: *Int. J. Mach. Tools Manuf.*, 2004, **44**, 677–694.
315. K. C. P. Lum, S. L. Ng and I. Black: *Opt. Laser Technol.*, 2000, **32**, 67–76.
316. S. L. Ng, K. C. P. Lum and I. Black: *Opt. Laser Technol.*, 2000, **32**, 77–87.
317. D. K. Shanmugam, F. L. Chen, E. Siores and M. Brandt: *Compos. Struct.*, 2002, **57**, 289–296.
318. E. Uhlmann, G. Spur, H. Hocheng, S. Liebelt and C. T. Pan: *Int. J. Mach. Tools Manuf.*, 1999, **39**, 639–650.
319. T. J. Matthams and T. W. Clyne: *Compos. Sci. Technol.*, 1999, **59**, 1169–1180.
320. I. Black: *J. Mater. Sci. Lett.*, 1998, **17**, 1531–1533.
321. A. A. Cenna and P. Mathew: *Int. J. Mach. Tools Manuf.*, 1997, **37**, 723–736.
322. W. S. Lau, T. M. Yue, T. C. Lee and W. B. Lee: *J. Mater. Process. Technol.*, 1995, **48**, 199–205.
323. P. Sheng and G. Chryssolouris: *J. Mater. Process. Technol.*, 1994, **43**, 145–163.
324. J. M. P. Coelho, M. A. Abreu and F. C. Rodrigues: *J. Opt. Lasers Eng.*, 2004, **42**, 27–39.
325. R. F. de Graaf and J. Meijer: *J. Mater. Process. Technol.*, 2000, **103**, 23–28.
326. D. K. Y. Low, L. Li and P. J. Byrd: *Opt. Laser Technol.*, 2000, **32**, 347–354.
327. J. Cheng, E. J. Kahlen and A. Kar: Proc. ICALEO'00 Conf., Dearborn, MI, USA, October 2000, Laser Institute of America, 232–238.
328. C. Lehan and H. S. Kwok: *Appl. Phys. A*, 2001, **73A**, 45–48.
329. X. Zhu, D. M. Villeneuve, Yu. A. Naumov, S. Nikumb and P. B. Corkum: *Appl. Surf. Sci.*, 1999, **152**, 138–148.
330. Q. Wu, Y. Ma, J. Jie, B. Miao, R. Fang and X. Chen: Proc. ICALEO'01 Conf., Jacksonville, FL, USA, October 2001, Laser Institute of America, 150.
331. A. Corcoran, L. Sexton, B. Seaman, P. Ryan and G. Byrne: Proc. ICALEO'00 Conf., Dearborn, MI, USA, October 2000, Laser Institute of America, 100.
332. T. Beck, S. Seidel and H. Weber: *Laser Mater. Process.*, 2000, **87**, (Part 1), C70–C79.
333. C. Y. Yeo, S. C. Tam, S. Jana, Michael and W. S. Lau: *J. Mater. Process. Technol.*, 1994, **42**, 15–49.
334. M. Boutinguiza, J. Pou, F. Lusquinos, F. Quintero, R. Soto, M. Perez-Amor, K. Watkins and W. M. Steen: *Opt. Lasers Eng.*, 2002, **37**, 15–25.
335. T. Kurita, T. Ono and N. Morita: *J. Mater. Process. Technol.*, 2000, **97**, 168–173.
336. J. K. Lumpp and S. D. Allen: *IEEE Trans. Compon. Packag. Manuf. Technol.*, 1997, **20**, 241–246.
337. H. Zheng, E. Gan and G. C. Lim: *Opt. Lasers Eng.*, 2001, **36**, 355–371.
338. N. Katsarakis, E. Chatzitheodoridis, G. Kiriakidis, M. M. Sigalas, C. M. Soukoulis, W. Y. Leung and G. Tuttle: *Appl. Phys. Lett.*, 1999, **74**, 3263–3265.
339. Z. Illyefalvi-Vitez: *Microelectron. Reliab.*, 2001, **41**, 563–570.
340. P. P. Pronko, S. K. Dutta, J. Squier, J. V. Rudd, D. Du and G. Mourou: *Opt. Commun.*, 1995, **114**, 106–110.
341. S. Nikumb, Q. Chen, C. Li, H. Reshef, H. Y. Zheng, H. Qiu and D. Low: *Thin Solid Films*, 2005, **477**, 216–221.
342. S. Zoppel, M. Farsari, R. Merz and J. Zehetner, G. Stangl, G. A. Reider and C. Fotakis: *Microelectron. Eng.*, 2006, **83**, 1400–1402.
343. P. Psyllaki and R. Oltra: *Mater. Sci. Eng. A*, 2000, **A282**, 145–152.
344. Y. F. Lu, Y. P. Lee and M. S. Zhou: *J. Appl. Phys.*, 1998, **83**, 1677–1684.

345. A. Tsunemi, K. Hagiwara, N. Saito, K. Nagasaka, Y. Miyamoto, O. Suto and H. Tashiro: *Appl. Phys. A*, 1996, **63A**, 435–439.
346. M. J. J. Schmidt, L. Li and J. T. Spencer: *Surf. Coat. Technol.*, 2001, **141**, 40–47.
347. J. M. Lee, C. Curran and K. G. Watkins: *Appl. Phys. A*, 2001, **73A**, 219–224.
348. P. Meja, M. Autric, P. Alloncle, P. Pasquet, R. Oltra and J. P. Boquillon: *Appl. Phys. A*, 1999, **69A**, 687–690.
349. K. Rubahn and J. Ihlemann: *J. Appl. Phys.*, 1999, **86**, 2847–2855.
350. J. Zhang, K. Sugioka, S. Wada, H. Tashiro and K. Toyoda: *Appl. Phys. A*, 1997, **64A**, 367–371.
351. M. Verdier, S. Costil, C. Coddet, R. Oltra and O. Perret: *Appl. Surf. Sci.*, 2003, **205**, 3–21.
352. J. Philip, T. Jayakumar and B. Raj: *Mater. Sci. Eng. A*, 2002, **A338**, 17–23.
353. M. Kawasaki and H. Gijutsu: *J. Surf. Finish. Soc. Jpn*, 2002, **53**, 490–496.
354. H. Funakoshi, K. Jodan, K. Matsumaru and K. Ishizaki: *Interceram*, 2001, **50**, 466–468.
355. Y. Feng, Z. Q. Liu and X. S. Yi: *J. Mater. Sci. Lett.*, 2001, **20**, 517–519.
356. Y. S. Xiao and F. Yi: *J. Mater. Sci. Lett.*, 1999, **18**, 245–247.
357. G. Allcock, P. E. Dyer, G. Elliner and H. V. Snelling: *J. Appl. Phys.*, 1995, **78**, 7295–7303.
358. D. Beu-Zion, A. Inberg, N. Croitoru and S. A. Shalem Katzir: *Opt. Eng.*, 2000, **39**, 1384–1390.
359. L. Monaco, E. Gallus and M. Bianco: *Lamiera*, 2002, **39**, 58–63.
360. J. Bosman: *Materialen (Netherlands)*, 2001, **17**, 4–6.
361. G. S. Buckley and C. M. Roland: *Polym. Eng. Sci.*, 1997, **37**, 138–142.
362. D. R. Alexander and M. S. Khelif: *Opt. Lasers Eng.*, 1996, **25**, 55–70.
363. J. K. Lupp and S. D. Allen: *IEEE Trans. Compon. Packag. Manuf. Technol.*, 1997, **20**, 241–246.
364. S. V. Ponnaluri, R. Cherukuri and P. A. Molian: *J. Mater. Process. Technol.*, 2001, **112**, 199–204.
365. D. Raybould, M. Meola, R. Bye and S. K. Das: *Mater. Sci. Eng. A*, 1998, **A241**, 191–201.
366. A. D. Compaan, I. Matulionis and S. Nakade: *Opt. Lasers Eng.*, 2000, **34**, 15–45.
367. M. D. Berry, B. C. Stuart, P. S. Banks, M. D. Feit, V. Yanovsky and A. M. Rubenchik: *J. Appl. Phys.*, 1999, **85**, 6803–6810.
368. S. Ameer-Beg, W. Perrie, S. Rathbone, J. Wright, W. Weaver and H. Champoux: *Appl. Surf. Sci.*, 1998, **875**, 127–129.
369. R. Osellame, G. D. Valle, N. Chiodo, S. Taccheo, P. Laporta, O. Svelto and G. Cerullo: *Appl. Phys. A*, 2008, **93A**, 885–891.
370. C. Hosokawa, S. N. Kudoh, A. Kiyohara, Y. Hosokawa, K. Okano, H. Masuhara and T. Taguchi: *Appl. Phys. A*, 2008, **93A**, 57–63.
371. F. Dausinger, H. Hugel and V. I. Konov: *Proc. SPIE*, 2003, **5147**, 106–115.
372. M. Tyunina and S. Leppavuori: *J. Appl. Phys.*, 2000, **87**, 8132–8135.
373. J. Cheng, W. Perrie, S. P. Edwardson, E. Fearon, G. Dearded and K. G. Watkins: *Appl. Surf. Sci.*, 2009, **256**, 1514–1520.
374. P. A. Molian: in 'Surface modification technologies – an engineers guide', (ed. T. S. Sudarshan), 421; 1989, New York, Marcel Dekker Inc.
375. C. W. Draper and J. M. Poate: *Int. Met. Rev.*, 1985, **30**, 85–108.
376. C. W. Draper and C. A. Ewing: *J. Mater. Sci.*, 1984, **19**, 3815–3825.
377. C. W. Draper: in 'Laser and electron beam processing of materials', (ed. C. W. White and P. S. Peercy), 721; 1980, New York, Academic Press.
378. J. H. Perepezko and W. J. Boettinger: in 'Surface alloying by ion, electron and laser beams', (ed. L. E. Rehn *et al.*), 51; 1987, Metals Park, OH, USA, ASM.
379. J. Mazumder: *J. Met.*, 1983, **35**, 18–26.
380. E. Kennedy, G. Byrne and D. N. Collins: *J. Mater. Process. Technol.*, 2004, **155–156**, 1855–1860.
381. J. Grum and J. M. Slabe: *Appl. Surf. Sci.*, 2005, **247**, 458–465.
382. M. Heidkamp, O. Kessler, F. Hoffmann and P. Mayr: *Surf. Coat. Technol.*, 2004, **188–189**, 294–298.
383. M. Kulka and A. Pertek: *Mater. Charact.*, 2007, **58**, 461–470.
384. J. Grum and T. Kek: *Thin Solid Films*, 2004, **453–454**, 94–99.
385. M. Heitkemper, C. Bohne, A. Pyzalla and A. Fischer: *Int. J. Fatigue*, 2003, **25**, 101–106.
386. T. M. Shao, M. Hua and T. Y. Tam: *Wear*, 2003, **255**, 444–455.
387. I. Iordanova, V. Antonov and S. Gurkovsky: *Surf. Coat. Technol.*, 2002, **153**, 267–275.
388. M. F. Ashby and K. E. Easterling: *Acta Metall.*, 1984, **32**, 1935–1948.
389. S. Z. Shuja, A. F. M. Arif and B. S. Yilbas: *J. Manuf. Sci. Eng.*, 2002, **124**, 595–604.
390. L. W. Tsay, T. Y. Yang and M. C. Young: *Mater. Sci. Eng. A*, 2001, **A311**, 64–73.
391. A. Amulevicius, K. Mazeika and A. Daugvila: *J. Phys. D: Appl. Phys.*, 2000, **33**, 1985–1989.
392. S. L. Chen and D. Shen: *Int. J. Adv. Manuf. Technol.*, 1999, **15**, 70–78.
393. M. J. Tobar, C. Álvarez, J. M. Amado, A. Ramil, E. Saavedra and A. Yáñez: *Surf. Coat. Technol.*, 2006, **200**, 6362–6367.
394. P. A. Molian: *Mater. Sci. Eng. A*, 1981, **A51**, 253–260.
395. A. Roy and I. Manna: *Mater. Sci. Eng. A*, 2001, **A297**, 85–93.
396. R. S. Lakhkar, Y. C. Shin and M. J. M. Krane: *Mater. Sci. Eng. A*, 2008, **A480**, 209–217.
397. X. F. Wang, X. D. Lu, G. N. Chen, Sh. G. Hu and Y. P. Su: *Opt. Laser Technol.*, 2006, **38**, 8–13.
398. K. H. Lo, F. T. Cheng and H. C. Man: *Surf. Coat. Technol.*, 2003, **173**, 96–104.
399. K. H. Lo, F. T. Cheng, C. T. Kwok and H. C. Man: *Mater. Lett.*, 2003, **58**, 88–93.
400. Y.-Q. Xia, W.-M. Liu and Q.-J. Xue: *Wear*, 2002, **253**, 752–758.
401. A. Yanez, J. C. Alvarez, A. J. Lopez, G. Nicolas, J. A. Perez, A. Ramel and E. Saavedra: *Appl. Surf. Sci.*, 2002, **186**, 611–616.
402. N. Parvathavarthini, R. V. Subbarao, S. Kumar, R. K. Dayal and H. S. Khatak: *J. Mater. Eng. Perform.*, 2001, **10**, 5–13.
403. A. Conde, I. Garcia and J. de Damborene: *Corros. Sci.*, 2001, **43**, 817–828.
404. M. S. Trtica, B. M. Gacovic, T. M. Nenadovic and M. M. Mitrovic: *Appl. Surf. Sci.*, 2001, **177**, 48–57.
405. P. Gopalakrishnan, P. Shankar, R. V. Subba Rao, M. Sundar and S. S. Ramakrishnan: *Scr. Mater.*, 2001, **44**, 707–712.
406. G. Dumitru, V. Romano, H. P. Weber, H. Haefke, Y. Gerbig and E. Pflüger: *Appl. Phys. A*, 2000, **70A**, 485–487.
407. A. Amulevicius, A. Daugvila, R. Davidonis and K. Mazeika: *Acta Phys. Pol.*, 2003, **A104**, 537–548.
408. P. Peyre, X. Scherpereel, L. Berthe, C. Carboni, R. Fabbro, G. Beranger and C. Lemaitre: *Mater. Sci. Eng. A*, 2000, **A280**, 294–302.
409. K. Sun, B. Ke, L. F. Sun, X. Han and K. C. Wang: *Appl. Laser*, 1993, **13(1)**, 5–7.
410. J. Lawrence and L. Li: *J. Phys. D: Appl. Phys.*, 1999, **32**, 2311–2318.
411. R. Colação, C. Pina and R. Vilar: *Scr. Mater.*, 1999, **41**, 715–721.
412. I. Nikitin and I. Altenberger: *Mater. Sci. Eng. A*, 2007, **A465**, 176–182.
413. J. P. Chu, J. M. Rigsbee, G. Banas and H. E. Elsayed-Ali: *Mater. Sci. Eng. A*, 1999, **A260**, 260–268.
414. A. C. Agudelo, J. R. Gancedo, J. F. Marco, M. F. Creus, E. Gallego-Lluesma, J. Desimoni and R. C. Mercader: *Appl. Surf. Sci.*, 1999, **148**, 171–182.
415. Q. Y. Pan, W. D. Huang, R. G. Song, Y. H. Zhou and G. H. Zhang: *Surf. Coat. Technol.*, 1998, **102**, 245–255.
416. L. W. Tsay and Z. W. Lin: *Fatigue Fract. Eng. Mater. Struct.*, 1998, **21**, 1549–1558.
417. R. Colação and R. Vilar: *J. Mater. Sci. Lett.*, 1998, **17**, 563–567.
418. R. Colação and R. Vilar: *Scr. Mater.*, 1998, **38**, 107–113.
419. U. K. Mudali, M. G. Pujar and R. K. Dayal: *J. Mater. Eng. Perform.*, 1998, **7**, 214–220.
420. R. J. Di-Melfi, P. G. Sanders, B. Hunter, J. A. Eastman, K. J. Sawley, K. H. Leong and J. M. Kramer: *Surf. Coat. Technol.*, 1998, **106**, 30–43.
421. X. M. Zhang, H. C. Man and H. D. Li: *J. Mater. Process. Technol.*, 1997, **69**, 162–166.
422. D. K. Das: *Mater. Charact.*, 1997, **38**, 135–141.
423. Q. Song and L. Shen: *Scr. Mater.*, 1997, **36**, 531–534.
424. O. V. Akgun, M. Urgan and A. F. Cakir: *Mater. Sci. Eng. A*, 1995, **A203**, 324–331.
425. S. P. Gadag and M. N. Srinivasan: *J. Mater. Process. Technol.*, 1995, **51**, 150–163.
426. H. M. Wang and H. W. Bergmann: *Scr. Metall. Mater.*, 1995, **32**, 1149–1152.
427. B. E. Wilde, M. Manohar and C. E. Albright: *Mater. Sci. Eng. A*, 1995, **A198**, 43–49.
428. S. P. Gadag and M. N. Srinivasan: *Wear*, 1994, **175**, 21–29.
429. D. Yang, X. Zhang and Q. Xue: *Wear*, 1994, **171**, 13–18.
430. S. C. Tjong and H. C. Tsang: *Surf. Coat. Technol.*, 1993, **57**, 139–144.

431. V. K. Saxena, A. Bharti, G. Malakondaiah and V. M. Radhakrishnan: *Int. J. Fatigue*, 1993, **15**, 369–375.
432. D. R. K. Rao, B. Venkataraman, M. K. Asundi and G. Sundarajan: *Surf. Coat. Technol.*, 1993, **58**, 85–92.
433. O. V. Akgun, O. T. Inal, M. L. Lovato and T. R. Jervi: *Scr. Metall.*, 1992, **27**, 191–194.
434. M. L. Escudero and J. M. Bello: *Mater. Sci. Eng. A*, 1992, **A158**, 227–233.
435. M. B. Karamis and B. S. Yilbas: *Surf. Coat. Technol.*, 1991, **45**, 399–402.
436. W. J. Tomilson and M. G. Talks: *Wear*, 1990, **139**, 269–284.
437. P. A. Molian and M. Baldwin: *Wear*, 1987, **118**, 319–327.
438. H. J. Hegge and J. Th. M. de Hosson: *Scr. Metall.*, 1987, **21**, 1737–1742.
439. P. A. Molian: *J. Eng. Mater. Technol.*, 1987, **109**, 179–187.
440. J. P. Chu, J. M. Riggsbee, G. Banas, F. V. Lawrence, Jr and H. E. Elsayed-Ali: *Metall. Mater. Trans. A*, 1995, **26A**, 1507.
441. R. Fabbro, J. Fournier, P. Ballard, D. Devaux and J. Virmont: *J. Appl. Phys.*, 1990, **68**, 775.
442. P. Ballard, J. Fournier, R. Fabbro, J. Frelat and L. Castex: *J. Phys. Colloq. C3 Suppl.*, 1988, **49**, C3–401.
443. A. H. Clauer, C. T. Walters and S. C. Ford: in 'Laser in materials processing', (ed. M. Bass), 7; 1983, Metals Parks, OH, American Society of Metals.
444. N. Iwamoto: *Surf. Coat. Technol.*, 1988, **34**, 59–67.
445. J. Singh, B. N. Bhat, R. Poorman, A. Kar and J. Mazumder: *Surf. Coat. Technol.*, 1996, **79**, 35–49.
446. F. T. Cheng, C. T. Kwok and H. C. Man: *Surf. Coat. Technol.*, 2001, **139**, 14–24.
447. S. R. Pillai, P. Shankar, R. V. Subba-Rao, N. B. Sivai and S. Kumaravel: *Mater. Sci. Technol.*, 2001, **17**, 1249–1252.
448. L. Renaud, F. Fouquet, A. Elhamdaoui, J. P. Millet, H. Mazille and J. L. Crolet: *Acta Metall. Mater.*, 1990, **38**, 1547–1553.
449. C. T. Kwok, F. T. Cheng and H. C. Man: *Surf. Coat. Technol.*, 2001, **145**, 206–214.
450. I. Manna, K. Kondala Rao, S. K. Roy and K. G. Watkins: in 'Surface engineering in materials science I', (ed. S. Seal *et al.*), 367–376; 2000, Warrendale, PA, TMS.
451. C. T. Kwok, F. T. Cheng and H. C. Man: *Mater. Sci. Eng. A*, 2000, **A290**, 74–88.
452. C. T. Kwok, F. T. Cheng and H. C. Man: *Mater. Sci. Eng. A*, 2000, **A290**, 55–73.
453. Y. Isshiki, J. Shi, H. Nakai and M. Hashimoto: *Appl. Phys. A*, 2000, **70A**, 651–656; 395–402.
454. X. L. Wu and Y. S. Hong: *Metall. Mater. Trans. A*, 2000, **31A**, 3123–3127.
455. A. Agarwal and N. B. Dahotre: *Wear*, 2000, **240**, 144–151; *Metall. Mater. Trans. A*, 2000, **31A**, 401–408.
456. A. Agarwal, L. R. Katipelli and N. B. Dahotre: *Metall. Mater. Trans. A*, 2000, **31A**, 461–473.
457. S. Tondur, T. Schnick, L. Pawlowski, B. Wielage, S. Steinhauser and L. Sabatier: *Surf. Coat. Technol.*, 2000, **123**, 247–251.
458. J. Dutta Majumdar and I. Manna: *Mater. Sci. Eng. A*, 1999, **A267**, 50–59.
459. C. Xie: *Surf. Coat. Technol.*, 1999, **113**, 1–4.
460. C. N. Panagopoulos, A. E. Markaki and P. E. Agathocleous: *Mater. Sci. Eng. A*, 1998, **A241**, 226–232.
461. A. Gutierrez and J. Damborenea: *Oxid. Met.*, 1997, **47**, 259–275.
462. T. R. Jarvis, M. Nastasi, A. J. Griffin, Jr, T. G. Zocco, T. N. Taylor and S. R. Folton: *Surf. Coat. Technol.*, 1997, **89**, 158–164.
463. J. Khedkar, A. S. Khanna and K. M. Gupta: *Wear*, 1997, **205**, 220–227.
464. M. C. Garcia-Alson, M. L. Escudero, V. Lopez and A. Macias: *Corros. Sci.*, 1996, **38**, 515–530.
465. C.-C. Huang, W.-T. Tsai and J.-T. Lee: *Surf. Coat. Technol.*, 1996, **79**, 67–70.
466. P. D. Ding, J. L. Liu, G. Q. Shi, S. Z. Zhou and P. J. Cao: *J. Mater. Process. Technol.*, 1996, **58**, 131–135.
467. E. Gemelli, A. Galerie and M. Caillet: *J. Mater. Sci.*, 1996, **31**, 6627–6630.
468. O. V. Akgun and O. T. Inal: *J. Mater. Sci.*, 1995, **30**, 6105–6112.
469. H. C. Chong, T. T. Wen and L. T. Ju: *Mater. Sci. Eng. A*, 1995, **A190**, 199–205.
470. W.-T. Tsai, T.-H. Lai and J.-T. Lee: *Mater. Sci. Eng. A*, 1994, **A183**, 239–245.
471. A. S. Khanna, R. K. Singh Raman, E. W. Kreutz and A. L. E. Terrance: *Corros. Sci.*, 1992, **33**, 949–958.
472. A. W. Batchelor, G. W. Stachowiak, G. B. Stachowiak, P. W. Leech and O. Reinhold: *Wear*, 1992, **152**, 127–150.
473. Y.-H. Tan and J.-L. Doong: *Wear*, 1991, **145**, 43–60.
474. S. Barnes, N. Timms, B. Bryden and I. Pashby: *J. Mater. Process. Technol.*, 2003, **138**, 411–416.
475. S. F. Corbin, E. Toyserkani and A. Khajepour: *Mater. Sci. Eng. A*, 2003, **A354**, 48–57.
476. A. Conde, F. Zubiri and J. de Damborenea: *Mater. Sci. Eng. A*, 2002, **A334**, 233–238.
477. A. S. C. M. d'Oliveira, R. Vilar and C. G. Feder: *Appl. Surf. Sci.*, 2002, **201**, 154–160.
478. J. Dutta Majumdar, B. Ramesh Chandra and I. Manna: *Tribol. Int.*, 2007, **40**, 146–152.
479. S. Ignata, P. Sallamand, A. Nichici, B. Vannes, D. Grevey and E. Cicala: *Intermetallics*, 2003, **11**, 931–938.
480. S. Shah and N. B. Dahotre: *Mater. Manuf. Process.*, 2002, **17**, 1–12.
481. W. Jiang and P. A. Molian: *Surf. Coat. Technol.*, 2001, **135**, 139–149.
482. M. H. McCay, N. B. Dahotre, J. A. Hopkins and T. D. McCay: *J. Mater. Sci.*, 1999, **34**, 5789–5802.
483. J. M. Pelletier, E. Sauger, Y. Gachon and A. B. Vannes: *J. Mater. Sci.*, 1999, **34**, 2955–2969.
484. D. W. Zhang, T. C. Lei, J. G. Zhang and J. H. Ouyang: *Surf. Coat. Technol.*, 1999, **115**, 176–183.
485. K. C. Chang, W. J. Wei and C. Chen: *Surf. Coat. Technol.*, 1998, **102**, 197–204.
486. A. L. Shyh, T. L. Tung and T. T. Wen: *Scr. Mater.*, 1998, **38**, 559–563.
487. M. A. Anjos, R. Vilar and Y. Y. Qiu: *Surf. Coat. Technol.*, 1997, **92**, 142–149.
488. H. So, C. T. Chen and Y. A. Chen: *Wear*, 1996, **192**, 78–84.
489. K. Nagarathnam and K. Komvopoulos: *Metall. Mater. Trans. A*, 1996, **27A**, 381–390.
490. A. A. Zhukov, J. Dutta Majumdar and I. Manna: *J. Mater. Sci. Lett.*, 1995, **14**, 828–829.
491. A. S. Khanna, R. K. Singh Raman, E. W. Kreutz and L. E. Terrance: *Corros. Sci.*, 1992, **33**, 949–958.
492. J. Khedkar, A. S. Khanna and K. M. Gupta: *Wear*, 1997, **205**, 220–227.
493. C. P. Chan, T. M. Yue and H. C. Man: *J. Mater. Sci.*, 2003, **38**, 2689–2702.
494. E. Sicard, C. Boulmer-Leborgnel, C. Andreazza-Vignolle and M. Frainais: *Appl. Phys. A*, 2001, **73A**, 55–60.
495. N. Hari Prasad and R. Balasubramaniam: *J. Mater. Process. Technol.*, 1997, **68**, 117–120.
496. H. Badekas, A. Koutsomichalis and C. Panagopoulos: *Surf. Coat. Technol.*, 1998, **34**, 365–371.
497. C. Meneau, P. Andreazza, C. Andreazza-Vignolle, P. Goudeau, J. P. Villain and C. Boulmer-Leborgne: *Surf. Coat. Technol.*, 1998, **100–101**, 12–16.
498. L. R. Katipelli, A. Agarwal and N. B. Dahotre: *Mater. Sci. Eng. A*, 2000, **289**, 34–40.
499. A. H. Wang, B. D. Zhu, Z. Y. Tao, X. Y. Ma, S. J. Deng and X. D. Cheng: *Surf. Coat. Technol.*, 1993, **57**, 169–172.
500. K. Uenishi and K. F. Kobayashi: *Intermetallics*, 1999, **7**, 553–559.
501. P. Kadolkar and N. B. Dahotre: *Mater. Sci. Eng. A*, 2003, **A342**, 183–191.
502. A. Almeida, M. Anjos, R. Vilar, R. Li, M. G. S. Ferreira, W. M. Steen and K. G. Watkins: *Surf. Coat. Technol.*, 1995, **70**, 221–229.
503. K. G. Watkins, Z. Liu, M. McMahon, R. Vilar and M. G. S. Ferreira: *Mater. Sci. Eng. A*, 1998, **A252**, 292–300.
504. M. G. Ferreira, R. Li and R. Vilar: *Corros. Sci.*, 1996, **38**, 2091–2094.
505. T. M. Yue, S. M. Mei, K. C. Chan and H. C. Man: *Mater. Lett.*, 1999, **39**, 274–279.
506. A. K. Jain, V. N. Kulkarni and D. K. Sood: *Thin Solid Films*, 1981, **86**, 1–9.
507. A. Almeida, P. Petrov, I. Nogueira and R. Vilar: *Mater. Sci. Eng. A*, 2001, **A303**, 273–280.
508. J. L. de Mol van Otterloo and J. Th. M. de Hosson: *Scr. Metall.*, 1994, **30**, 493–498.
509. M. A. Pinto, N. Cheung, M. C. F. Ierardi and A. Garcia: *Mater. Charact.*, 2003, **50**, 249–253.
510. M. Gremaud, M. Carrard and W. Kurz: *Acta Metall. Mater.*, 1990, **38**, 2587–2599.
511. T. T. Wong, G. Y. Liang and C. Y. Tang: *J. Mater. Process. Technol.*, 1997, **66**, 172–178.
512. T. M. Yue, Y. X. Wu and H. C. Man: *Surf. Coat. Technol.*, 1999, **114**, 13–18.

513. R. Li, M. G. S. Ferreira, A. Almeida, R. Vilar, K. G. Watkins, M. A. McMahon and W. M. Steen: *Surf. Coat. Technol.*, 1996, **81**, 290–296.
514. J. A. Vreeling, V. Ocelik, Y. T. Pei, D. T. L. van Agterveld and J. Th. M. de Hosson: *Acta Mater.*, 2000, **48**, 4225–4233.
515. E. Carpenne and P. Schaaf: *Appl. Surf. Sci.*, 2002, **186**, 100–104.
516. Y. B. Liu, J. D. Hu, Z. Y. Cao and P. K. Rohatgi: *Wear*, 1997, **206**, 83–86.
517. Y. Fu, A. W. Batchelor, H. Xing and Y. Gu: *Wear*, 1997, **210**, 157–164.
518. C. H. Tang, F. T. Cheng and H. C. Man: *Surf. Coat. Technol.*, 2006, **200**, 2606–2609.
519. J. Dutta Majumdar, B. Ramesh Chandra, A. K. Nath and I. Manna: *Surf. Coat. Technol.*, 2006, **201**, 1236–1242.
520. X. B. Zhou and J. Th. M. de Hosson: *Scr. Metall.*, 1993, **28**, 219–224.
521. Dubourg, D. Ursescu, F. Hlawka and A. Cornet: *Wear*, 2005, **258**, 1745–1754.
522. I. Manna, S. Abraham, G. Reddy, D. N. Bose, T. B. Ghosh and S. K. Pabi: *Scr. Metall. Mater.*, 1994, **31**, (6), 713–718.
523. A. Michaelides and C. Panagopoulos: *Surf. Coat. Technol.*, 1993, **57**, 173–177.
524. J. M. Vandenberg and C. W. Draper: *Mater. Lett.*, 1984, **2**, 386–391.
525. J. Dutta Majumdar and I. Manna: *Mater. Sci. Eng. A*, 1999, **A268**, 216–226 (Part 1); 227–235 (Part 2).
526. Z. Zhao, C. Wang, M. Li, L. Wang and L. Kong: *Appl. Surf. Sci.*, 2006, **252**, 4257–4263.
527. Y. S. Tian, C. Z. Chen, D. Y. Wang, Q. H. Huo and T. Q. Lei: *Appl. Surf. Sci.*, 2005, **250**, 223–227.
528. J. Dutta Majumdar, B. L. Mordike and I. Manna: *Wear*, 2000, **242**, 18–27.
529. J. Dutta Majumdar, B. L. Mordike, S. K. Roy and I. Manna: *Oxid. Met.*, 2002, **57**, 473–498.
530. J. Dutta Majumdar, A. Weisheit, B. L. Mordike and I. Manna: *Mater. Sci. Eng. A*, 1999, **A266**, 123–134.
531. I. Manna, W. M. Steen and K. G. Watkins: in 'Surface engineering in materials science I', (ed. S. Seal *et al.*), 377–384; 2000, Warrendale, PA, TMS.
532. I. Manna, W. M. Steen and K. G. Watkins: *Scr. Mater.*, 1997, **37**, 561–568.
533. K. G. Watkins, W. M. Steen, I. Manna, D. F. Williams, S. Rhodes, P. Mazzoldi, S. L. Russo, M. G. S. Ferreira, J. T. Rito, T. M. Silva and A. M. P. Simoes: in Proc. ICALEO'96 Conf., (ed. W. Duley *et al.*), Vol. 81A, 37–46; 1996, Detroit, MI, USA, Laser Institute of America.
534. B. S. Yilbas, M. S. J. Hashmi and S. Z. Shuja: *Surf. Coat. Technol.*, 2001, **140**, 244–250.
535. K. P. Cooper, P. L. Slebodnick, K. E. Lucas and E. A. Hogan: *J. Mater. Sci.*, 1998, **33**, 3805–3816.
536. L. Fouilland-Paille, S. Ettaqi, S. Benayoun and J. J. Hantzpergue: *Surf. Coat. Technol.*, 1997, **88**, 204–211.
537. V. M. Weerasinghe, D. R. F. West and J. de Damborenea: *J. Mater. Process. Technol.*, 1996, **58**, 79–86.
538. J. E. Masse and J. F. Mathieu: *Mater. Manuf. Process.*, 1996, **11**, 199–206.
539. D. Dube, M. Fiset, A. Couture and I. Nakatsugawa: *Mater. Sci. Eng. A*, 2001, **A299**, 38–45.
540. D. Schippman, A. Weisheit and B. L. Mordike: *Surf. Eng.*, 1999, **15**, 23–26.
541. T. M. Yue, A. H. Wang and H. C. Man: *Scr. Mater.*, 1998, **38**, 191–198.
542. G. Abbas, Z. Liu and P. Skeldon: *Appl. Surf. Sci.*, 2005, **247**, 347–353.
543. J. Dutta Majumdar, R. Galun, B. L. Mordike and I. Manna: *Mater. Sci. Eng. A*, 2003, **A361**, 119–129.
544. T. M. Yue, Q. W. Hu, Z. Mei and H. C. Man: *Mater. Lett.*, 2001, **47**, 165–170.
545. A. H. Wang and T. M. Yue: *Compos. Sci. Technol.*, 2001, **61**, 1549–1554.
546. S. Ignat, P. Sallamand, D. Grevey and M. Lambertin: *Appl. Surf. Sci.*, 2004, **225**, 124–134.
547. J. Dutta Majumdar, B. L. Mordike, R. Galun and I. Manna: *Lasers Eng.*, 2002, **12**, 147–170.
548. A. H. Wang and T. M. Yue: *J. Mater. Sci. Lett.*, 2001, **20**, 1965–1967.
549. K. Williams, N. Jaikumar, G. Saiprasanna, M. Hegler and J. Maxwell: Proc. Solid Freeform Fabrications Symp., Austin, TX, USA, August 1999, University of Texas at Austin, 543–551.
550. D. Bauerle: 'Chemical processing with lasers'; 71–93; 1986, Berlin, Springer-Verlag.
551. R. Banal, T. Kimura and T. Goto: *Mater. Trans.*, 2005, **46**, 2114–2116.
552. J. Christopher, W. K. S. Jensen and K. S. Chiu: *Surf. Coat. Technol.*, 2006, **201**, 2822–2828.
553. D. Jean, C. Duty, B. Fuhrman and W. J. Lackey: Proc. Solid Freeform Fabrication Symp., Austin, TX, USA, August 1999, University of Texas at Austin, 59–65.
554. M. S. Noel and D. Kover: *J. Mater. Sci.*, 2002, **37**, 689–697.
555. T. Goto and T. Kimura: *Thin Solid Films*, 2006, **515**, 46–52.
556. M. Jian, R. Johnson and W. J. Lackey: *J. Am. Ceram. Soc.*, 2006, **89**, (2), 519–526.
557. M. G. Heras, M. A. de Buergero, E. Rebollar, M. Oujja, M. Castillejo and R. Fort: *Appl. Surf. Sci.*, 2003, **219**, 290–299.
558. J. Lawrence and L. Li: *J. Mater. Process. Technol.*, 2003, **142**, 461–465.
559. H. M. Wang, Y. L. Yu and S. Q. Li: *Scr. Mater.*, 2002, **47**, 57–61.
560. L. Shepeleva, B. Medres, W. D. Kaplan, M. Bamberger, M. H. McCay, T. D. McCay and M. Sharp: *Surf. Coat. Technol.*, 2000, **125**, 40–44.
561. D. Sciti, C. Melandri and A. Bellosi: *J. Mater. Sci.*, 2000, **35**, 3799–3810.
562. L. Bradley, L. Li and F. H. Stott: *Mater. Sci. Eng. A*, 2000, **A278**, 204–212.
563. M. J. J. Schmidt and L. Li: *Appl. Surf. Sci.*, 2000, **168**, 9–12.
564. B. Stolz and R. Poprawe: *Surf. Coat. Technol.*, 1999, **112**, 394–400.
565. K. H. zum Gahr: *Wear*, 1996, **200**, 215–224.
566. M. I. Cooper, D. C. Emmony and J. Larson: *Opt. Laser Technol.*, 1995, **27**, 69–73.
567. D. H. Lowndes, M. DeSilva, M. J. Godbole, A. J. Pedraza, T. Thundat and R. J. Warmack: *Appl. Phys. Lett.*, 1994, **64**, (14), 1791–1793.
568. C. Buerhop, B. Blumenthal, R. Weissmann, N. Lutz and S. Biermann: *Appl. Surf. Sci.*, 1990, **46**, 430–434.
569. S. P. Harimkar and N. B. Dahotre: *Scr. Mater.*, 2008, **58**, 545–548.
570. W. Wong, K. Chan, K. W. Yeung and K. S. Lau: *J. Mater. Process. Technol.*, 2003, **132**, 114–118.
571. K. P. Adhi, R. L. Owings, T. A. Railkar, W. D. Brown and A. P. Malshe: *Appl. Surf. Sci.*, 2003, **218**, 17–23.
572. F. Huang, Q. Lou, J. Dong and Y. Wei: *Appl. Surf. Sci.*, 2001, **174**, 1–6.
573. M. Csete, O. Marti and Zs. Bor: *Appl. Phys. A*, 2001, **73A**, 521–526.
574. V. Drinek, H. Niino, J. Pola and A. Yabe: *Appl. Phys. A*, 2001, **73A**, 527–530.
575. Q. H. Lu, M. Li, J. Yin, Z. K. Zhu and Z. G. Wang: *J. Appl. Polym. Sci.*, 2001, **82**, 2739–2743.
576. M. Csete and Z. Bor: *Appl. Surf. Sci.*, 1998, **133**, 5–16.
577. A. Hartwig, G. Vittr and V. Schlett: *Int. J. Adhes. Adhes.*, 1997, **17**, 373–377.
578. H. Frerichs, J. Stricker, D. A. Wesner and E. W. Kreutz: *Appl. Surf. Sci.*, 1995, **86**, 405–410.
579. E. Arenholz, J. Heitz, M. Wagner, D. Bzuerle, H. Hibst and A. Hagemeyer: *Appl. Surf. Sci.*, 1993, **69**, 16–19.
580. H. J. Leamy, S. T. T. Rozgonyi and G. K. Celler: *Appl. Phys. Lett.*, 1978, **32**, 535–537.
581. P. A. Temple and M. J. Soileau: *IEEE J. Quantum Electron.*, 1981, **17**, 2067–2072.
582. H. M. van Driel, J. E. Sipe and J. F. Young: *J. Lumin.*, 1985, **30**, 446–471.
583. C. T. Walters: *Appl. Phys. Lett.*, 1974, **25**, 696–698.
584. M. Bolle and S. Lazare: *Appl. Surf. Sci.*, 1993, **69**, 31–37.
585. A. Buchman, H. Dodiuk, M. Rotal and J. Zahavi: *Int. J. Adhes. Adhes.*, 1991, **11**, 3–10.
586. <http://www.coherent.com/Applications>.